
CanSat 2024 Critical Design Review

2044
PWr Aerospace

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Paweł Kabacik, PhD Eng
Faculty Advisor



Jakub Kaszowski
Team leader
3rd bachelor



Hubert Kulik
Leader of Electronics
4th bachelor



Paweł Rak
Leader of Mechanics
3rd bachelor



Ula Wiśniewska
Leader of Management
2nd master



Julia Szymańska
Ground Station
3rd bachelor



Maciej Sikorski
Electronic
2nd bachelor



Szymon Jutrzenka
Mechanic
2nd bachelor



Tymoteusz Puślecki
Management
2nd bachelor



Maksymilian Tara
Electronic
3rd bachelor



Filip Ziolo
Electronic
2nd bachelor



Eunika Puślecka
Tour supervisor

A – Analysis

ABS – Acrylonitrile butadiene styrene

API – Application Programming Interface

ASA - Acrylonitrile styrene acrylate

CAD – Computer Aided Design

CDH – Communication and Data Handling

CDR – Critical Design Review

CONOP – Concept of Operations

CG – Center of Gravity

CP – Center of Pressure

CSV – Comma Separated Value

D – Demonstration

DCS – Descent Control System

DPS – Differential Pressure Sensor

EPS – Electrical Power Subsystem

FR-4 – Glass-reinforced epoxy laminate

FRR – Flight Readiness Review

FSW – Flight Software

GCS – Ground Control System

GPS – Global Positioning System

GS – Ground Station

HW – Hardware

I – Inspection

I2C, IIC – Inter-Integrated Circuit

IMU – Inertial Measurement Unit

LCO – Launch Control Officer

MCU – Microcontroller Unit

uC – Microcontroller

ADC – Analog to Digital Converter

RTC – Real Time Clock

SDIO – Secure digital input/output interface

NMEA – National Marine Electronics Association Protocol

PA-6 – Nylon - Polyamide 6

PDR – Preliminary Design Review

PCB – Printed Circuit Board

PETG – Polyethylene terephthalate

PFB – Pre-Flight Briefing

PFR – Post Flight Review

PWM – Pulse Width Modulation

RPM – Rotations Per Minute

RSO – Range Safety Officer

SOE – Sequence of Events

SPI – Serial Peripheral Interface

T – Test

TBD – To Be Determined

TBR – To Be Resolved

TBT – To Be Tested

UART – Universal Asynchronous Receiver-Transmitter

USB – Universal Serial Bus

VM –Verification method

Systems Overview

Paweł Rak

Main objectives

The CanSat shall be launched to a maximum altitude of 725 meters.

The CanSat shall operate as a nose cone during ascent and include a pitot tube.

The descent rate using an aero-braking heatshield shall be between 10 to 30 meters/sec.

At 100 meters, CanSat shall release the aero-braking heat shield and deploy a parachute to reduce the descent rate to less than 5 meters/sec.

The CanSat shall land with the egg intact.

A video recording camera shall be included capturing the horizontal view during ascent and landing. During descent, the camera shall point in one direction and maintain that direction.

CanSat shall collect the required telemetry at a 1 Hz sample rate and transmit the telemetry data to the ground station.

Bonus objective

A video camera shall be integrated into the CanSat and point aft. The camera shall capture the Cansat being deployed from the rocket and the release of the parachute. The video shall be in color with a minimum resolution of 640x480 pixels and a minimum of 30 frames per second. We made a decision to go for the bonus objective.

External objective

Create as a well-coordinated team. Make a video in a better resolution than last year.

We made **no major changes** since submitting PDR.

Small changes, mainly related to materials are introduced in:

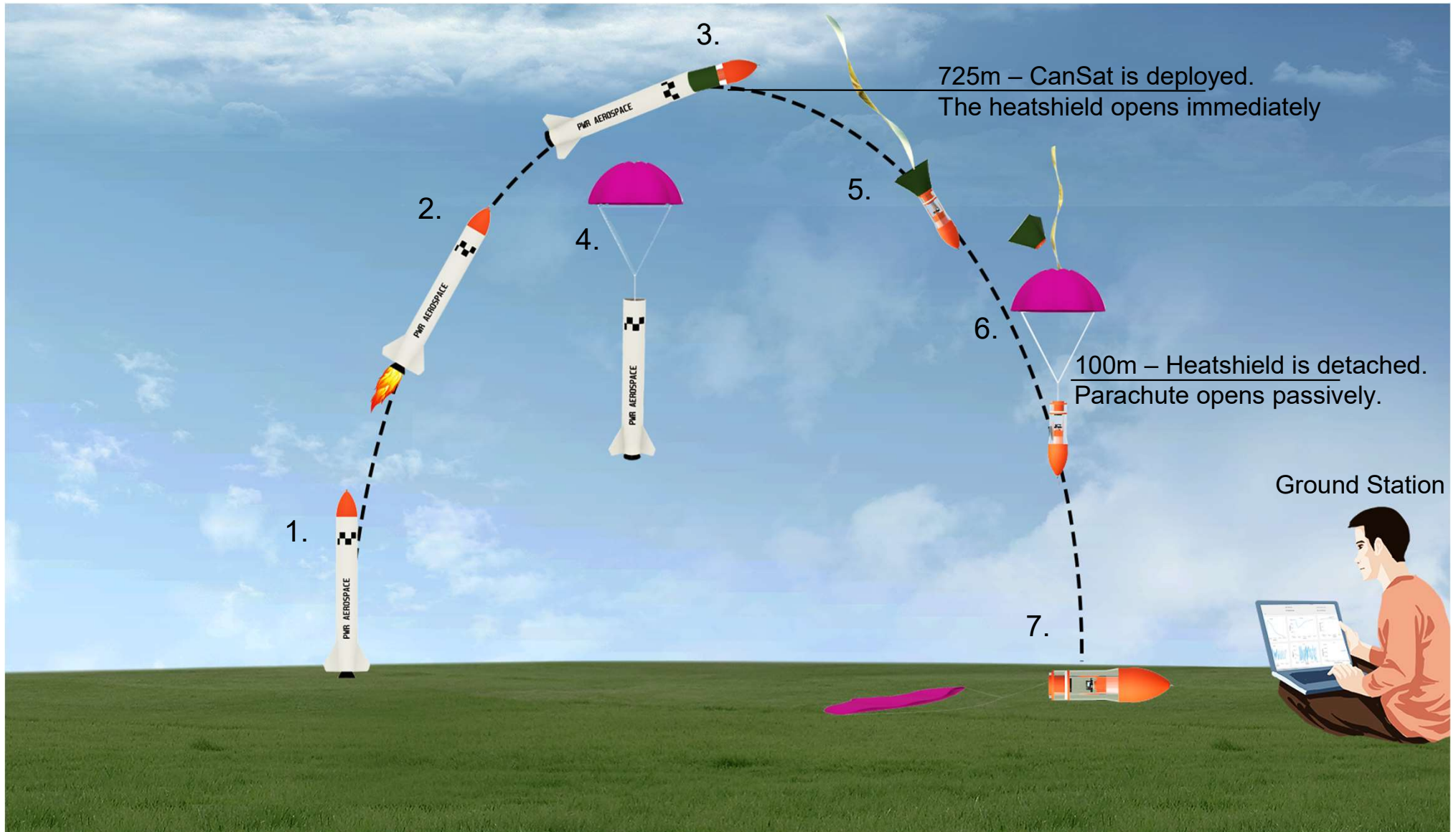
Mass Budget [52]

MSD [39]

Rn	Requirement	Verification			
		A	I	T	D
1	The Cansat shall be deployed from the rocket when the rocket motor ejection charge fires.	X		X	X
2	After deployment from the rocket, the Cansat shall deploy its heat shield/aerobraking mechanism.		X	X	X
3	At 100 meters, the Cansat shall deploy a parachute and release the heat shield.	X		X	X
4	During descent with the heat shield deployed, the descent rate shall be between 10 to 30 m/s.	X		X	
5	All telemetry shall be displayed in real time during descent on the ground station.	X			X
7	The heat shield shall be used as an aerobrake and limit the descent rate to 10 to 30 m/s.	X			X
8	At 100 meters, the Cansat shall release a parachute to reduce the descent rate to less than 5 m/s.	X		X	

Rn	Requirement	Verification			
		A	I	T	D
9	A silver or gold mylar streamer of 50 mm width and 1.5 meters length shall be connected to the CanSat and released at deployment. This will be used to locate and identify the CanSat.			X	X
10	Upon landing, the CanSat shall stop transmitting data.	X			X
11	Upon landing, the CanSat shall activate an audio beacon.	X			X
12	After the CanSat has separated from the rocket and if the nose cone portion of the CanSat is to be separated from the rest of the CanSat, the nose cone portion shall descend at less than 10 meters/second using any type of descent control device.	X			

System Concept of Operations (CONOPS) (1/2)



System Concept of Operations (CONOPS) (2/2)

Mission phase:

1. Pre-launch preparation.
2. Launch and ascend to apogee.
3. Probe separation from the rocket at around 700m.
4. Rocket descends on its parachute, then lands.
5. Deployment of probe's aerobraking mechanism. Descent at 10-30m/s.
6. At 100m the probe detaches the heat shield and its mechanism, opens the parachute and slows below 5m/s.
7. CanSat lands with the egg intact, end of telemetry and cameras recording. Audio and light beacon activation.

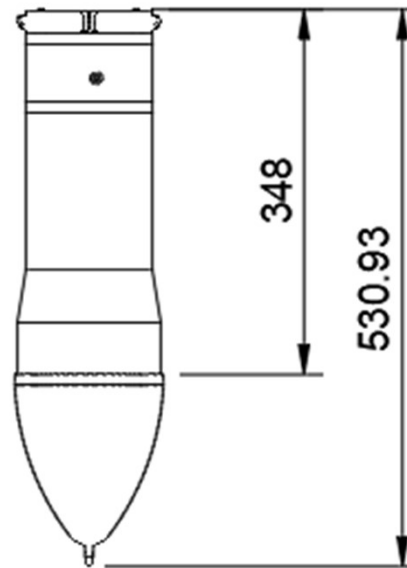
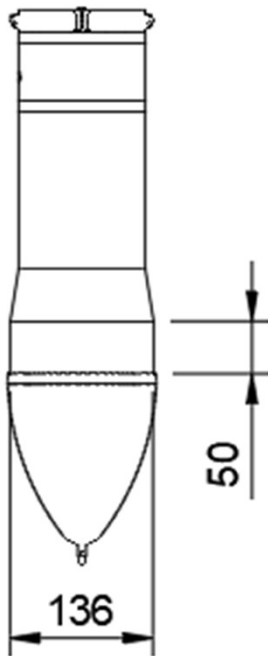
Recovery and data reduction:

1. Probe recovery using the latest known GPS position, audio and light beacon, fluorescent probe body color and observation of mylar streamer during descent.
2. Video data recovery from camera internal storage.
3. Sampled data analysis, formatting and data reduction.
4. PFR preparation.
5. PFR presentation.

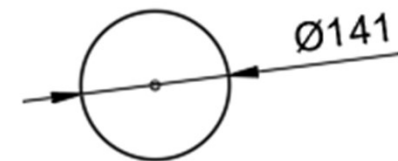
Role	Person	Responsibilities
Mission Control Officer (MCO)	Jakub Kaszowski	Communication with judges
Ground Station Crew (G)	Julia Szymańska, Tymoteusz Puślecki, Filip Ziolo	Ground Station preparation, Receiving telemetry, Issuing commands to CanSat
Recovery Crew (R)	Maciej Sikorski, Szymon Jutrzenka	Finding CanSat after mission
CanSat Crew (C)	Paweł Rak, Maksymilian Tara, Hubert Kulik	Preparing the CanSat, integrating it into the rocket and verifying its status

The most important dimensions:

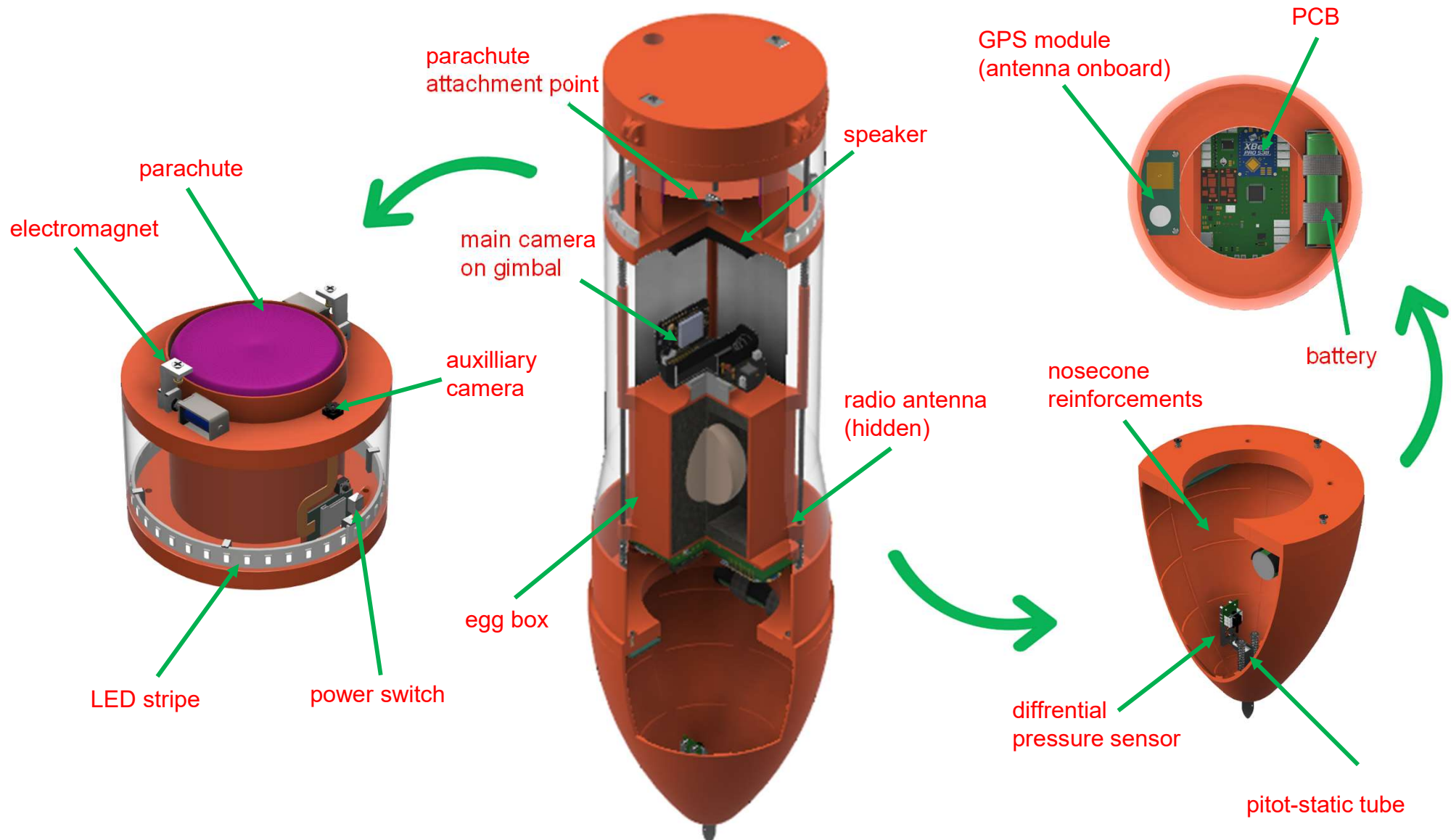
Shoulder:



Nosecone:

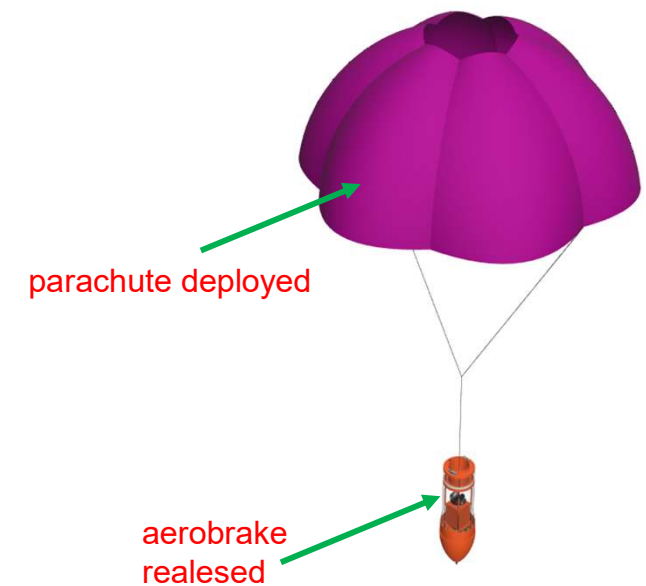
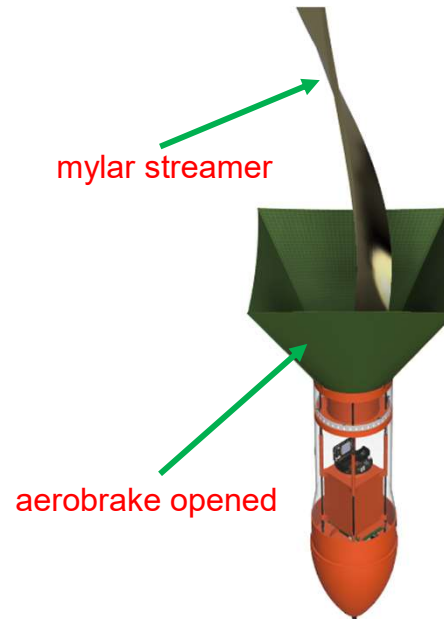
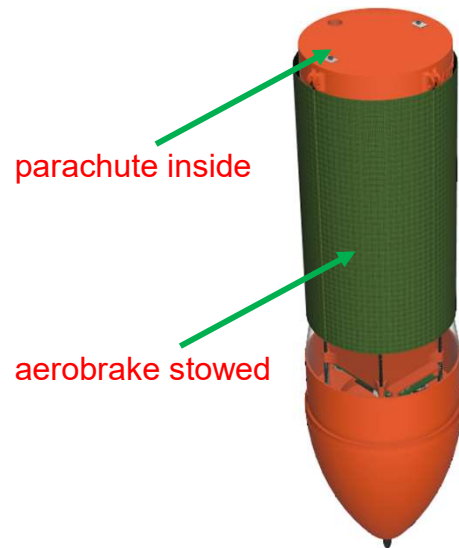


Note: All dimensions are in mm



Configurations in flight stages

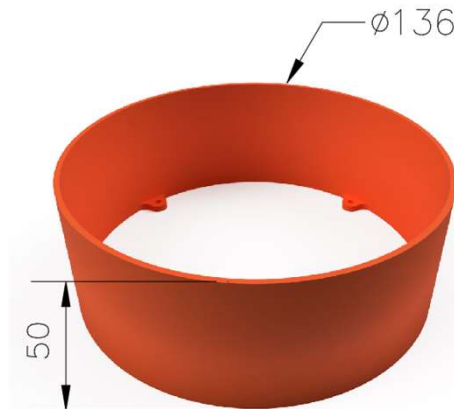
Stage 1 (pre-deployment CanSat configuration)	Stage 2 (heatshield descent configuration)	Stage 3 (parachute descent configuration)
While inside the rocket, the CanSat is in the position shown below. Parachute is in the storage. The heatshield structure is in a closed position.	Heatshield opens immediately after deployment. The mylar streamer is unfolded and stabilizes the flight. In this configuration, the CanSat reaches a velocity between 10 to 30 m/s.	At the height of 100m, the heatshield is detached from the CanSat by electromagnets. It also triggers the parachute opening. The CanSat slows down to the landing speed.



Accurate dimensions

Shoulder and nose cone dimensions are **exactly as provided by the Mission Guide**. It allows CanSat to fit properly in the Payload section of the rocket. The length of the payload, from the top of the airframe is 350 mm. For the Rocket to fly straight the nose cone is symmetrical around the vertical axis. The heatshield has a diameter of 129 mm when closed, this leaves some margin for easy deployment.

Dimensions	Payload	Shoulder	Nose cone
Diameter	129 mm	136 mm	141 mm
Length	348 mm	50 mm	183.5 mm



Protrusions

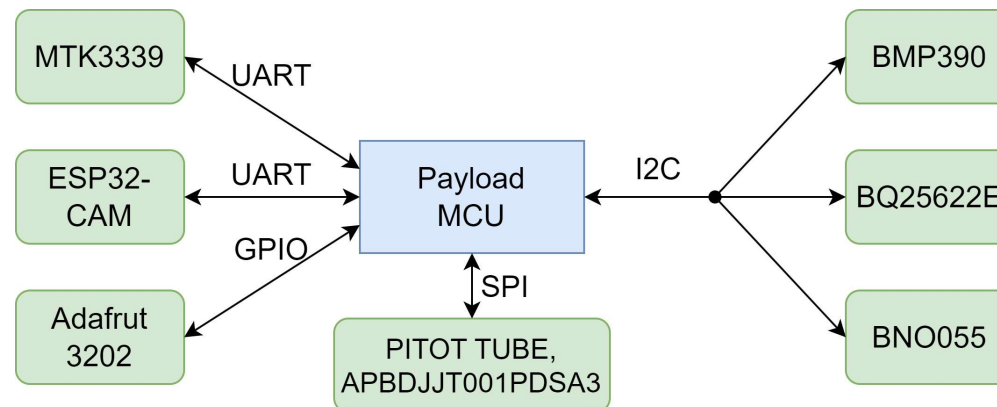
Edges will be rounded up to minimize the risk of failure during deployment. No additional protrusions are present. The parachute is located in the parachute compartment.

Note: All dimensions are in mm

Sensor Subsystem Design

Jakub Kaszowski

TYPE	MODEL	FUNCTIONS	INTERFACE
AIR PRESSURE AIR TEMPERATURE	BMP390	Air temperature and pressure (altitude) measurement	I2C
GPS	MTK3339	Tracking the payload position	UART
VOLTAGE	BQ25622E	Monitoring the battery condition	I2C
GYROSCOPE	BNO055	Tracking the payload tilt and rotation	I2C
SPEED	PITOT TUBE, ABPDJJT001PDSA3	Tracking the payload speed	SPI
CAMERA	ESP32-CAM	Capturing the video during flight	UART
BONUS CAMERA	Adafruit 3202	Capturing the view behind the CanSat	GPIO



We made **no changes** since submitting PDR in the SSD section.

Payload Air Pressure Sensor Summary

Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	Range [hPa]	Resolution [Pa]	Accuracy [hPa]	Interface	Price [\$]
BMP390	2 2 0.75	0.9	1.8 ÷ 3.6	0.730	300 ÷ 1250	0.016	±0.03	I2C	4

After enabling the pressure measurement, the pressure registers are periodically read and combined into one variable (*formula 1*). Then, the compensation formula from the datasheet is applied (*formula 2*), and the result is divided by 256 as a **floating point** (*formula 3*). **It gives the measured pressure in Pa.**

Formula 1:

$pressure = data[0] \ll 12 \mid data[1] \ll 4 \mid data[2] \gg 4;$

Formula 2:

$pressure_compensate = \text{datasheet page 56.}$

Formula 3:

$pressure_Pa = (float)pressure_compensate / 256;$



The data is stored in IEEE 754 float.

To convert pressure to altitude we use the formula:
Where p_0 is the air pressure measured at the launch pad.

$$altitude = 44330 * \left(1 - \left(\frac{p}{p_0} \right)^{\frac{1}{5.255}} \right)$$

We expect raw relative altitude accuracy of ±1m, after filtration the accuracy can be as low as ±0.05m.

p – measured pressure value at altitude [Pa]
 p_0 – reference pressure value at reference level [Pa]

Payload Air Temperature Sensor Summary

Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	Range [°C]	Resolution [°C]	Accuracy [°C]	Interface	Price [\$]
BMP390	2 2 0.75	0.9	1.8 ÷ 3.6	0.320	-40 ÷ 85	0.005	±0.5	I2C	4

After enabling the temperature measurement, the temperature registers are periodically read and combined into one variable (*formula 1*). Then, the compensation formula from the datasheet is applied (*formula 2*), and the result is divided by a 100 as a **floating point** (*formula 3*). **It gives the measured temperature in °C.**

Formula 1:

temperature = data[0] << 12 | data[1] << 4 | data[2] >> 4;

Formula 2:

temperature_compensate = datasheet page 45.

Formula 3:

temperature_C = (float) temperature_compensate / 100

The data is stored in IEEE 754 float.



Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	Sensitivity [dBm]	Resolution [m]	Update rate [Hz]	Interfaces	Price [\$]
MTK3339	23 51 7	8.8	3 ÷ 5.5	20	-165	<2.5	1÷10	UART	38

The GPS communicates through UART using the NMEA protocol. To determine the position of the payload, the **\$GPGGA** command received from the GPS sensor is parsed.

Command template:

\$GPGGA,hhmmss.ss,llll.ll,a,yyyy.yy,a,x,xx,x.x,x.x,
M,x.x,M,x.x,xxxx*hh

The table on the right explains all the segments of the command in order. Bolded values are required by the mission

Time resolution is **0.01s**.

Altitude resolution is **0.1m**

Latitude/Longitude resolution is **0.0001°**.

Number of satellites is an **integer**.

No.	Description
1	UTC of Position
2	Latitude
3	N or S
4	Longitude
5	E or W
6	GPS quality indicator
7	Number of satellites in use
8	Horizontal dilution of position
9	Antenna altitude above/below mean sea level
10	Meters (Antenna height unit)
11	Geoidal separation
12	Meters (Units of geoidal separation)
13	Update age
14	Diff. reference station ID#
15	Checksum

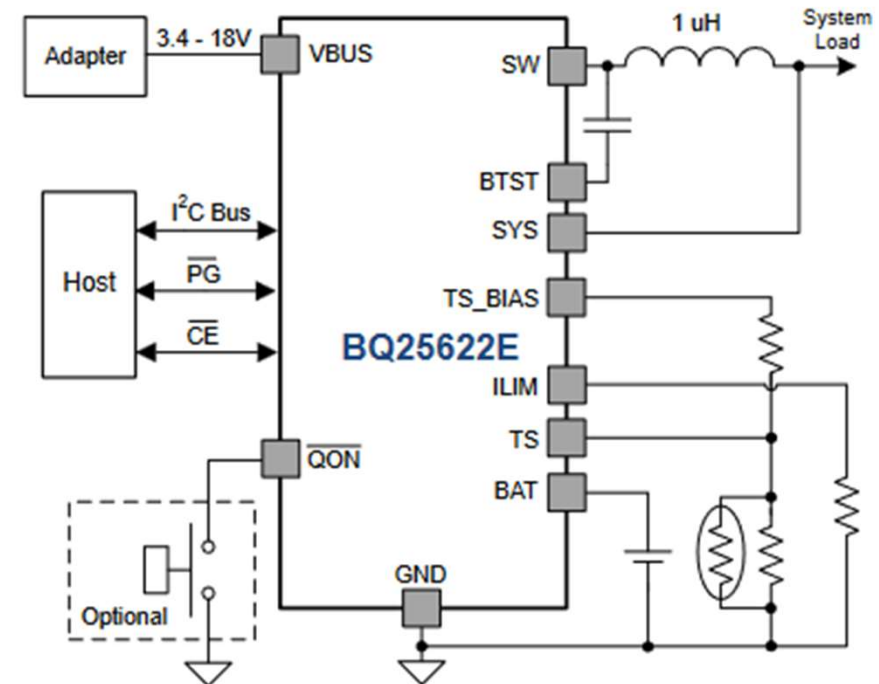
Name	Size [mm]	Mass [g]	Operating current [mA]	Range [V]	Resolution [V]	Interface	Price [\$]
BQ25622E	2.5 3	2	0.260	0 ÷ 18	0.004	I2C	2

The PMIC provides an integrated 12-bit ADC for monitoring the battery voltage. To convert the ADC reading to the voltage of the CanSat main voltage bus (formula 1).

Formula 1:

$$Voltage = \frac{ADC_{reading} * Voltage_{reference}}{2^{12}}$$

The result of conversion is **stored as a float**, since the Voltage reference is a constant floating point number. Keeping the voltage an integer that is a ratio of ADC reading to maximum voltage is also possible to increase performance on slower MCUs, but in our case the floating point form will be needed for telemetry anyway.



Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	Range [m/s]	Resolution [m/s]	Interfaces	Price [\$]
ABPDJJT001 PDSA3	8 11 7	0.8	3 ÷ 3.6	3.1	-107 ÷ 107	0.013	SPI	12

The output received from the sensor is an **integer**. It can be expressed by a transfer function (formula 1) that is solved for Pressure to obtain the reading in **psi** (formula 2). Then, the Pressure reading is converted to Pa (formula 3) so that we can calculate the speed (formula 4) of the payload in m/s. It is stored as a **floating-point** value.

Formula 1:

$$Output = \frac{Output_{max} - Output_{min}}{Pressure_{max} - Pressure_{min}} * (Pressure - Pressure_{min}) + Output_{min}$$

Formula 2:

$$Pressure = \frac{(Output - Output_{min}) * (Pressure_{max} - Pressure_{min})}{(Output_{max} - Output_{min})} + Pressure_{min}$$

Formula 3:

$$Pressure_{read} = Pressure * 6894.757$$

Formula 4:

$$V = \sqrt{\frac{2 * (Pressure_{read} - Pressure_{static})}{Density}}$$

Air density is compensated using thermometer.



ABPDJJT001PDSA3

Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	Range [°]	Resolution [°]	Interfaces	Price [\$]
BNO055	3.8 5.2 1	0.15	2.4 ÷ 3.6	11	180*	0.005*	I2C	13

The formula for tilt in an N-axis is:

$$TiltN(t) = TiltN(0) + \int_0^t RotationN(t) * dt$$

The integral can be substituted by a Newton-Cotes formula.

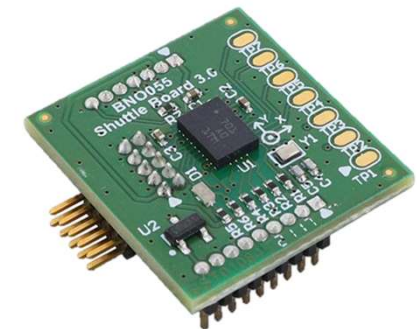
The C constant influence can be suppressed using the Madwick filter.

The internal Signal Processor is set to do these operations, the final value will be directly read from it.

If the read drifts, we will add filtering in the MCU.

Thanks to the built-in sensor fusion algorithm, the sensor provides us with compensated data that can be read from the appropriate registers as a **Signed Integer** in **degrees**.

The sensor will be initialized on the launch pad, and automatically run a calibration algorithm so that the default position (0 degrees tilt) is defined as when the axes are perpendicular to the Z axis (which is towards the center of gravity of the earth).

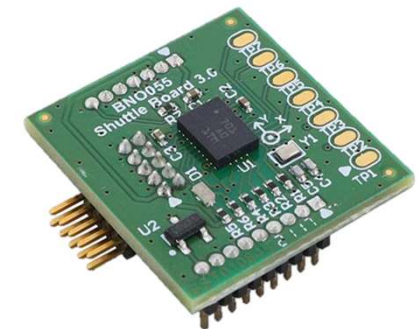


BNO055

Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	Range [°/s]	Resolution [°/s]	Interfaces	Price [\$]
BNO055	3.8 5.2 1	0.15	2.4 ÷ 3.6	11	125, 250, 500, 1000, 2000	0.001	I2C	13

As the rotation and tilt are measured by the same sensor, naturally it will follow the same initialization procedure and run the similar algorithm. The data will be periodically read from the sensors registers.

Again, thanks to the built-in sensor fusion algorithm, the sensor will provide us with compensated data that can be read from the appropriate registers as a **Signed Integer** in **degrees per second**.



BNO055

Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	microSD card	Resolution [pixels]	Field of view [°]	Frames per second	Interfaces	Price [\$]
ESP32-CAM	40.5 27 4.5	20	3.3÷5.0	140	Yes	800x600	65	30	UART	10

The camera will be turned on and start recording the video, on the launch pad, to record during both the ascent and landing of the payload. It will capture the **horizontal view** in one direction.

The video is stored on a microSD card in **800x600 resolution, 30fps and in color**, so that it meets the requirements.



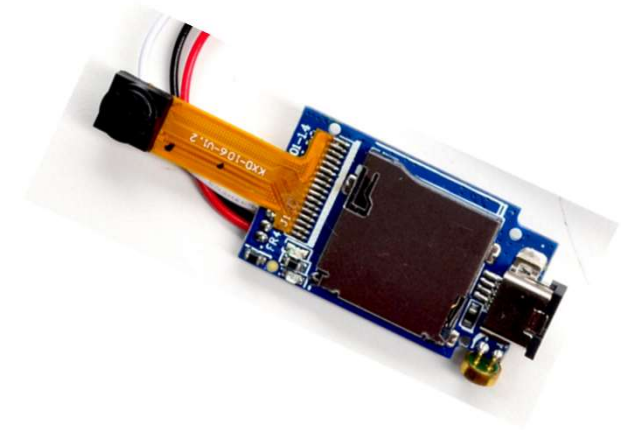
ESP-32 CAM

Name	Size [mm]	Mass [g]	Operating voltage [V]	Operating current [mA]	microSD card	Resolution [pixels]	Field of view [°]	Frames per second	Interfaces	Price [\$]
Adafruit 3202	29 17 5	2.8	3.7÷5.0	110	Yes	640x480	120	30	GPIO	16

The bonus camera will be recording the view behind the CanSat from the start, in order to capture its release from the rocket and the release of the parachute.

The video is stored on a microSD card in **640x480 resolution, 30fps and in color**, so that it meets the requirements.

The communication with the module is simple – if the communication line is pulled down for a second, a photo is taken. If the line is pulled down for longer, the camera records a video.



Adafruit 3202

Descent Control Design

Paweł Rak

The CanSat descent is divided into 2 sections:

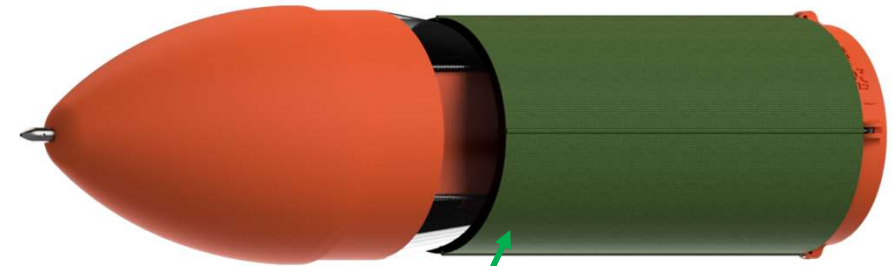
1. This stage starts when the CanSat is deployed from the rocket at an altitude of under **725m**. The **heatshield and streamer** develop immediately after rocket separation.

During this stage, the velocity is in the range of **10 to 30 m/s**.

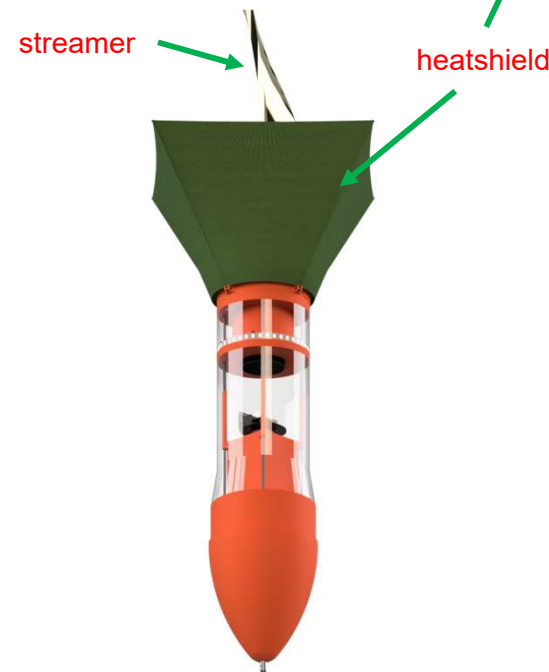
2. At an altitude of **100 m** CanSat is going to release the heatshield with upper cover. The parachute will be pulled out by a secondary streamer and it will develop immediately.

During this stage, the velocity is less than **5 m/s**.

0. pre-deployment CanSat configuration



1. heatshield descent configuration



2. parachute descent configuration



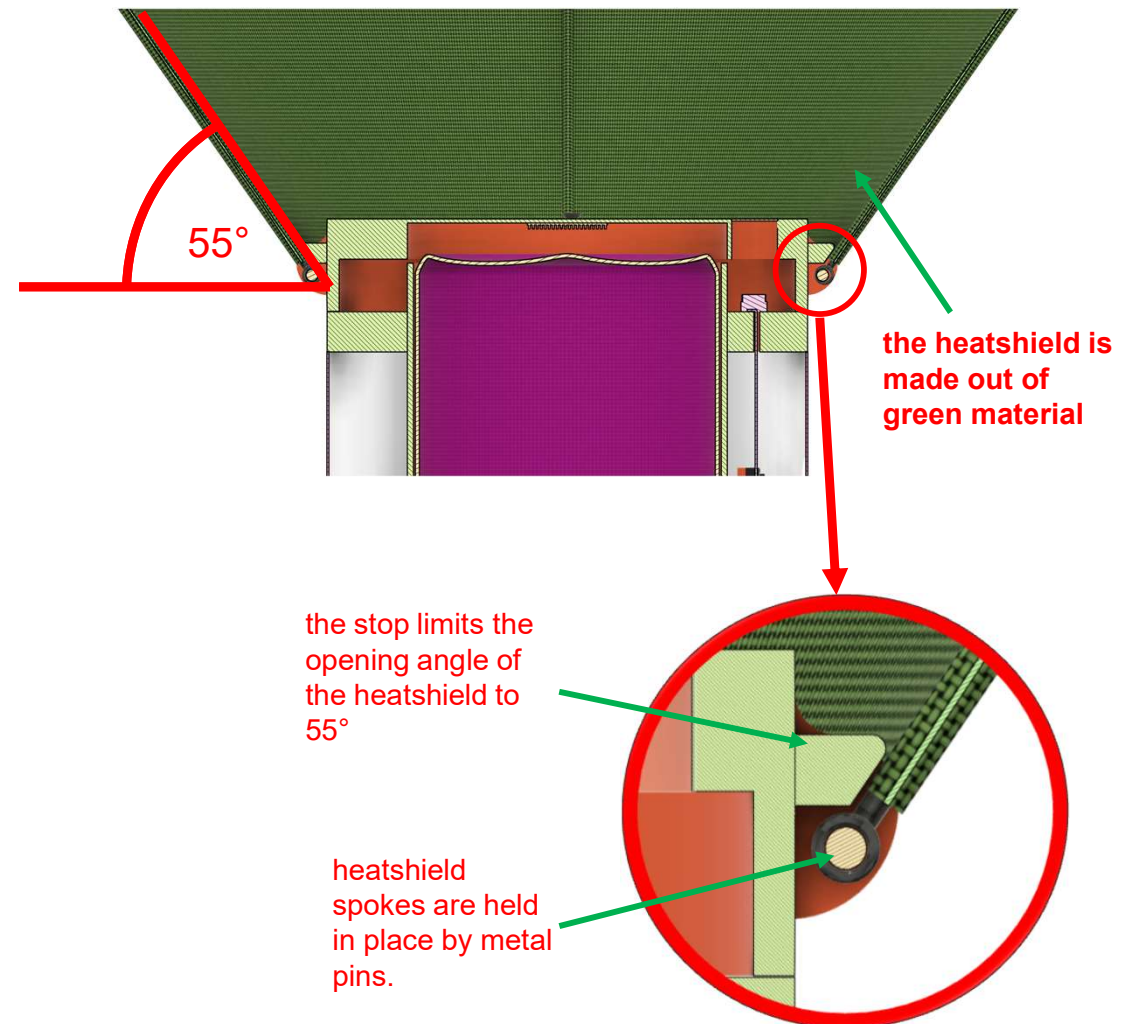
We made **no changes** since submitting PDR in the DCD section.

Prototype testing

Full test of the descent control system is yet to be performed. Currently, the parachute container and the upper cover have been tested for fit. The parachute mounting point has been checked for adequate strength.

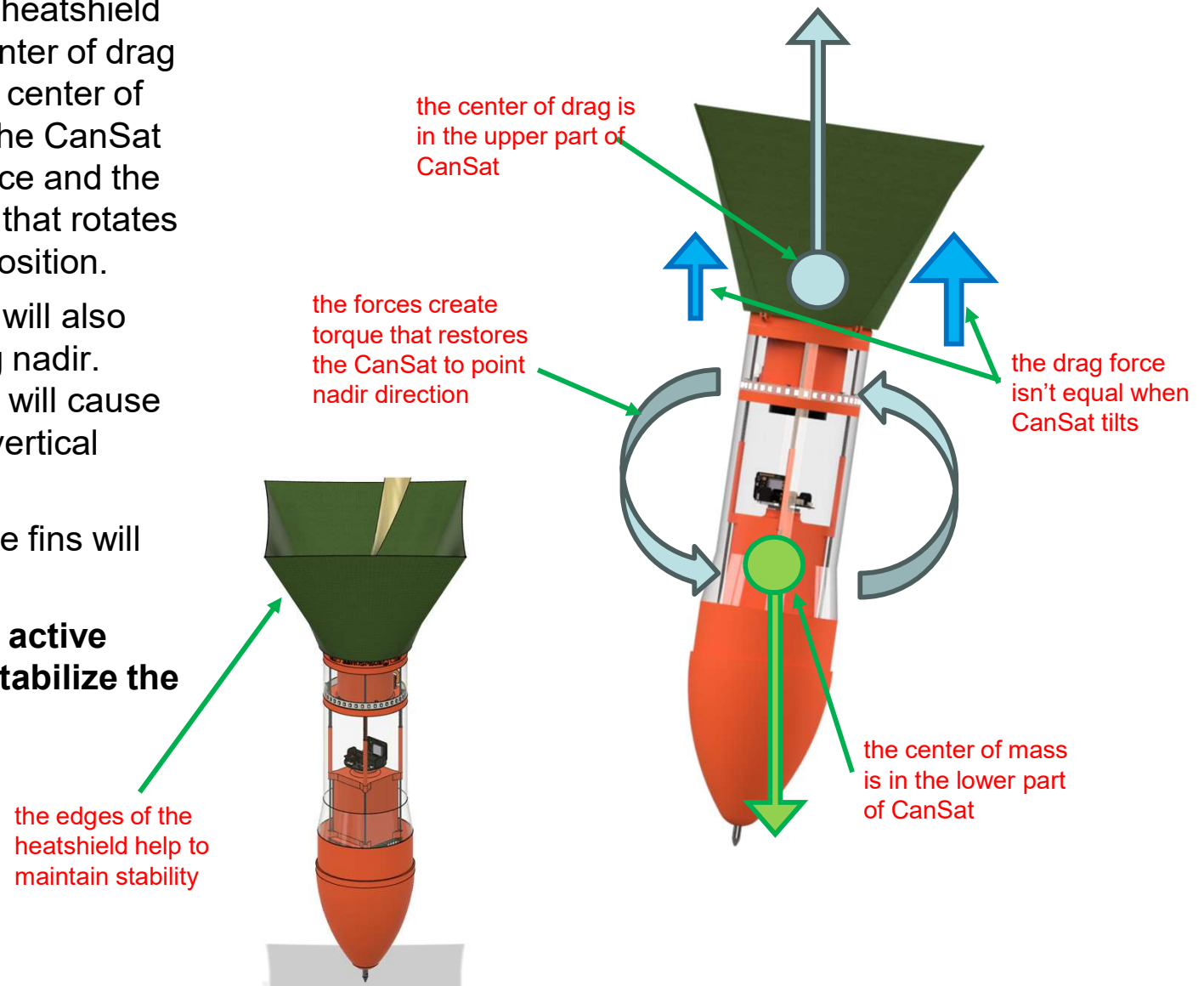
Payload Aerobraking Descent Control Hardware Summary

- The heatshield in our design is made out of durable green nylon stretched over four rods. The headshield is a square when fully stretched. This shape is used due to simplicity and our experience with it.
- The rods are mounted to the upper cover only by a single yet durable hinge. During ascend, the heatshield will be held closed by rocket payload section walls. When the CanSat is deployed the heatshield is opened by the air flowing around the CanSat. The angle of the heatshield opening is limited to **55°** by using a simple limiter that is part of the upper cover.
- The heatshield material is **square in shape and it is 0.58m x 0.58m in size**. This allows for a steady descent with a speed of over **11 m/s**.
- This design is passive. We do not use any sensors or actuators.

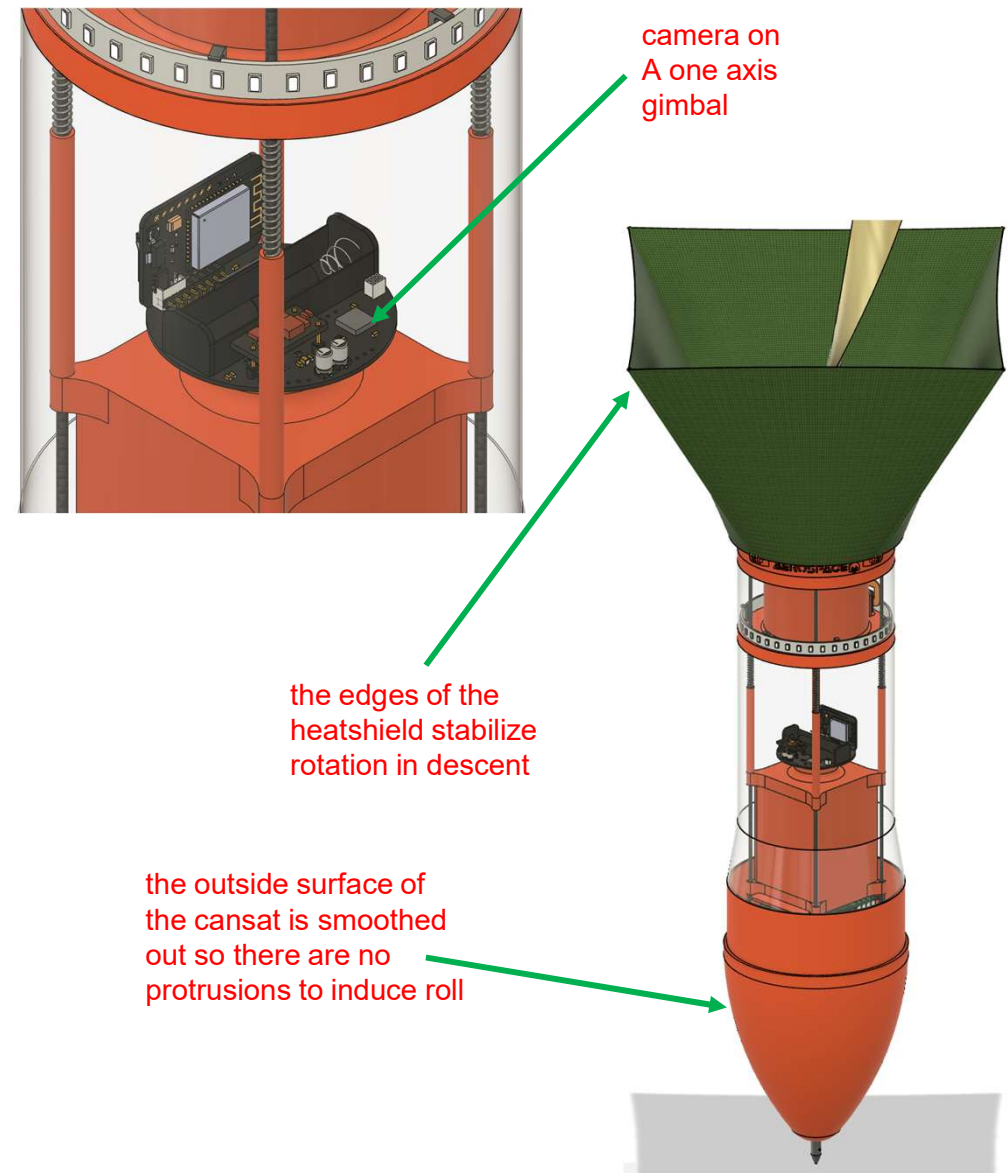


Payload Descent Stability Control Design

- Thanks to the placing of the heatshield on top of the CanSat, the center of drag will be much higher than the center of mass. Thanks to this when the CanSat is not vertical, the gravity force and the drag force will create torque that rotates the probe back to the right position.
- The shape of the heatshield will also help keep the probe pointing nadir. When swaying, greater drag will cause the probe to go back to the vertical position.
- Heatshield rods behaving like fins will further stabilize the CanSat.
- **This design is passive. No active mechanisms are used to stabilize the CanSat.**

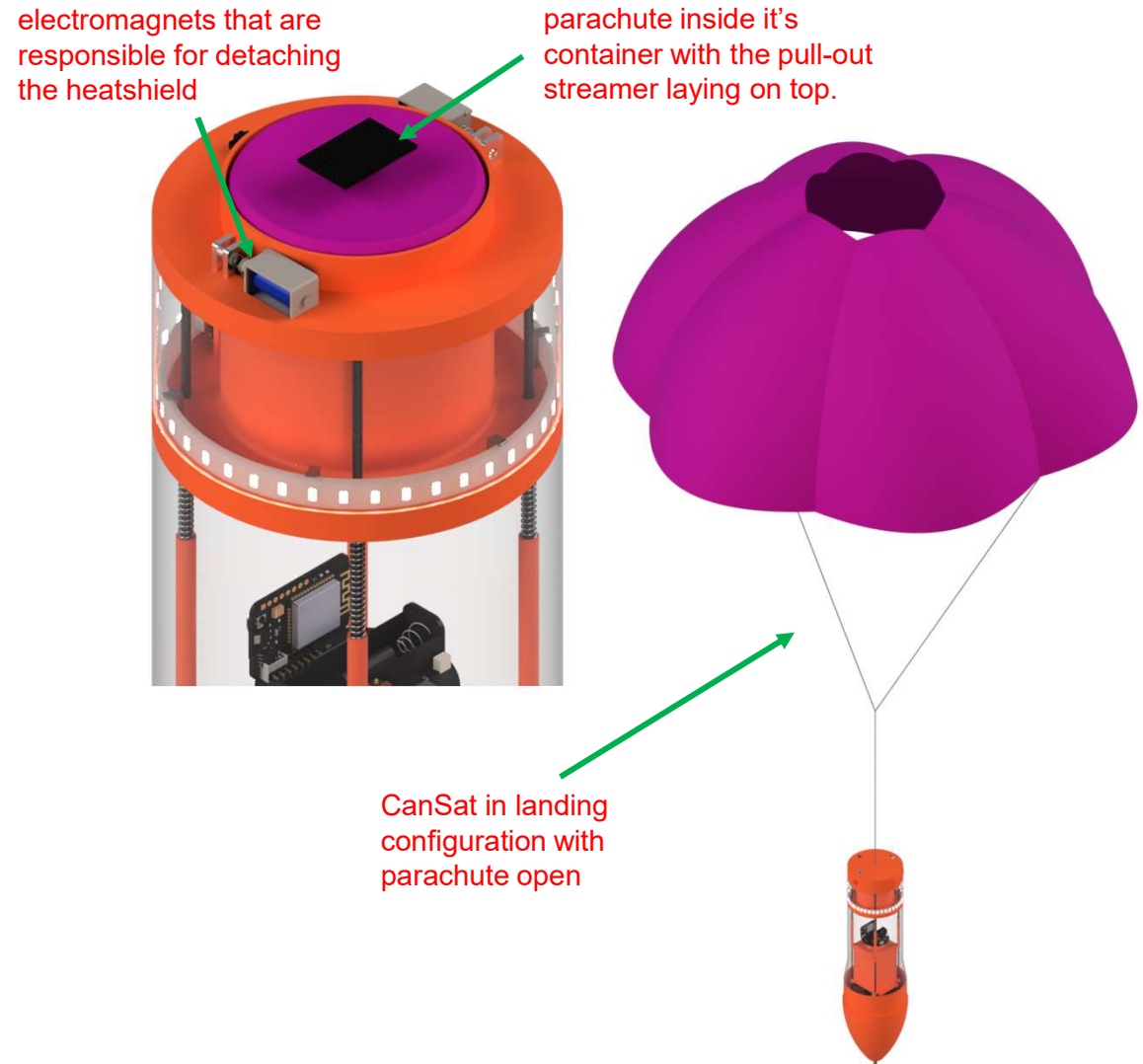


- During preliminary testing we have concluded that the geometry of our heatshield, parachute and CanSat body provide good enough stability in rotation to satisfy our needs. **We do not use any active mechanisms to stabilize our CanSat and rotation control is passive.**
- We will still use a one axis gimbal for our camera to counteract any rotation that may arise during the flight. **Active stabilization of our camera is used to ensure good recording quality.**



Payload Parachute Descent Control Hardware Summary

- The payload parachute is stowed away in a special container. This container is closed by the heatshield assembly. When the CanSat descends to an altitude of **100m** electromagnets release the brackets that hold the heatshield assembly thus releasing it. When the heatshield assembly detaches, at first a streamer is pulled. **The streamer then pulls the parachute.** This setup ensures reliable operation of the parachute opening mechanism. After the parachute opening the CanSat will descend until landing with a speed of **4.28 m/s** until landing.
- The parachute will be **Purple and hexagonal**. It will be made out of nylon. **The diameter of the inscribed circle of the hexagonal parachute will be 1.125 meter.**



Assumptions (all situations):

ρ - air density at standard temperature: $1.12 \text{ (kg/m}^3\text{)}$

C_d - drag coefficient for parachute: 0.75

C_D - drag coefficient for heat shield 0.4

g - gravitational acceleration 9.81 m/s^2

Aerobraking descent: *Extra assumption: streamer is omitted*

Parameters:

$$\frac{1}{2} \frac{\rho}{B} v^2 = g + \frac{dv}{dt}$$

m - payload mass $0.955 \text{ kg (+/- 0.01)}$

S - aerobraking area 0.34 m^2

$$B = \frac{m}{C_D S}$$

$\frac{dv}{dt}$ - tends to zero

B - ballistic coefficient

**Estimated payload aerobraking
descent rate = 11.2 m/s**

Parachute descent:

Parameters:

d - parachute diameter
(diameter of inscribed circle) 1.125 m

m - payload mass $0.860 \text{ kg (+/-0.01)}$

$$S_p = \frac{2mg}{v^2 \rho C_d}$$

$$d = \sqrt{\frac{2S_p}{\sqrt{3}}}$$

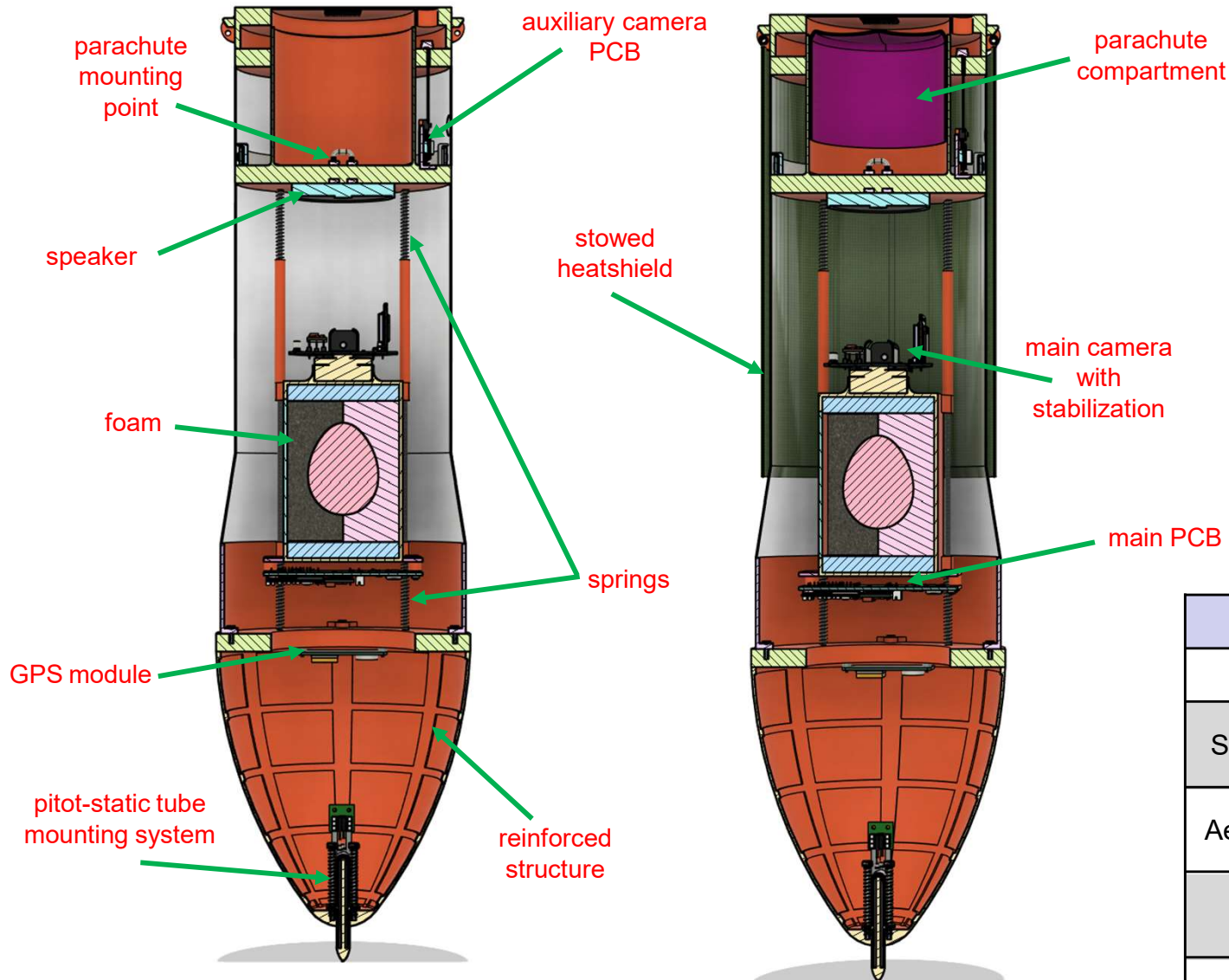
**Estimated payload aerobraking
descent rate = 4.28 m/s**

Flight phase	Speed of descent [m/s]
Payload descent with heatshield	11.2
Payload descent with parachute	4.28

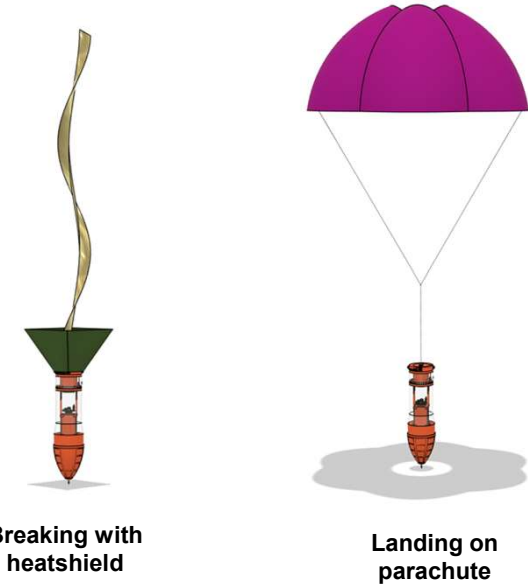
Calculated values are within the scope of the competition!

Mechanical Subsystem Design

Paweł Rak



Mechanical Configurations:



Part	Material
Structural Struts	Carbon fiber
Structural Elements	3D printed ABS, PET-G, Nylon
Aerodynamic Fairing	Laminating foil
Heatshield and parachute	Nylon, paracord
Egg suspension	Poliurethane foam

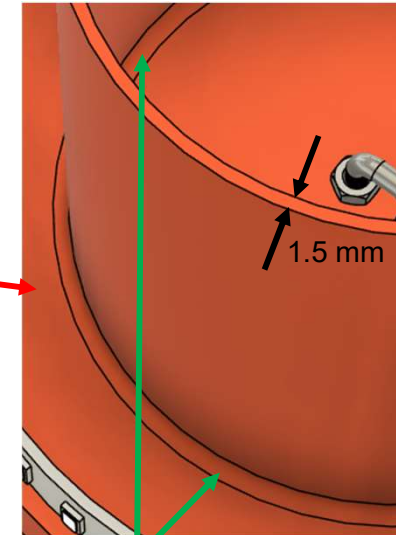
Mechanical Subsystem Changes Since PDR



reconstructed gimbal



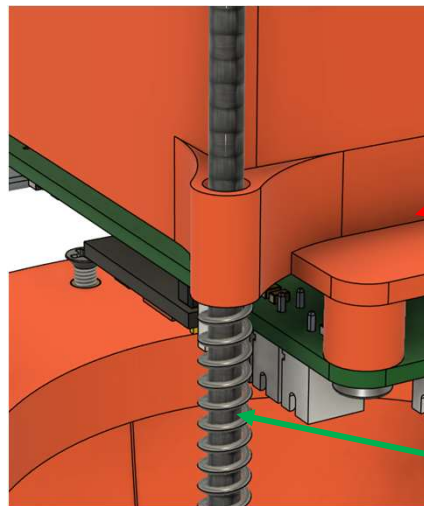
(aerobrake not visible)



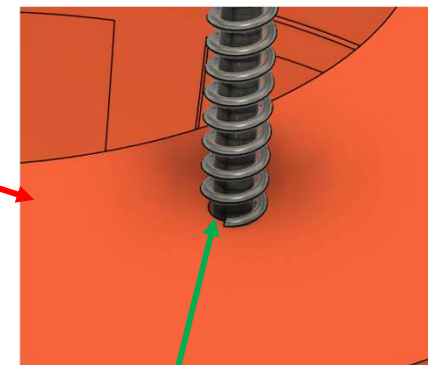
thickened walls
of parachute
container

1.5 mm

fillets for container
reinforcement



threaded
aluminium struts,
replaced by
carbon fibre



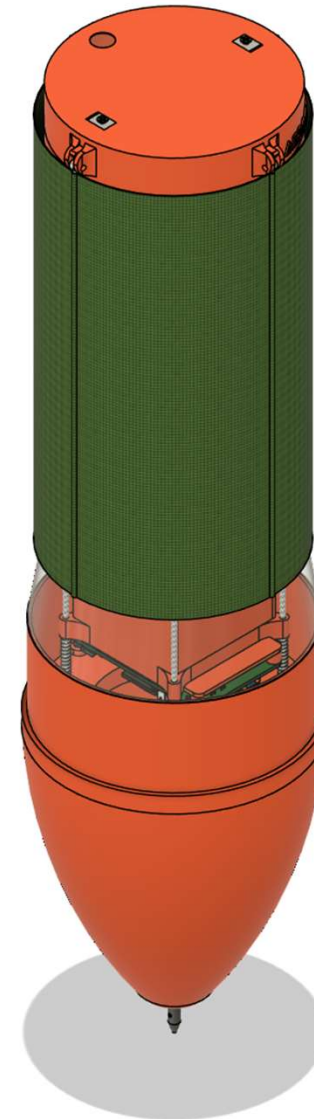
bolt nuts removed,
struts glued in place

Payload Mechanical Layout of Components (1/5)

Overall design summary and CAD model

Design features

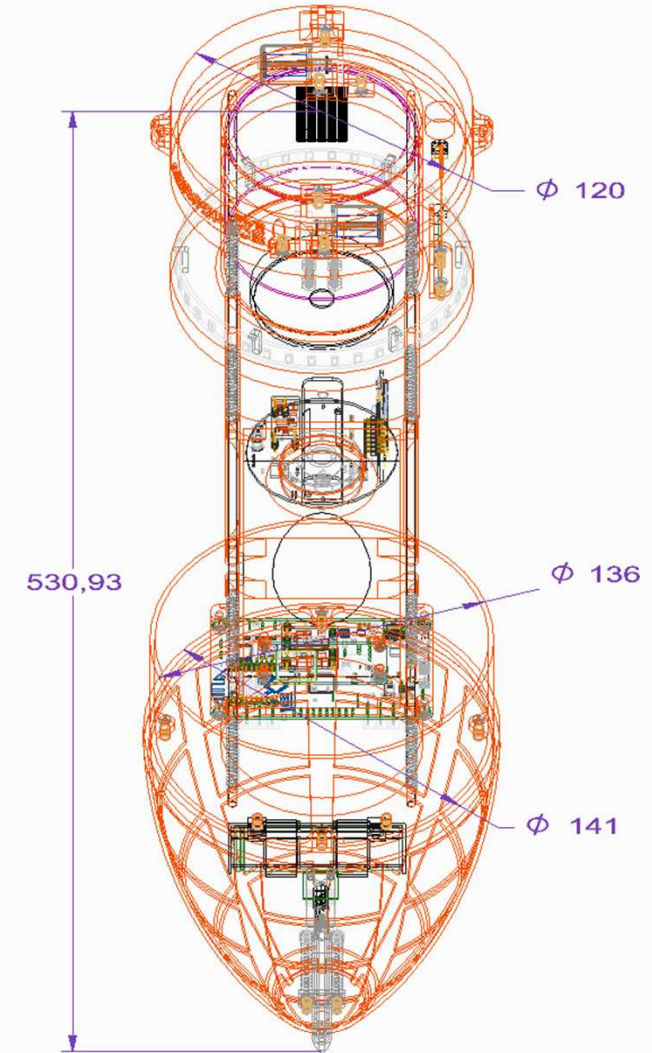
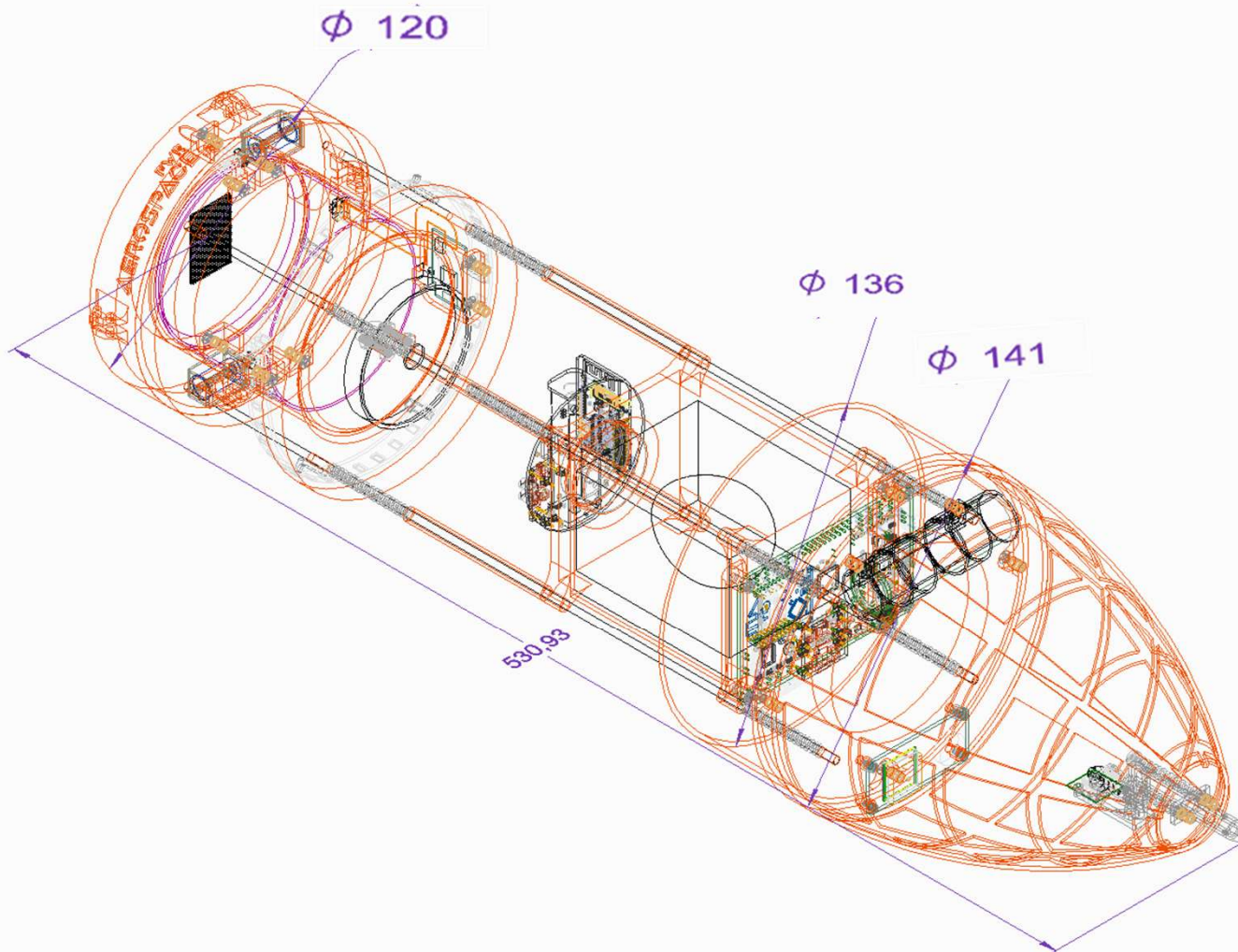
- **No pyrotechnical, chemical or heat based actuators are used**
- **Nose cone is not a part of heatshield**
- 3D printed structure with vertical carbon struts
- Bottom heavy design for increased stability
- Custom electronics board
- Heatshield opened by aerodynamic pressure
- Parachute pulled from its compartment by detaching upper assembly with the heatshield.
- Egg suspended on struts with springs, enclosed in foam.
- Main camera single axis stabilization
- Auxilliary camera pointing upwards
- Self retracting pitot-static tube



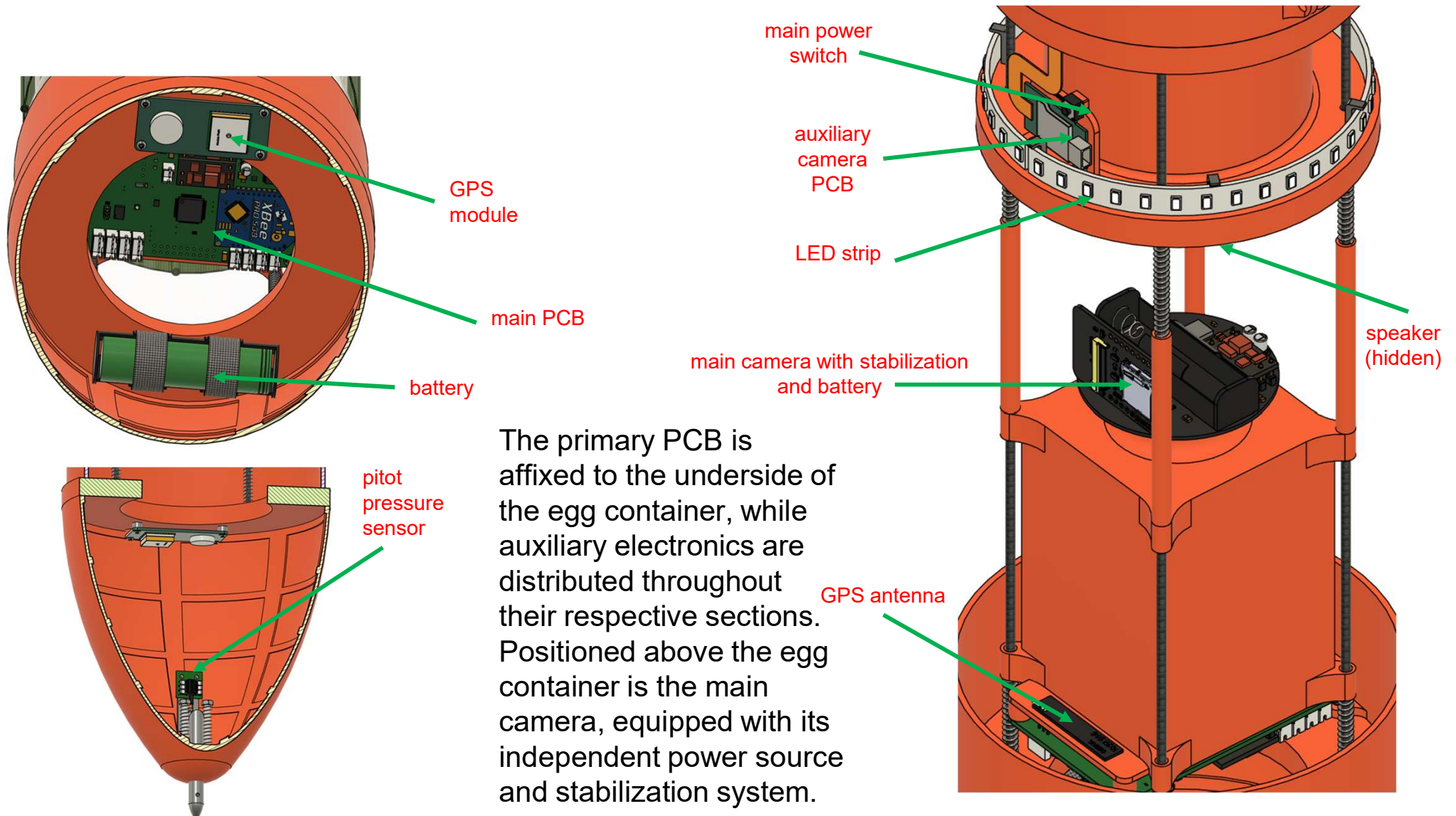
Payload Mechanical Layout of Components (2/5)

Overall structure and dimensions drawing

Note: All dimensions in mm

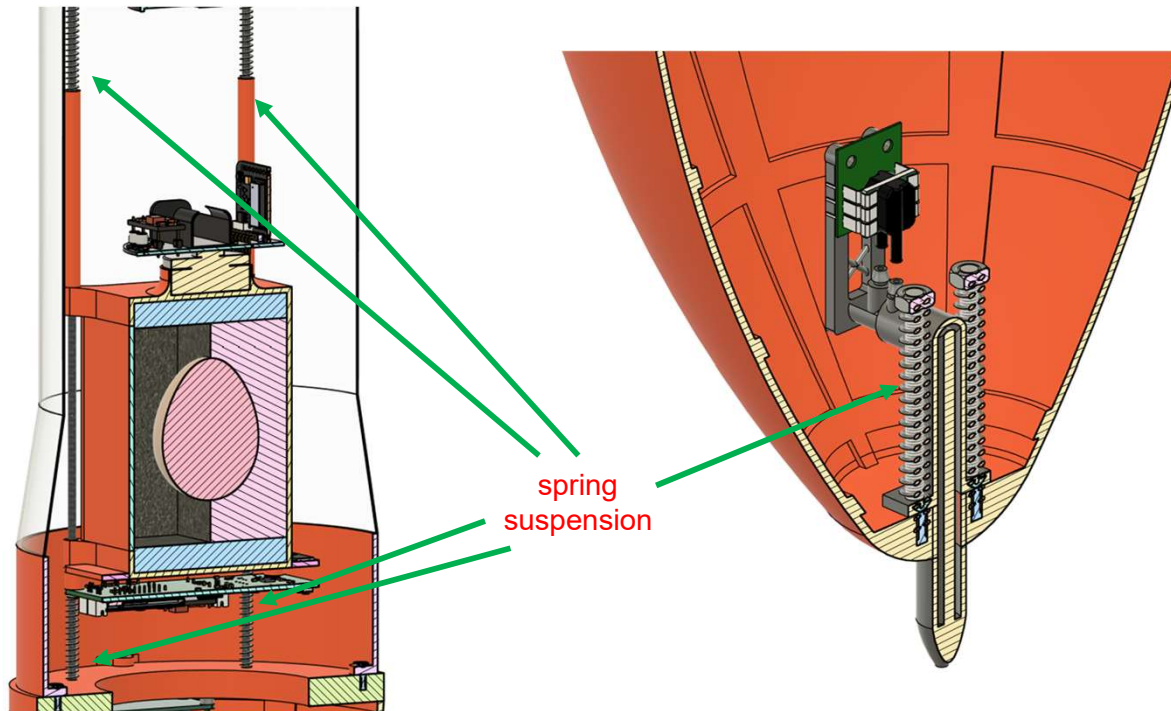


Location of electronics

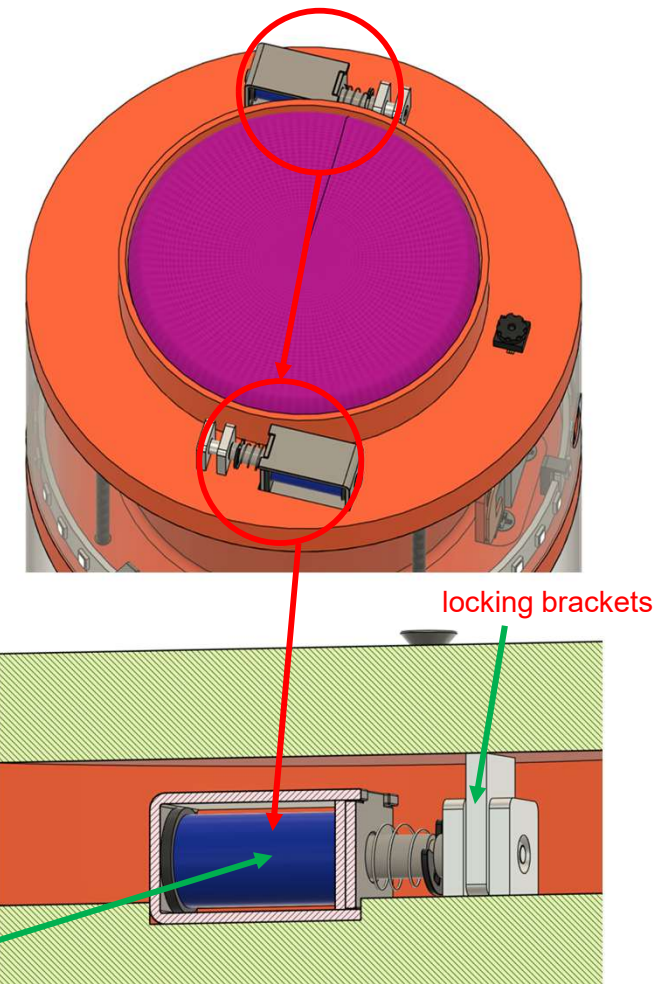


Payload Mechanical Layout of Components (4/5)

Location and description of mechanical components



The egg containment unit is positioned atop the main struts, featuring a spring suspension system. The upper section, housing the heat shield, is secured using locking brackets and pins. These pins are operated by solenoids, facilitating the detachment of the heat shield to expose the parachute. Additionally, the pitot-static tube is suspended on springs, allowing for automatic retraction to prevent damage during touchdown.



Payload Mechanical Layout of Components (5/5)

Structural material selection(s)

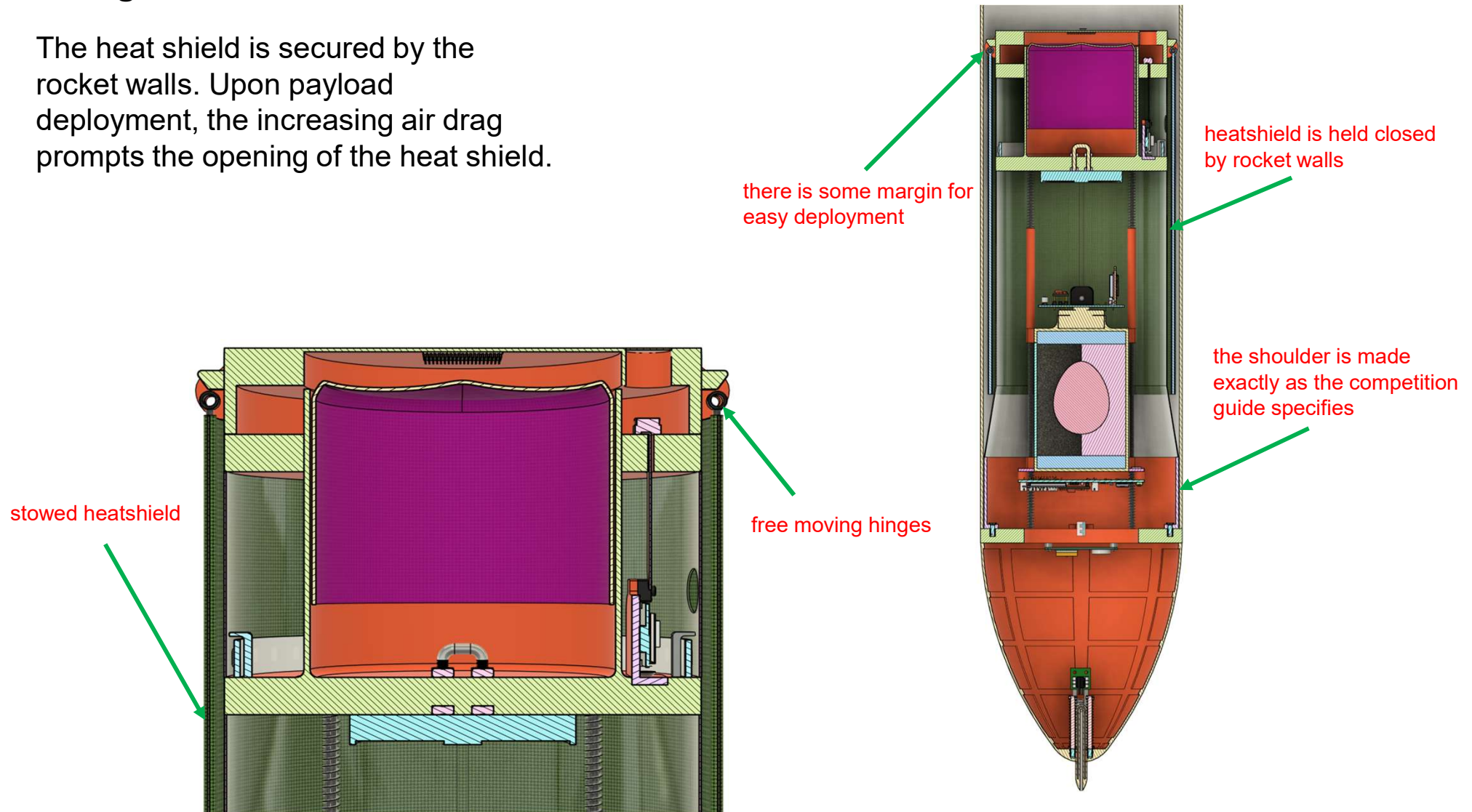
Part	Material	Explanation
Structural Struts	Carbon	Lightweight and more durable than their metal counterparts
Structural Elements	3D printed ABS, PET-G, Nylon	It is easy to make prototypes and make any changes to make the model more effective, different materials are selected for their own tasks, efficiency and complexity of production criteria.
Aerodynamic Fairing	Laminating foil	Light weight
Heatshield and parachute	Nylon, paracord	Durability and the experience of using this material
Egg suspension	Poliurethane foam	Light weight and excellent shock-absorbing properties

Payload Aerobraking Pre Deployment Configuration

Pre deployment configuration

The heat shield is secured by the rocket walls. Upon payload deployment, the increasing air drag prompts the opening of the heat shield.

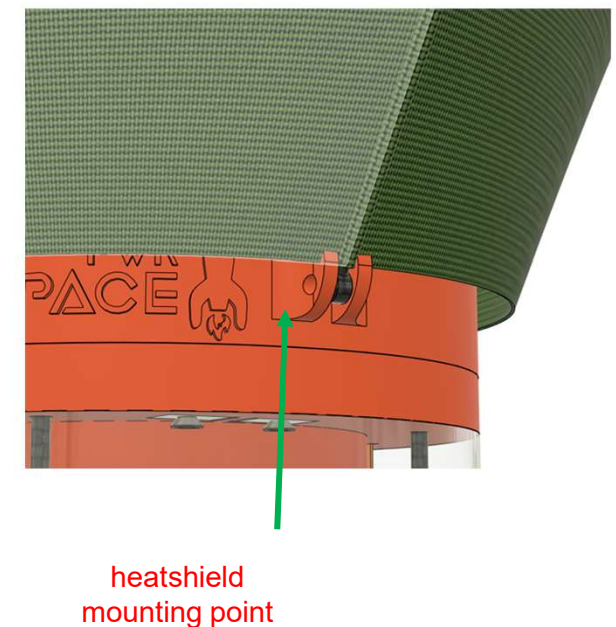
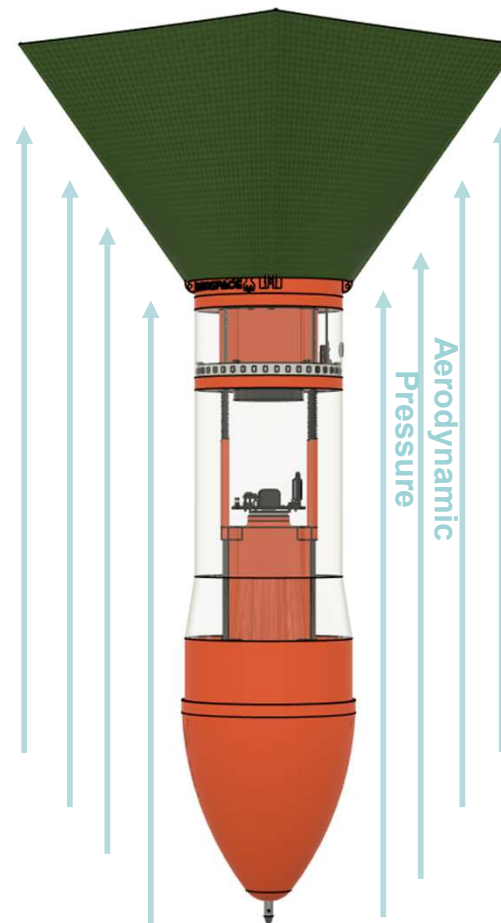
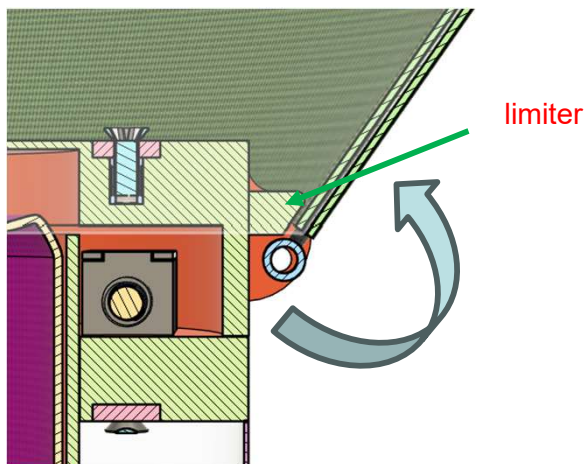
Visualization of CanSat inside the rocket



Payload Aerobraking Deployment Configuration

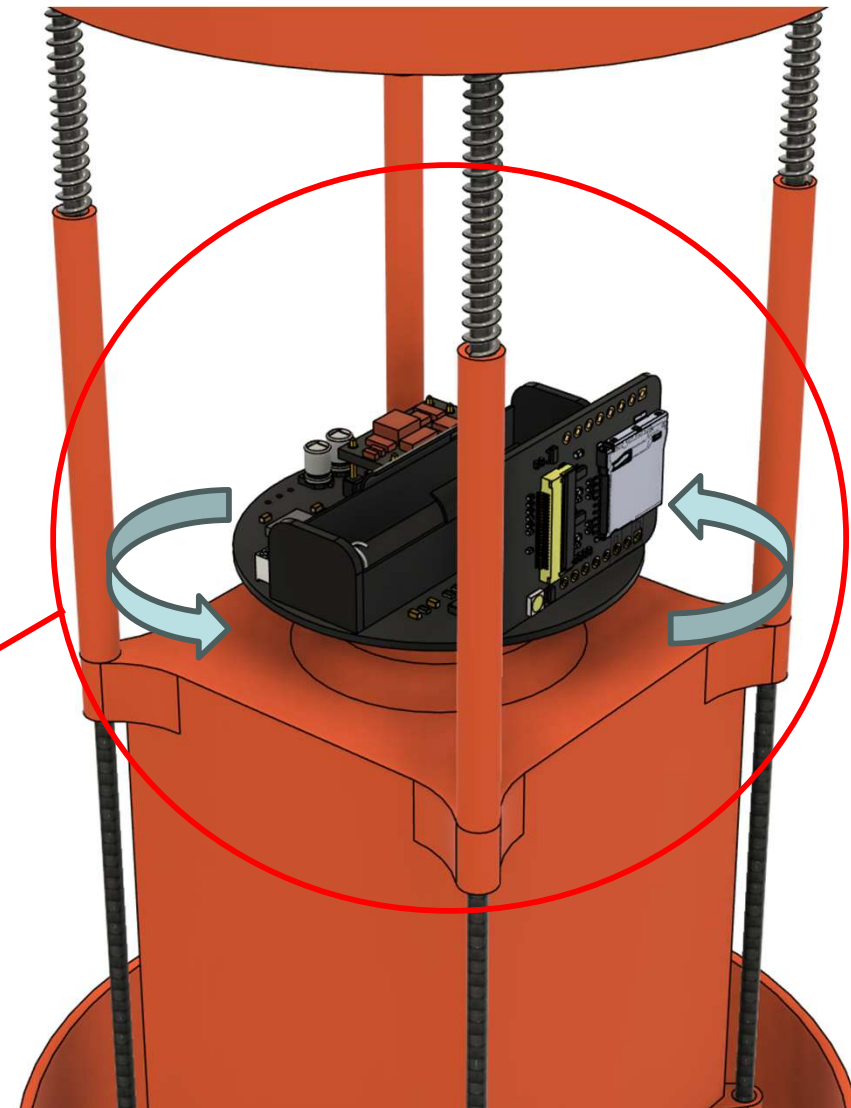
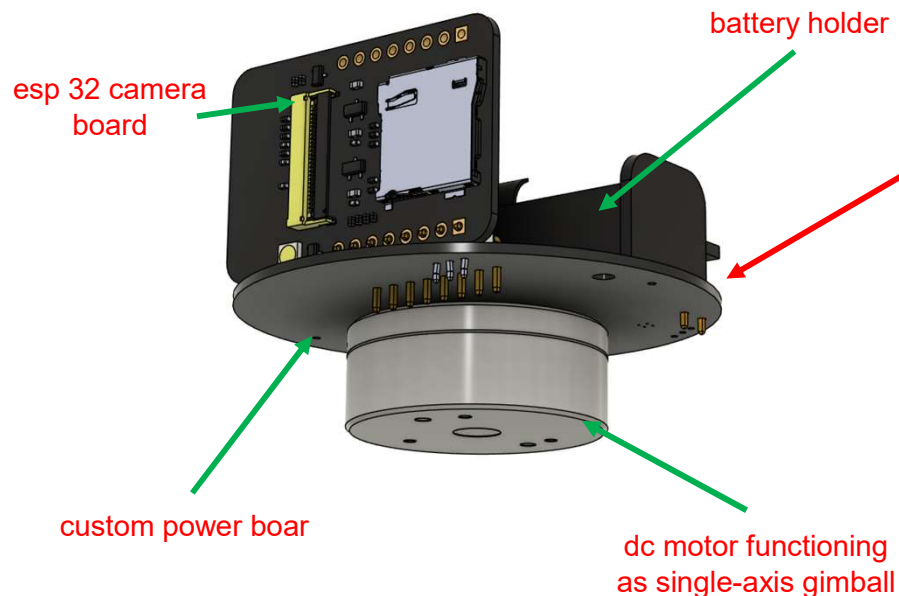
Aerobraking Deployment

The aerobrake deploys upon separation from the starting container, activated by aerodynamic pressure. Mechanical limiters individually restrict the angle of each supporting rod, ensuring controlled deployment. This design permits longer aerobrake rods, necessitating smaller rod angles for the same braking area, thereby reducing strain on the mounting points of the rods.



Camera Pointing Control

Camera pointing control is accomplished by a DC motor functioning as a single axis gimbal. To overcome problems with cables, decision was made to integrate main camera, its battery, power board and the engine into a single self sufficient module. Isolating the motor and camera equipment from the main electronics greatly reduced the impact of EMI for the main PCB.



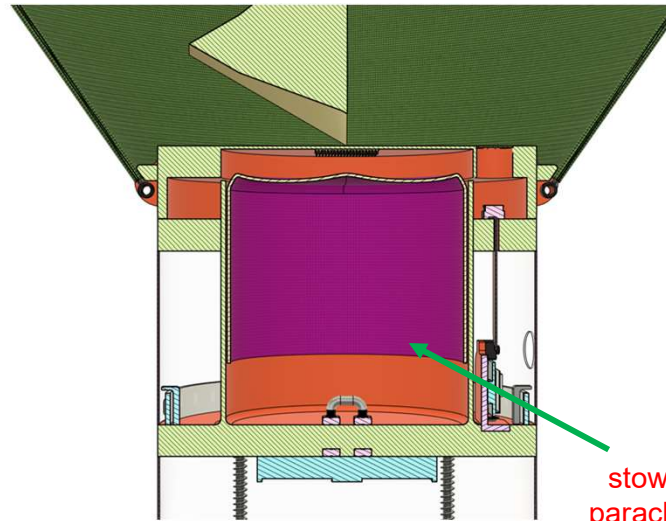
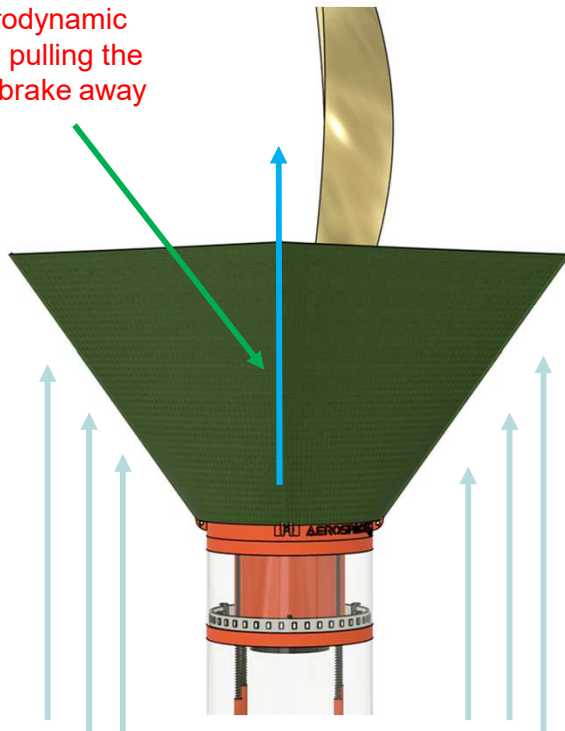
(aerobrake and aerodynamic fairings not visible)

Payload Parachute Deployment Configuration (1/2)

Parachute Deployment

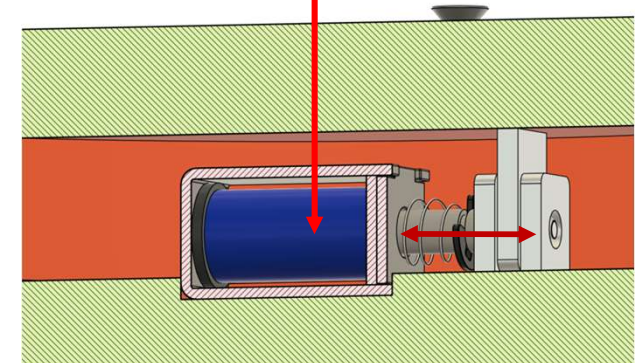
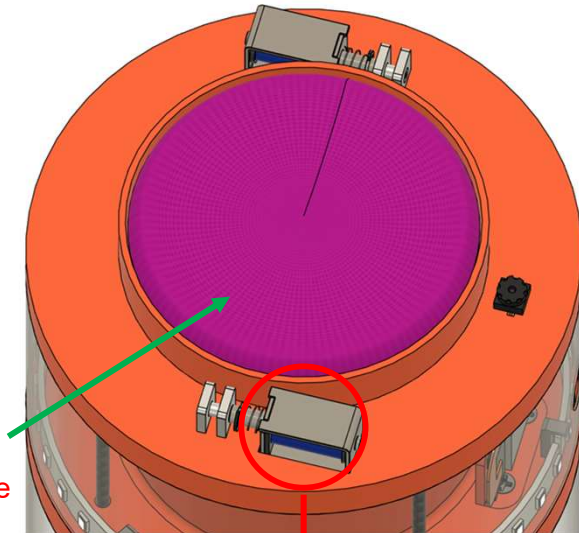
The parachute is safely stowed inside the container, securely fastened for deployment when needed.

aerodynamic drag pulling the aerobrake away



stowed parachute

The aerobrake release mechanism employs metal pins and solenoids to securely lock the aerobrake assembly in place. Detachment and separation are facilitated by aerodynamic drag. Solenoids, known for their fast response, ease of control, and high reliability, ensure efficient operation of the release mechanism.

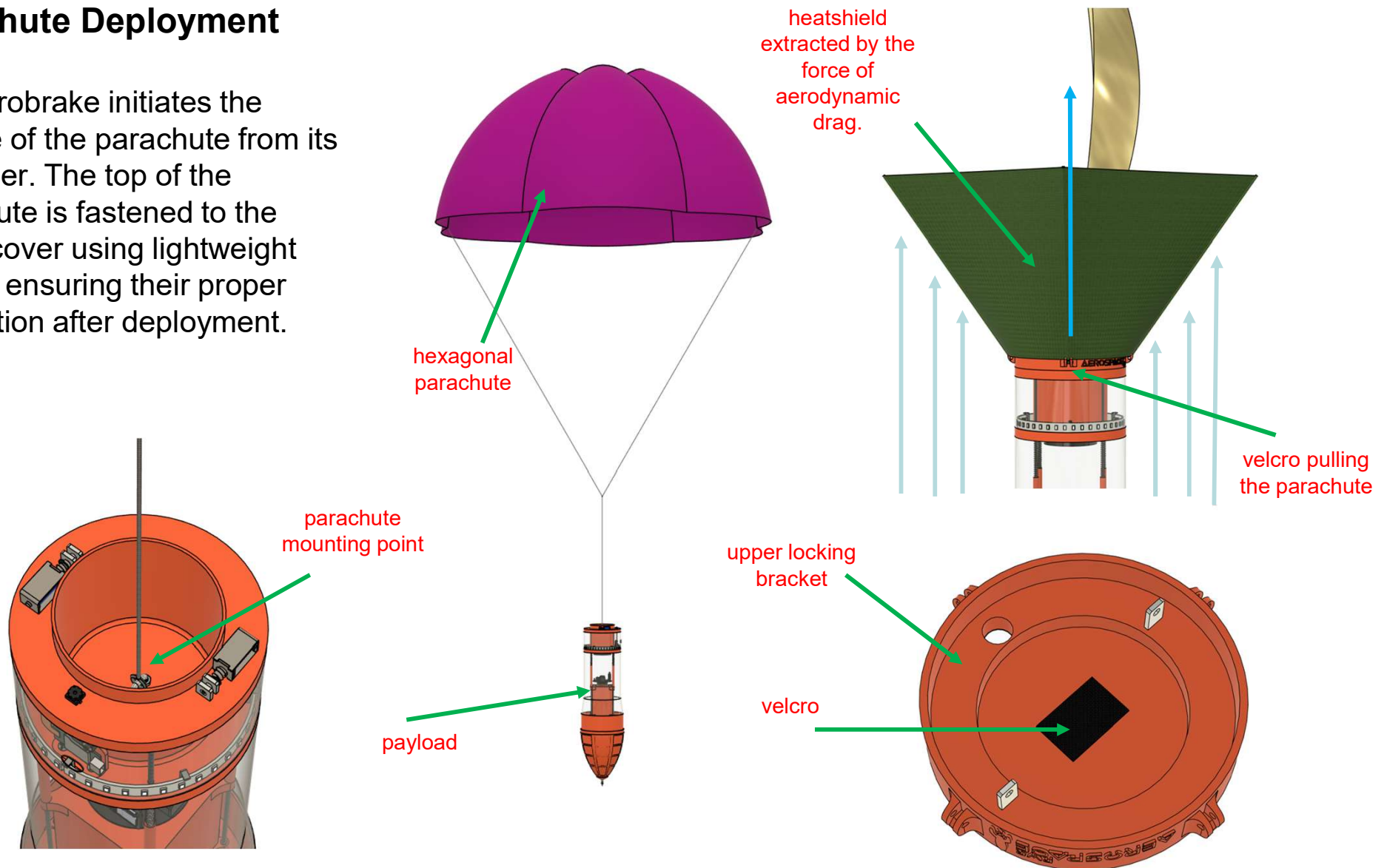


After separation from CanSat, **Heat shield descends with rate lower than 10m/s**

Payload Parachute Deployment Configuration (2/2)

Parachute Deployment

The aerobrake initiates the release of the parachute from its container. The top of the parachute is fastened to the upper cover using lightweight Velcro, ensuring their proper separation after deployment.

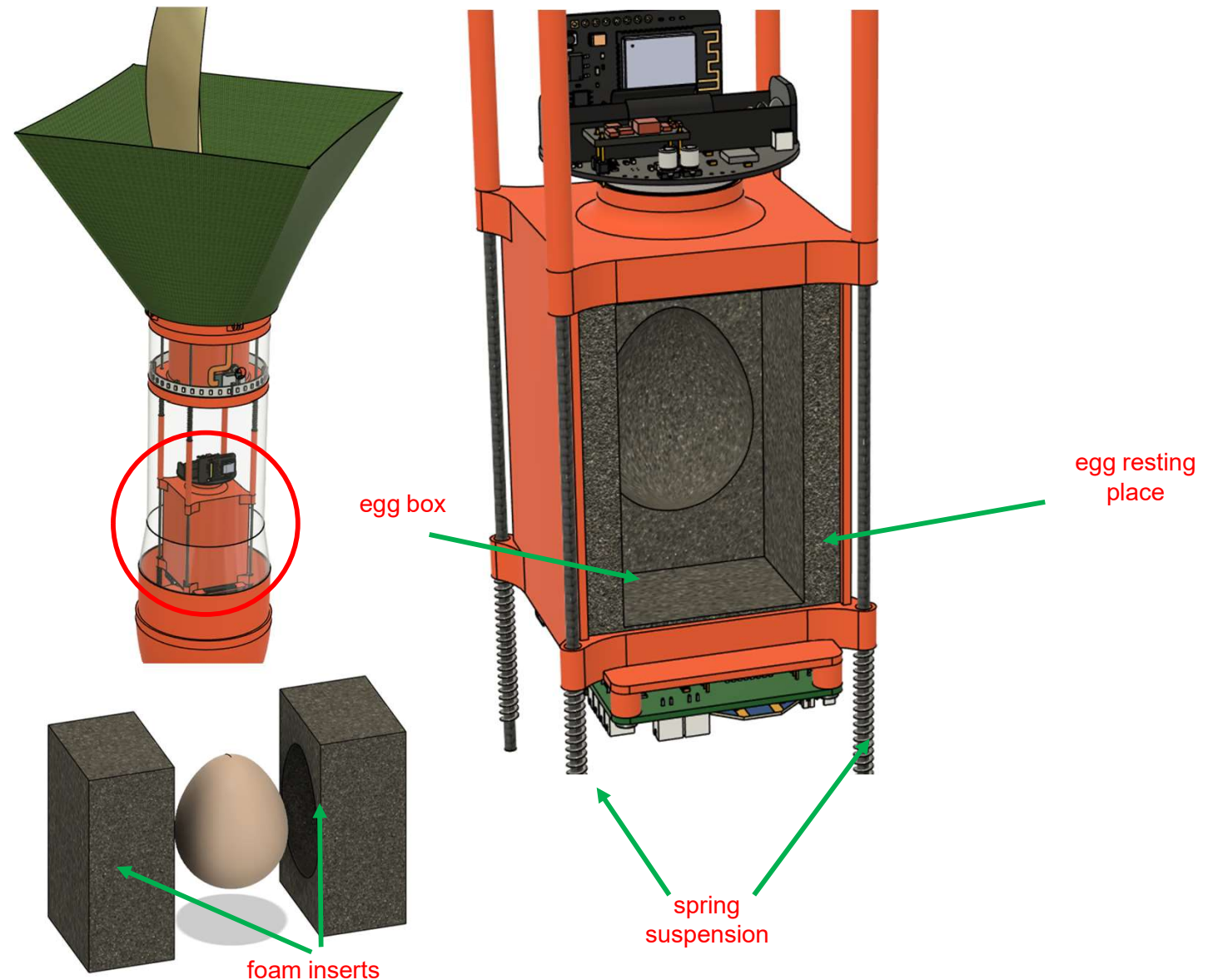


Payload Egg Containment Configuration

Egg Containment

The hen's egg is encased in soft polyethylene foam, which is then placed inside a plastic box. This box is suspended on carbon struts, utilizing a spring suspension system to provide additional shock protection.

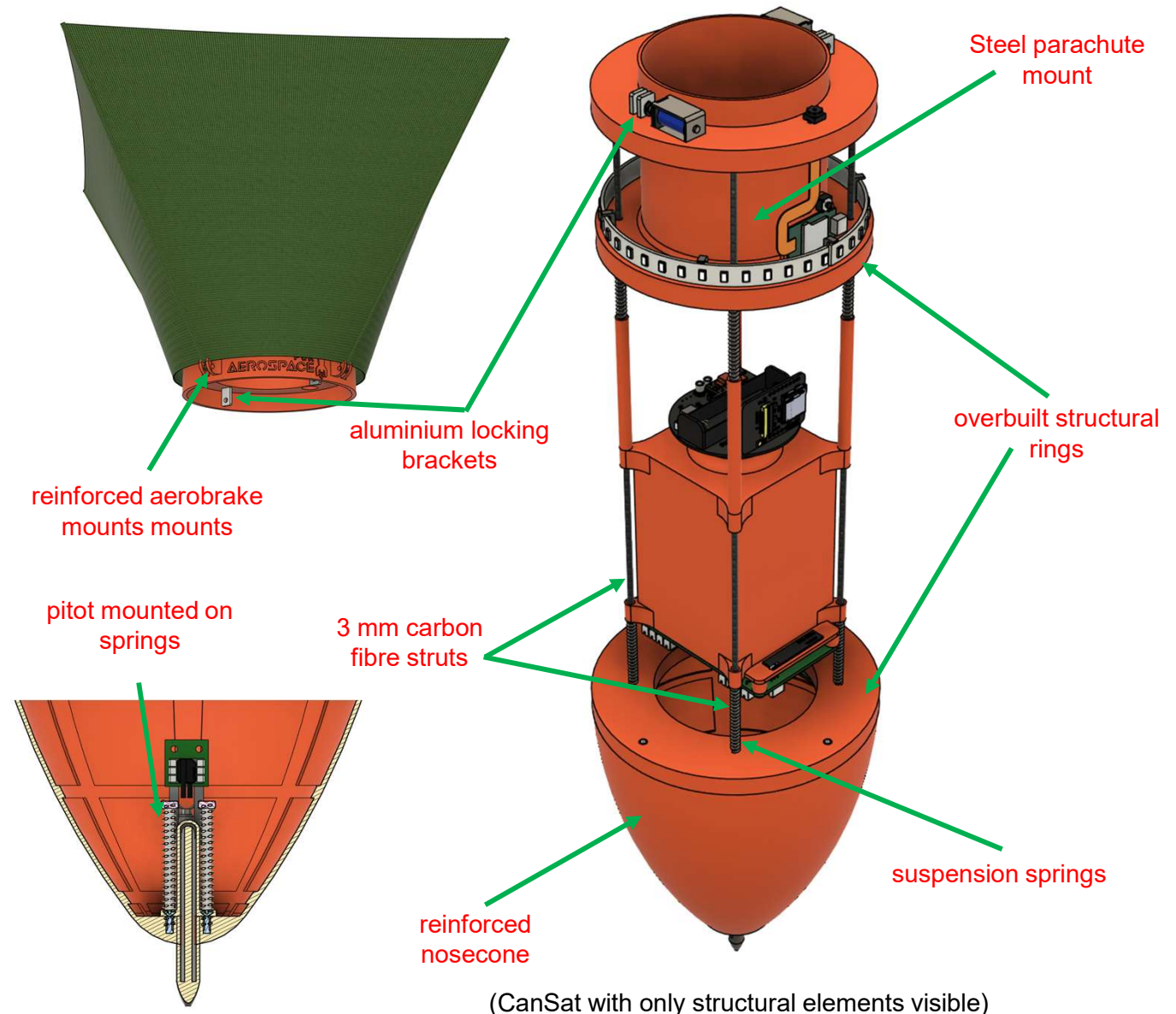
The egg is inserted within the foam and then encased in a plastic box, which is sealed using a door secured with Velcro.



Structure Survivability

All structural elements of the payload were **already tested to ensure their survivability**. Four 3 mm thick carbon fibre struts are the main structural elements of the payload. All non-structural elements are mounted on overbuilt platforms glued onto the carbon fibre struts. Nosecone is additionally reinforced to withstand a 6 m/s impact on hard surface. Centrally mounted Pitot tube is mounted on springs, so it will retract on impact, protecting its integrity. Parachute mounting point is a 3 mm steel u-shaped bracket secured by nuts. Eggbox assembly is shock protected by springs. Batteries are soldered in place and secured by velcro straps. All non structural elements were mounted using M3 screws with brass threaded inserts. Aerobrake locking brackets are laser cut aluminum, with steel locking pins. All threads are secured by a medium threadlock glue.

Cansat will survive 30 G shock and 15 G vibration.



Subsystem	Component	Material / Part number	Quantity	Mass piece [g]	Mass summary [g]	Source
Nose cone Structure	Structural ring	ASA	1	46±1	46±1	Measured
	Aerodynamic fairings	PET	1	26.4±2	26.4±2	Estimated
	Nose cone shoulder	ASA	1	42±1	42±1	Measured
	Nose cone fairing	Nylon	1	99±1	99±1	Measured
	Pitot tube	PET-G	1	1.41±0.1	1.41±0.1	Calculated
	Pitot mounting struts	Steel	2	2.26±0.1	4.52±0.2	Estimated
	Pitot dampening springs	Steel	2	0.88±0.05	1.76±0.1	Estimated
	Pitot air pipe	Plastic	2	2±0.2	4±0.4	Measured
	M3 x 5,5 mm brass inserts	Brass	8	0.27±0.0125	2.16±0.1	Estimated
	M3 5.5 2.4 ISO nuts	Steel	2	0.37±0.02	0.74±0.04	Estimated
	M3 x 8 mm screw	Steel	8	0.46±0.025	3.68±0.2	Estimated

Summary mass = 231.67 ± 6.14 g

Subsystem	Component	Material / Part number	Quantity	Mass piece [g]	Mass summary [g]	Source
Upper Structure (heatshield and parachute mechanism)	Upper cover	ASA	1	50±1	50±1	Measured
	Camera glass	Glass	1	0.14±0.01	0.14±0.01	Estimated
	Detach mechanism structure	ASA	1	30±1	30±1	Measured
	Parachute compartment structure	ASA	1	66±1	66±1	Measured
	Auxiliary camera mount	ASA	1	2.69±0.1	2.69±0.1	Calculated
	Long brackets	Steel	2	7.44±0.25	14.88±0.5	Estimated
	Short brackets	Steel	4	3.41±0.1	13.64±0.4	Estimated
	Parachute mounting U-Bolt	Steel	1	1.95±0.05	1.95±0.05	Estimated
	M3 x 5,5 mm brass inserts	Brass	8	0.27±0.0125	2.16±0.1	Estimated
	M3 5.5 2.4 ISO nuts	Steel	4	0.37±0.02	1.48±0.08	Estimated
	M3 x 8 mm screw	Steel	4	0.46±0.025	1.84±0.1	Estimated
	LED holders	Aluminium	6	0.22±0.01	1.32±0.06	Estimated

Summary mass = 186.1 ± 4.4 g

Subsystem	Component	Material / Part number	Quantity	Mass piece [g]	Mass summary [g]	Source
Struts, springs and shock absorbers	Main struts	Carbon	4	3.5±0.1	14±0.4	Measured
	Spring distances	PET	4	0.75±0.02	3±0.08	Measured
	Dampening spring	Steel	8	0.85±0.02	6.8±0.16	Estimated

Summary mass = 23.8 ± 0.64 g

Subsystem	Component	Material / Part number	Quantity	Mass piece [g]	Mass summary [g]	Source
Descent control system	50mm x 2m mylar streamer	PP	2	1.85±0.025	3.7±0.05	Calculated
	Heatshield material	Ripstop nylon	1	48±2	48±2	Calculated
	Parachute material	Ripstop nylon	1	64±3	64±3	Calculated
	Nylon thread	Nylon	4m	0.4±0.01	1.6±0.04	Estimated

Summary mass = 117.3 ± 5.09 g

Subsystem	Component	Material / Part numer	Quantity	Mass piece [g]	Mass summary [g]	Source
Egg box structure	Egg box	ASA	1	75±1	75±1	Measured
	Foam insert	Foam	1	10.0±1	10±1	Estimated
	Egg box door	ASA	1	10±1	10±1	Calculated
	M3 x 8 mm screw	Steel	4	0.46±0.025	1.84±0.1	Estimated

Summary mass = 96.84 ± 3.1 g

Subsystem	Component	Material / Part number	Quantity	Mass piece [g]	Mass summary [g]	Source
Main electronics board	Board	-	1	12±0.2	12±0.2	Measured
	Intergrated circuits	-	1	7.2±0.1	7.2±0.1	Estimated
	Passive components	-	-	18±0.2	18±0.2	Measured
	Connectors	-	4	1±0.2	4±0.8	Estimated
	XBee 900MHz	XBP9B-XCST-001	1	10±0.2	10±0.2	Datasheet

Summary mass = 51.2 ± 1.5 g

Subsystem	Component	Material / Part number	Quantity	Mass piece [g]	Mass summary [g]	Source
Cameras and stabilisation mechanism	BLDC	PM3505	1	40±1	40±1	Measured
	Main Camera	ESP32-CAM	1	7±0.3	7±0.3	Datasheet
	Main Camera PCB	PCB	1	10±1	10±1	Measured
	Battery	Li-Ion 14500	1	21±0.1	21±0.1	Datasheet
	Secondary Camera	Adafrut 3202	1	3±0.3	3±0.3	Datasheet

Summary mass = 81 ± 2.7 g

Subsystem	Component	Material / Part number	Quantity	Mass piece [g]	Mass summary [g]	Source
Other elements	GPS	MTK3339	1	6±0.1	6±0.1	Measured
	Battery	Li-Ion 18650	1	59±0.1	59±0.1	Datasheet
	Pitot's tube sensor	ABPDJJT001PDSA3	1	5±0.3	5±0.3	Datasheet
	Speaker	57CS50G-75ND	1	9±0.1	9±0.1	Measured
	Electromagnets	JF-0530B	2	12±1	24±2	Measured
	LED stripe	-	0.4m	12±1	4.8±0.4	Estimated
	Battery holder	-	1	9±0.2	9±0.2	Measured

Summary mass = 116.8 ± 3.2 g

Component	Subsystem	Mass summary [g]
Payload	Nose cone structure	231.67 ± 6.14
	Upper Structure (heatshield and parachute mechanism)	186.1 ± 4.4
	Struts, springs and shock absorbers	23.8 ± 0.64
	Descent control system	117.3 ± 5.09
	Egg box structure	96.84 ± 3.1
	Main electronics board	51.2 ± 1.5
	Cameras and main camera stabilization mechanism	81 ± 2.7
	Other elements	116.8 ± 3.2

Summary mass: 904.71 ± 26.77 g

Margin: 904.71 - 900 = 4.71 g

The CanSat mass is within the given tolerance of ±10g. The heaviest components were measured, and we are quite sure our CanSat will not exceed the weight requirements when fully assembled.

Correction methods if the mass falls out of spec (900 ± 10g)

- In case the mass will be greater than 910g:
 - The infill of 3D printed components will be reduced
 - lighter parachute material will be used
- In case the mass will be less than 890g:
 - The infill of 3D printed components will be increased
 - Ballast may be added to CanSat

Communication and Data Handling (CDH) Subsystem Design

Jakub Kaszowski

Component	Type	Functions
STM32F412RET6	Processor	Executing received commands. Reading and processing data from sensors. Progressing through FSM, enabling actuators.
Internal STM backup registers	Memory	Storage of software state in case of reset.
XBee: XBP9B-XCST-001	Radio	Transmitting data and commands. XBee 900 MHz.
ANTX150P118B09153	Antenna	Amplifying signal range.
STM32 internal RTC	RTC	Measurement of mission time.

We made **no changes** since submitting PDR in the CDH section.

Payload Processor & Memory Selection

Selected CPU:	Boot time	Processor speed [MHz]	Data interfaces	Flash memory[kB]	RAM[kB]
STM32F412RET6	~3ms	100	4x I2C, 4x UART, 5x SPI/I2S, 1x SDIO, 1x USB 2.0	1024	256

Selected memory:	Capacity	Interfaces	Package
Internal STM backup registers	80B	APB1	-
Goodram M1A0 microCard 32GB	32GB	SPI SDIO	microSD

STM32 has almost everything we need integrated. Extra external memory (SD Card) is to store all telemetry data. This way the team can prepare PFR even if the radio link malfunctions.



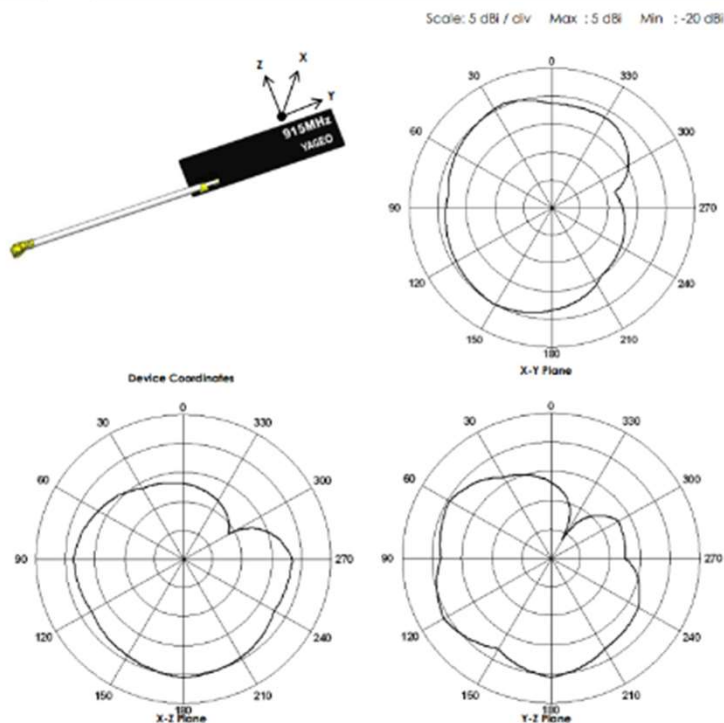
Name	Interface	Reset tolerance	Size	Weight
STM32 internal RTC	APB1	Resilient due to coin battery power backup	Integrated into STM32	-

STM32F412 has a **suitable RTC onboard**. It requires only an external crystal and small battery. Moreover, it is highly customizable and provides a number of calibration methods. The documentation is excellent and it is easy to program, as its internal registers can be accessed via debugger.

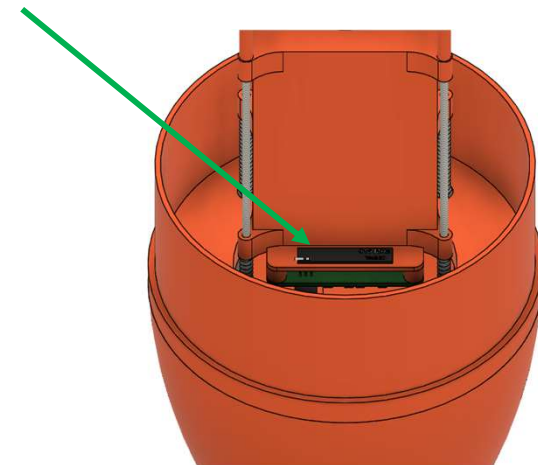


Name	Connector type	Antenna type	Frequency range(s) [MHz]	Weight [g]	Peak gain [dBi]	Efficiency [%]	Range [km]
ANTX150P118B091 53	I-PEX (U.fl compatible)	PCB strip	890-925	0.675	1.9	>55	~0.6 (worst case) ~9 (best case)

Directional characteristics:



Antenna position



XBee Radio selection:

XBee XBP9B-XCST-001 has been selected to be used in both payload and Ground Station.
The device operates up to **250 mW** of transmit power in **~915MHz** band.

XBee Configuration

XBees will operate in the same network in AT (transparent) mode.
The radio will operate in **unicast mode**, and it will communicate with the MCU using UART interface.

PANID/NETID is set to **2044** in accordance with requirement **X2**.

Transmission Control

Before the start, the radio will be waiting for configuration (CAL, CX).

After receiving the CX command, a data frame will be transmitted every second (**1Hz** frequency).

When landing is detected, the packet transmission is stopped.

Data field	Description	Resolution
TEAM_ID	Four digit team identification number.	-
MISSION_TIME	UTC time in format hh:mm:ss.	1s
PACKET_COUNT	Total count of transmitted packets maintained through processor reset.	1
MODE	'F' for flight mode, 'S' for simulation mode.	-
STATE	The operating state of the software (BOOT, IDLE, READY, FLIGHTUP, MAIN, PARACHUTE, TOUCHDOWN).	-
ALTITUDE	Altitude relative to ground level in meters.	0.1m
AIR_SPEED	Air speed measured with the pitot tube during ascent and descent in meters per second.	1m/s
HS_DEPLOYED	'P' if heat shield is deployed, 'N' otherwise.	-
PC_DEPLOYED	'C' if parachute is deployed, 'N' otherwise.	-
TEMPERATURE	Temperature in degrees Celcius.	0.1°C
PRESSURE	Measured air pressure in kPa.	0.1kPa
VOLTAGE	Voltage of the Cansat power bus in volts.	0.1V
GPS_TIME	UTC time from the GPS receiver in format hh:mm:ss.	1s

Data field	Description	Resolution
GPS_ALTITUDE	Altitude above sea level from the GPS receiver in meters.	0.1m
GPS_LATITUDE	Latitude from the GPS receiver in decimal degrees.	0.0001°
GPS_LONGTITUDE	Longitude from the GPS receiver in decimal degrees.	0.0001°
GPS_SATS	Number of GPS satellites being tracked by the GPS receiver.	1
TILT_X	X angle of Cansat. Perpendicular to Y axis and gravitation force vector.	0.01°
TILT_Y	Y angle of Cansat. Perpendicular to X axis and gravitation force vector.	0.01°
ROT_Z	Rotation rate of Cansat in degrees per second.	0.1°/s
CMD_ECHO	Last command received and processed by Cansat.	-
CAMERA1_STATE	State of first camera. States are: 'S' - sleeping, 'W' - waiting, 'R' - recording, 'E' - error	-
CAMERA2_STATE	State of second camera. States are: 'S' - sleeping, 'W' - waiting, 'R' - recording, 'E' - error	-

The data will be transmitted at **1Hz** frequency with each field delimited by a comma. Every packet is terminated by a single carriage return character. Additional telemetry fields are **CAMERA1_STATE** and **CAMERA2_STATE**.

Telemetry packet format

TEAM_ID, MISSION_TIME, PACKET_COUNT, MODE, STATE, ALTITUDE,
AIR_SPEED, HS_DEPLOYED, PC_DEPLOYED, TEMPERATURE, VOLTAGE,
PRESSURE, GPS_TIME, GPS_ALTITUDE, GPS_LATITUDE, GPS_LONGITUDE,
GPS_SATS, TILT_X, TILT_Y, ROT_Z, CMD_ECHO, CAMERA1_STATE,
CAMERA2_STATE

Telemetry packet example

2044,13:37:37,42,S,READY,0.0,0.0,N,N,11.1,4.15,101.6,
13:37:37,153.3,51.1090,17.0327,7,0.21,0.83,0.0,CX,R,R

The telemetry data shall be named **Flight_2044.csv**.

Competition requirements are met!

Command name	Format	Description	Example
CX - Payload Telemetry On/Off Command	CMD,<TEAM_ID>, CX,<ON_OFF>	1. CMD and CX are static text. 2. <TEAM ID> is the assigned team identification. 3. <ON_OFF> is the string 'ON' to activate the payload telemetry transmissions and 'OFF' to turn off the transmissions.	CMD,2044,CX,ON Activate payload telemetry transmission.
ST – Set Time	CMD,<TEAM_ID>, ST, <UTC_TIME> GPS	1. CMD and ST are static text. 2. <TEAM ID> is the assigned team identification. 3. <UTC_TIME> GPS is UTC time in the format hh:mm:ss or 'GPS' string which sets the flight software time to the current time read from the GPS module.	CMD,2044,ST,GPS Set flight software time to the current GPS time.
SIM - Simulation Mode Control Command	CMD,<TEAM_ID>, SIM,<MODE>	1. CMD and SIM are static text. 2. <TEAM_ID> is the assigned team identification. 3. <MODE> is the string 'ENABLE' to enable the simulation mode, 'ACTIVATE' to activate the simulation mode, or 'DISABLE' which both disables and deactivates the simulation mode.	CMD,2044,SIM,ENABLE Enable simulation mode.
SIMP - Simulated Pressure Data	CMD, <TEAM ID>, SIMP, <PRESSURE>	1. CMD and SIMP are static text. 2. <TEAM ID> is the assigned team identification. 3. <PRESSURE> is the simulated atmospheric pressure data in units of pascals with a resolution of one Pascal. Only used in simulation mode.	CMD,2044,SIMP,101325 Set simulated pressure reading to 101325Pa.

Command name	Format	Description	Example
CAL - Calibrate Altitude to Zero	CMD, <TEAM ID>,CAL	1. CMD and CAL are static text. 2. <TEAM ID> is the assigned team identification.	CMD,2044,CAL Sets altitude of relative altitude of Cansat to 0 meters.
BCN - Control Audio Beacon	CMD, <TEAM ID>, BCN,ON OFF	1. CMD and BCN are static text. 2. <TEAM ID> is the assigned team identification. 3. <ON OFF> are static strings “ON” or “OFF” that control the audio beacon.	CMD,2044,BCN,ON Turn on audio beacon.
RST* – System Reset	CMD, <TEAM ID>, RST, <TYPE>	1. CMD and RST are static text. 2. <TEAM ID> is the assigned team identification. 3. <TYPE> is the string ‘HARD’ to simulate power loss of MCU or ‘SOFT’ to reset RTC, packet count and relative altitude to default values.	CMD,2044,RST,HARD Simulate hard reset of MCU.

* - optional command

Competition requirements are met!

Electrical Power Subsystem (EPS) Design

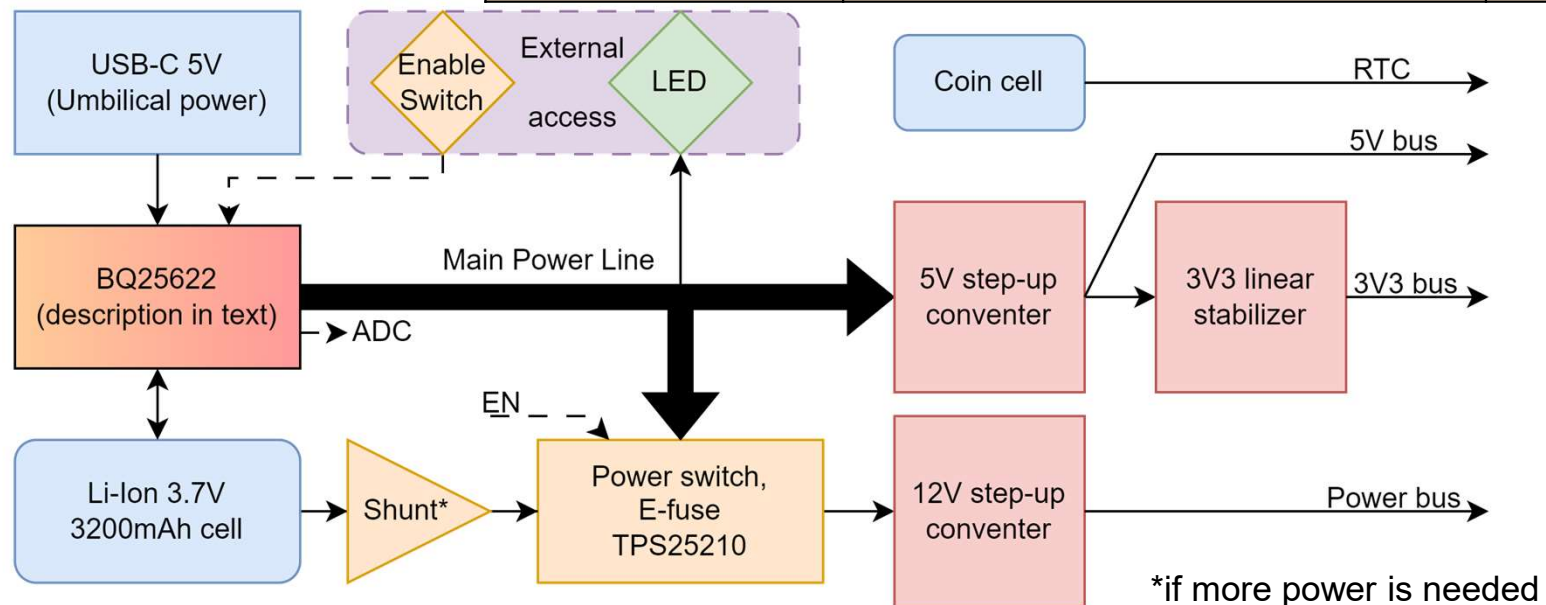
Jakub Kaszowski

The power section is based on BQ25622 IC. It is a highly integrated single-cell manager, capable of:

- switching up to 7A
- charging the cell from USB
- communicating over IIC
- measuring cell voltage
- fuel estimation (Columb counter)
- measuring load and temperature

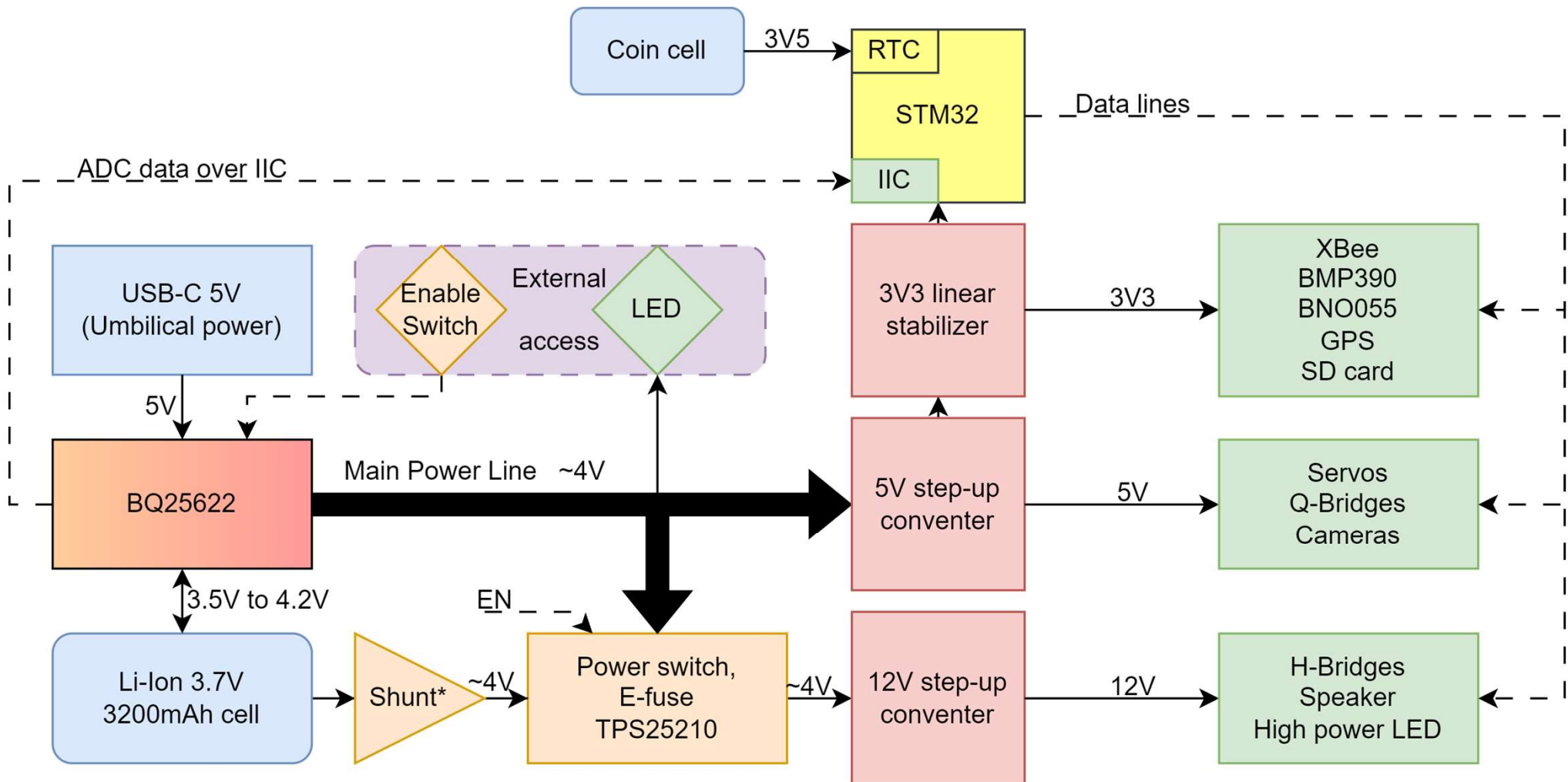
Required voltages are created by smps circuits, and RTC has its own cell to count time during unwanted restart. There is an external switch to enable CanSat, and a well visible LED to show it is enabled.

Component	Description	Type
USB-C connector	Umbilical power source	Source
3.7V Li-Ion cell	Main power source	Source
Coin cell	Prevents RTC from resetting	Source
5V step-up smps	Power supply for sensors and servos	Converter
3V3 linear stabilizer	Power supply for sensors and MCU	Converter
12V step-up smps	Power supply for power elements	Converter
External switch	Turns on/off main power line	Management
BQ25622	Highly integrated power manager IC	Management
TPS25210	Power switch with protection	Management



We made **no changes** since submitting PDR in the EPS section.

The external switch will allow us to switch on the circuit. LED provides feedback on the power state independent of the MCU. The main bus voltage is measured by BQ25622.



Part	Dimensions [mm]	Weight [g]	Voltage [V]	Capacity [mAh]	Current max [mA]	Type	Charge cycles	Integrated fuses	Cost [\$]
Panasonic NCR18650B	∅18.5 x 65.3	48.5	3.6	3200	6400	Li-Ion	500	No	7.39

Selected battery:	Reasons:
Panasonic NCR18650B	• High capacity • Small dimensions • High energy density • Low internal resistance • Easy to replace • Rechargeable



*Panasonic
NCR18650B*

The battery will be placed in a custom holder that is a part of the CanSat cone.
No spring contacts are used, the cell will be spot-welded to wires.
A single cell **does not have a configuration. Li-Poly cells are not used!**

Payload Power Budget (1/2)

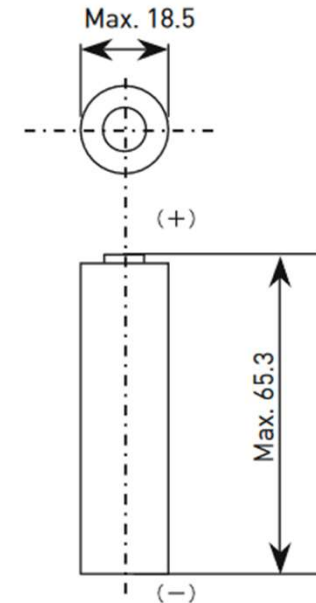
Component	Model	Voltage [V]	Current [mA]	Power [mW]	Duty cycle [s]	Duty Cycle [%]	Required Energy [Wh]	Source
uC+memory	STM32F412	3.3	30	99	7200	100%	0.198	Datasheet
Radio (TX)	XBP9B-XCST-001	3.3	215	709.50	1080	15%	0.213	Datasheet
Radio (RX)		3.3	26	85.80	6120	85%	0.146	Datasheet
GPS	MTK3339	3.3	20	66.00	7200	100%	0.132	Datasheet
Pressure	BMP390	3.3	1	3.30	7200	100%	0.007	Datasheet
Temperature								
IMU 9axis	BNO055	3.3	11	36.30	7200	100%	0.073	Datasheet
Pitot DPS	ABPDJJT001PDSA3	3.3	3.1	10.23	1080	15%	0.003	Datasheet
LED Strip	30/m	12	100	1200	1080	15%	0.360	Calculated
LED indicators	Multiple LEDs	3.3	30	99	7200	100%	0.198	Datasheet
SD card	GoodRam 32GB	3.3	(30) 100	(99) 330	(5760) 1440	(80%) 20%	0.290	Datasheet
3V3 stabilizer	AZ1117-3V3	5	0.06	0.30	7200	100%	0.001	Datasheet
5V SMPS /w main power IC	JENOR20200908 BQ25622E	3.7	3	11.10	7200	100%	0.022	Datasheet
12V SMPS	JENOR20200908	3.3	2	6.60	300	4.2%	0.001	Datasheet
Speaker	57CS50G-75ND	12	500	6000	360	5%	0.600	Datasheet
Electromagnets	JF-0530B	12	1000	12000	30	0.5%	0.100	Datasheet
Gimbal	PM3505	5	600	3000	240	3.3%	0.200	Calculated
Camera	ESP32-CAM	5	140	700	240	3.3%	0.047	Datasheet
Bonus Camera	Adafruit 3202	5	110	550	240	3.3%	0.037	Datasheet

Note: The value inside brackets () is standby or idle value.

Summary:

Energy required:	2.626 Wh
Energy required assuming efficiency 50%	~ 5.3 Wh
Power source:	One Li-ion cell
Total power available:	$3.6 \text{ V} * 3200 \text{ mAh} = 11520 \text{ mWh} = 11.52 \text{ Wh}$
Margin:	$11.52 \text{ Wh} - 5.3 \text{ Wh} = 6.22 \text{ Wh}$
Percent margin:	$5.3 \text{ Wh} / 11.52 \text{ Wh} * 100\% = 46\%$

DIMENSIONS (MM)



SPECIFICATIONS

Model number	NCR-18650B
Nominal voltage (V)	3.6
Nominal capacity*1 - Minimum (mAh)	3,250
Nominal capacity*1 - Typical (mAh)	3,350
Dimensions - Diameter (mm)	18.5
Dimensions - Height (mm)	65.3
Approx. weight (g)	47.5

With the selected battery, **the payload can operate for approximately 4 hours** when integrated into the rocket.

For safety we assumed that combined efficiency is 50%.

Flight Software (FSW) Design

Jakub Kaszowski

Overview

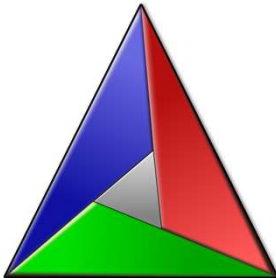
Flight software operates on the state machine concept. Using collected data, the FSW evaluates different states and manages events, such as reaching the proper altitude. Additionally, it is responsible for transmitting data to the Ground Station (GS) via XBee and restoring the correct state and time in the event of an unwanted restart.

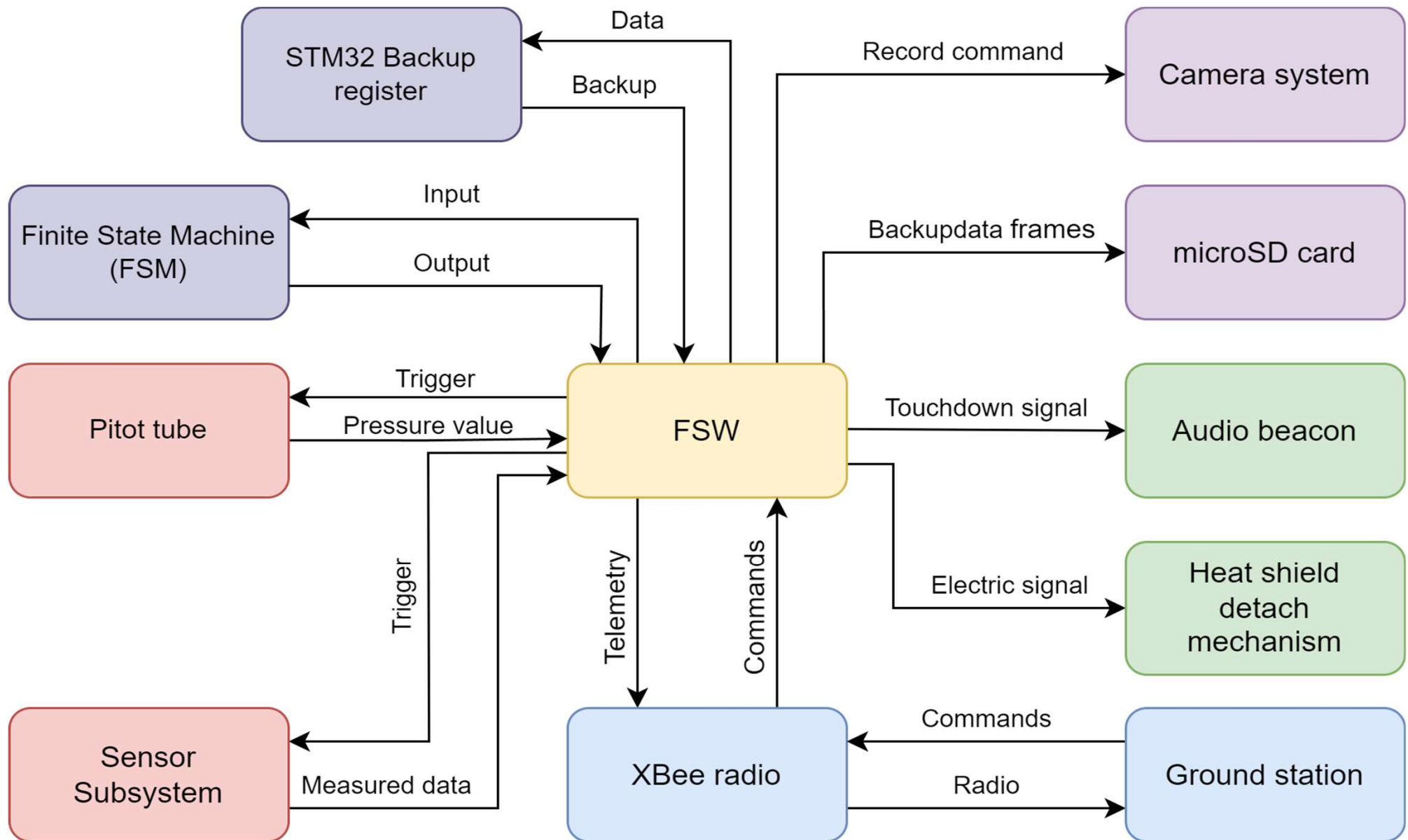
Programming language

Our software is based on C/C++ language.

Developing environment

CubeIDE, arm-none-eabi (compiler and debugger)

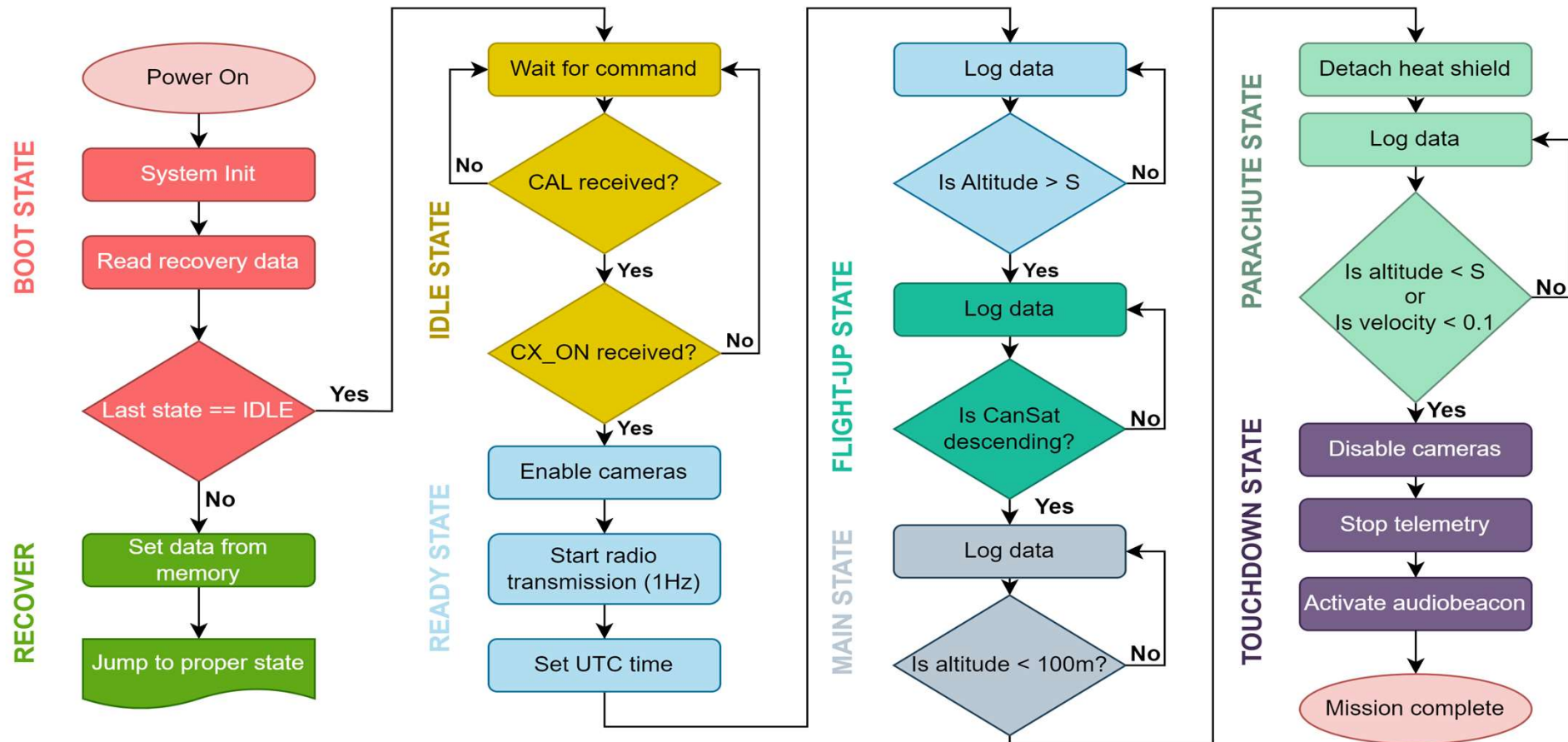




FSW Summary Tasks

- Recording raw sensor values and translating them into standardized engineering units
- Sensor calibration
- Collect and analyze commands from the Ground Station (e.g. to start transmission or perform reset)
- Formatting data packets in the necessary format for transmission
- Swapping pressure readings for received values in simulation mode
- Backing-up telemetry data on SD cards
- Activating the audio beacon
- Advancing through Flight Software (FSW) States
- Detaching the heat shield at the appropriate moment
- Controlling video recording
- Transmitting telemetry packets to Ground Station via XBee
- Saving recovery information on MCU battery backed-up registers

We made **no changes** since submitting PDR in the FSW section.



Sampling of Sensors	Communications	Extra data storage
The sensors are sampled with a frequency of 1 Hz.	XBee transmit data at a rate of 1 Hz.	Telemetry data is also saved on an SD card.

S - safe altitude over the launchpad that cannot be reached before flight

Unwanted restart mechanism

- In the event of an unwanted restart, correct values are retrieved from non-volatile memory (STM32 backup registers). The values are saved every time they change. These values are:
 - internal packet counter
 - software state number
 - altitude of the launchpad

- Reasons for reset are:
 - RST command
 - watchdog timer error
 - momentary power loss

To restore the state after a restart, the MCU will examine the backup registers. If data is available, it will be loaded into variables, and the state selector will transition to the appropriate state. Otherwise, the state selector will transition to the Idle state.

Simulation mode serves testing and demonstration needs. To activate simulation mode, the Ground Station must transmit two commands (SIM ENABLE and SIM ACTIVATE) to the Payload. Requiring two commands minimizes the risk of unintentional activation.

Once initiated, the Ground Station will periodically dispatch air pressure values as barometric pressure sensor commands (SIMP) at **one-second** intervals. FSW will use received values instead of real data from pressure sensor. It will let FSW to compute a simulated altitude for use in software logic.

SIM - Simulation Mode Control Command

- CMD,2044,SIM,ENABLE - enable the simulation mode
- CMD,2044,SIM,ACTIVATE - activate the simulation mode
- CMD,2044,SIM,DISABLE - both disables and deactivates the simulation mode

SIMP - Simulated Pressure Data

- CMD,2044,SIMP,<PRESSURE> - simulated atmospheric pressure data in Pascals (1Pa resolution)

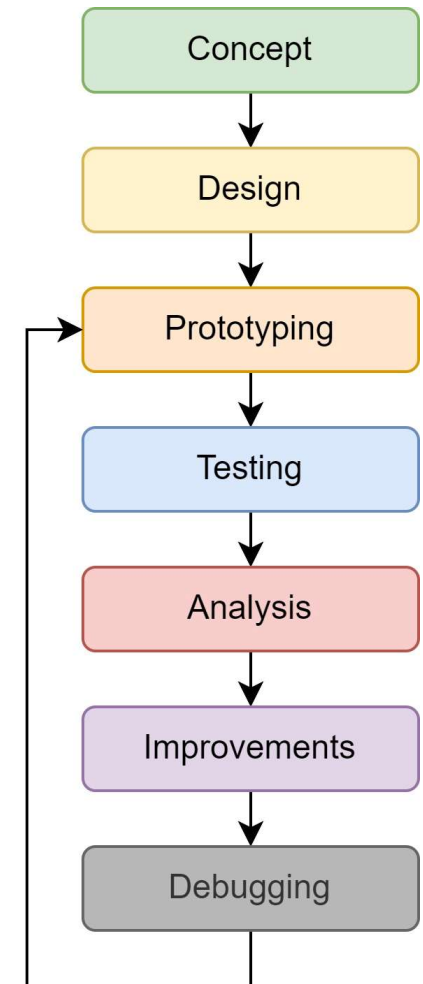
Prototyping and prototyping environments

For prototyping, breakout modules and breadboards are employed. Each sensor undergoes individual testing. Upon successful completion of the tests and meeting the specified requirements, all components are integrated into a custom printed circuit board (PCB). This approach facilitates the development of the entire system in a cohesive manner.

Software developing

We use Git to store and manage our codebase. This tool enables collaborative work within the team, issue tracking, and facilitates code reviews. To maintain the continuity of development, we plan to conduct weekly meetings to address and discuss any problems or challenges. We recreated XBee which used a USB connector instead of an antenna to transmit data

Development team: Hubert Kulik, Jakub Kaszowski, Filip Ziolo, Maksymilian Tara



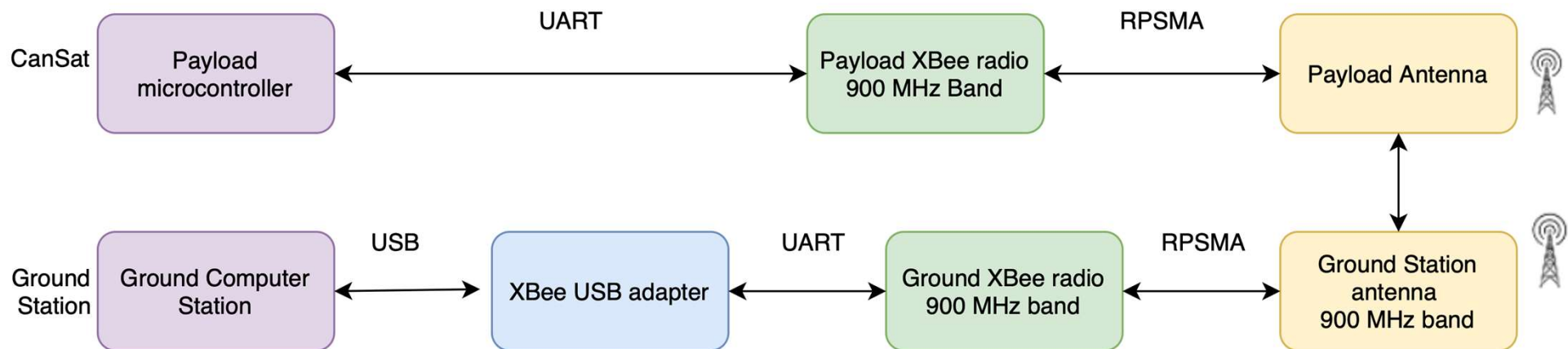
Since handing over the CDR, we have purchased all the necessary electronic parts, such as sensors and PCBs. Now we focus on programming sensors and communication. Below is a table with tasks.

Progress since PDR	
Programming Clocks, startup, logging	Done
Backing-up telemetry data on SD cards	Done
Communication between payload and GS	Done
Detaching the heat shield at the appropriate moment	In progress
Libraries for each sensor	Done
Camera	In progress

Ground Control System (GCS) Design

Jakub Kaszowski

We will configure our own Ground Control System, consisting of the following components:

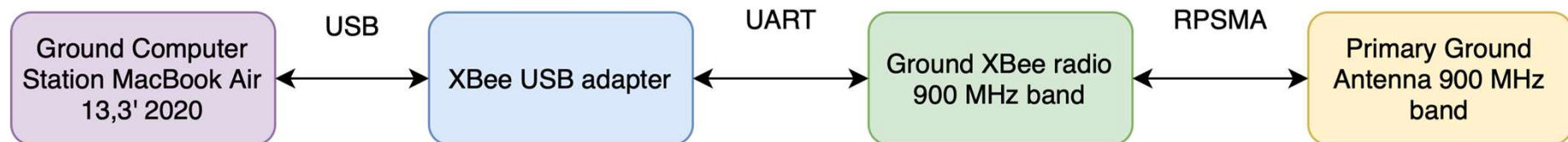


The whole GCS setup is powered from battery and easy to carry.

We made **no changes** since submitting PDR in the GCS section.

Specifications

Operation time	GCS will be able to operate for at least 3 hours with no external power supply.
Overheating mitigation	We will protect the computer from sunlight with an umbrella.
Screen reflection mitigation	The umbrella will be covered with a blanket. The computer will operate in light mode (black text on white background) with size of 15 points. The plots will be made using bold colors on light background
Autoupdate mitigation	A Macbook Air is going to be used with auto updates turned off, and macOS does not enforce auto updates when they are disabled.



Model	Type	Frequency range [MHz]	Gain [dBi]	Size [mm]	Direction	Range [km]
Interline HORIZON 900 8V	Dipole	900-950	8	8600x260	Omnidirectional	-5

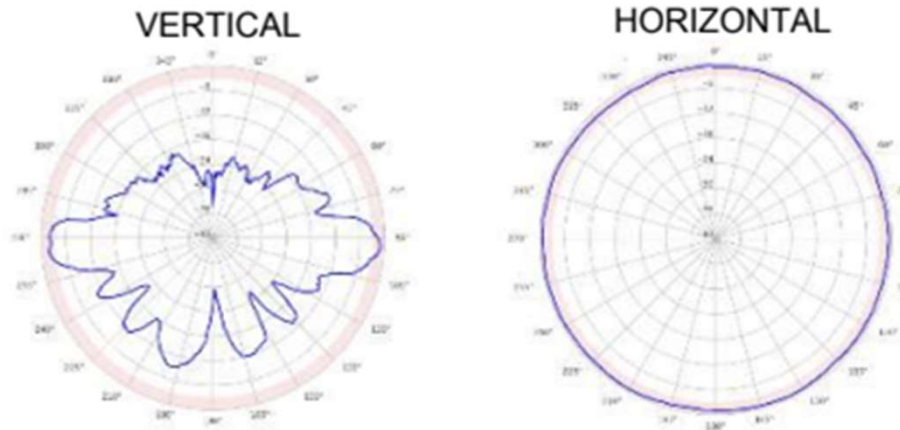


Interline HORIZON 900 8V

The dipole antenna offers excellent signal reception coverage. Furthermore, this particular antenna model underwent testing in real-world scenarios during past CanSat competitions.

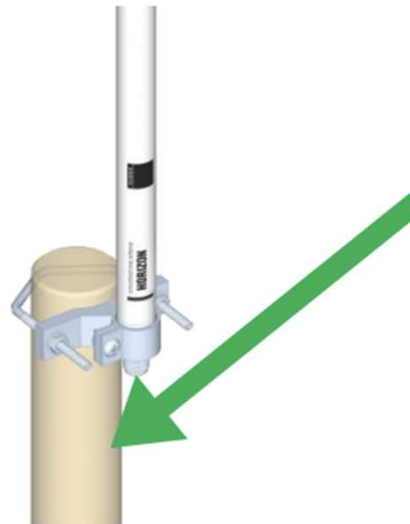
Radiation patterns comparison

Interline HORIZON 900 8V (selected)



Antenna mounting design

Our primary transmitter is the Interline HORIZON 900 8V antenna. We'll mount it onto a handle and manually adjust its direction to align with the signal sent from the CanSat. Attaching the Interline HORIZON 900 8V dipole antenna to a PVC pipe via its mounting holes allows for convenient handling and manipulation by the operator. This configuration facilitates easy maneuvering of the antenna during operation. **This is a handheld antenna.**



Interline HORIZON 900 8V

Path loss

Predicted distance: $d = 3500m = 3.5 km$

Center frequency: $f = 915 MHz$

Center frequency wavelength: $d = 0.327 = 32.7 cm$

$$L_{FS}[dB] = 20 \log_{10} \left(\frac{4\pi d}{\lambda} \right) = 20 \log_{10} \left(\frac{4\pi \cdot 3500}{0.327\lambda} \right) = 102.7 [dB]$$

Link Budget (Z)

Receiver sensitivity (min) = $-106 dBm @ (10kbps)$
 $-77.7 dBm > -106 dBm$

Receiver satisfies requirements

Link Budget

$$P_{rx} = ?$$

$$P_{tx} = 20 dBm (XBEE)$$

$$G_{tx} \approx 2 dBi$$

$$P_{rx} = P_{tx} + G_{tx} - L_{fs} - L_{ADD} + G_{rx}$$

$$G_{rx} = 8 dBi$$

$$L_{fs} = 102.8 dBi$$

$$P_{rx} = 20 + 2 - 102.7 - 5 + 8 = -77.7 dBm$$

$$L_{ADD} = 5 DBi$$

P_{rx} - Power @ receiver input

P_{tx} - Transmitter power

G_{tx} - Gain of transmit antenna

G_{rx} - Gain of receiver antenna

L_{fs} - Free space losses

L_{ADD} - Additional losses (cable etc.)

- **Software packages**

The GCS software will be built based on the QT framework and the C++ standard library. QT will be used for the frontend in order to achieve real-time plotting. The QT backend will also process data received on the serial port from the XBee radio.

- **Telemetry**

Telemetry will be live plotted during flight, each field on separate chart. All data will be displayed using engineering units. The units will be indicated on the displayed charts. After the flight, telemetry data from all sensors will be saved to a Flight_2044.csv. Telemetry data file will include mission time with 1 second resolution. Then, CSV files are going to be handed over to judges on USB stick. All plots will be made using bold colours on white background, text will be black with size at least 14 points. The program will count received packets independently from packet counter value of the received packets.

- **Simulation mode description**

The Ground Station operator can enable the simulation mode by sending two commands: SIMULATION ENABLE and SIMULATION ACTIVATE. A file with simulation data will be loaded beforehand. After broadcasting a packet that enables simulation mode, the operator will send an activation command, after which the Ground Station will start sending successive values from the CSV file at a frequency of 1 Hz as SIMP command.

- **Changes since PDR**

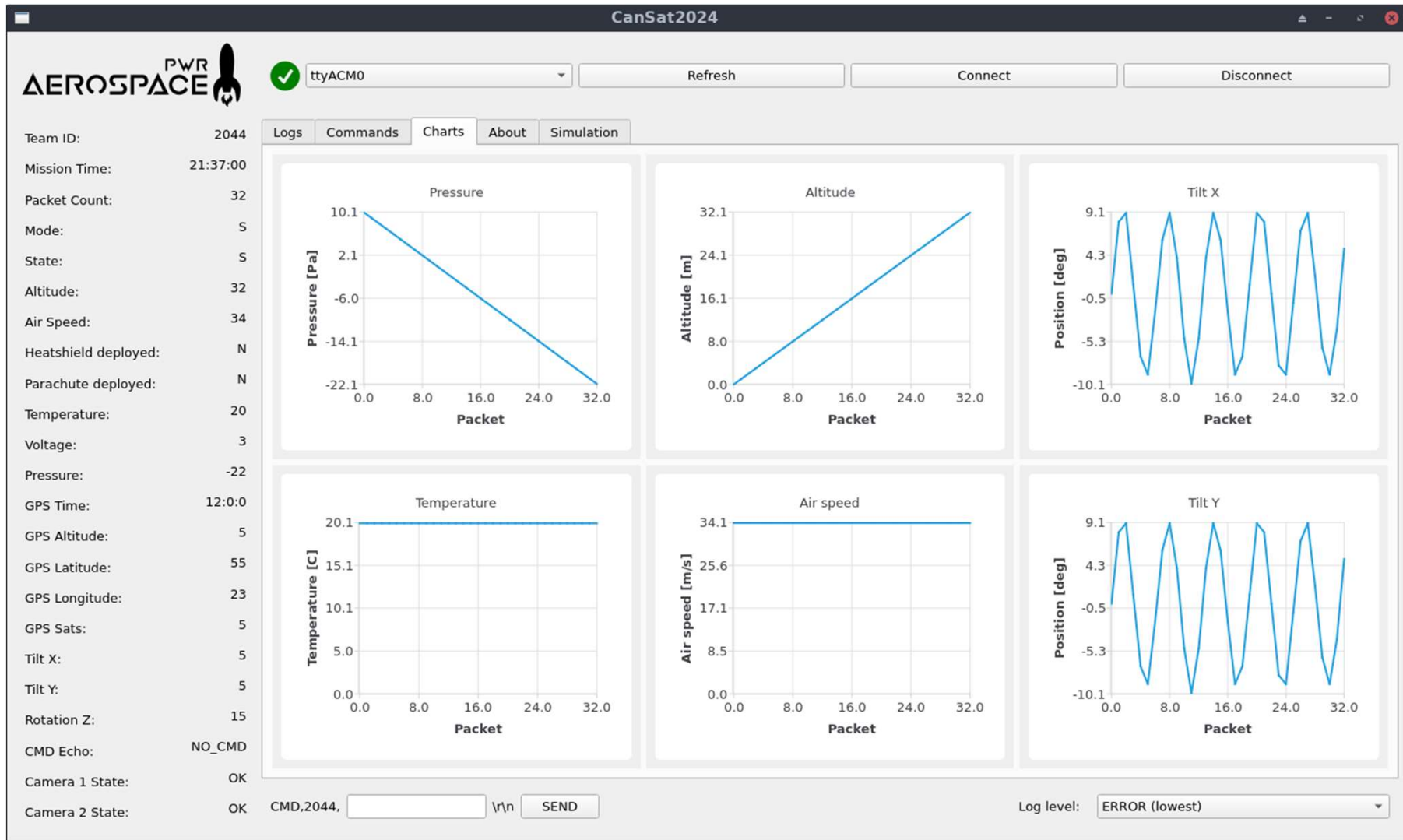
The code that has been written so far is responsible for parsing the input data and plotting it. It uses data structure and methods to transform text into data objects. The parser validates the data and returns the parsed information, allowing further actions with it. LineChart class has been added which creates a line chart in the Qt UI. The line chart can be dynamically updated. Moreover, sending of arbitrary commands and sending data from csv was also implemented since PDR.

- **Commands**

The Ground Station operator can send a calibration command for the launchpad altitude calibration and the tilt sensor. Calibration command will be transmitted by using XBee radio link.

The results of calibration can be verified in the GCS interface in the field "last command received". Verification before launch: altitude shown as 0, tilt shown as 0 and should be steady.

Current state of Ground Control System interface:

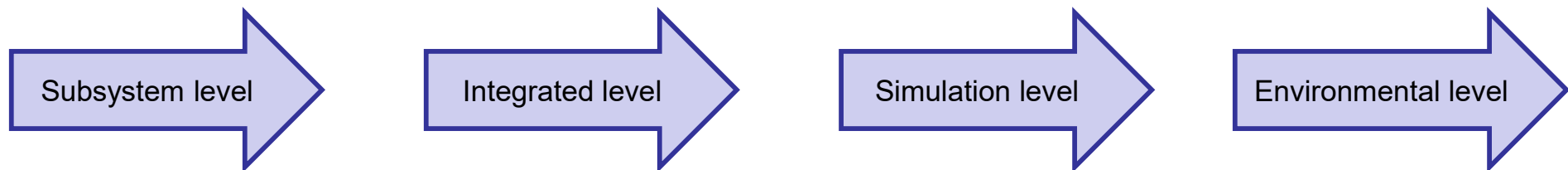


CanSat Integration and Test

Jakub Kaszowski

CanSat Integration and Test Overview

Our CanSat will be tested by four level testing		
Subsystem level	All the subsystems will be tested separately	Sensors, CDH, EPS, Radio Communications, FSW, Mechanical, Descent Control
Integrated level	Subsystems will be integrated and tested together	Descent Testing, Communications, Mechanisms, Deployment
Simulation level	Integrated subsystems will be tested in simulation mode to test flight level software logic	Flight Software Logic
Environment level	Whole device will be subjected to extreme conditions and final mechanical checks will be performed.	Drop test, Thermal test, Vibration test, Fit Check, Vacuum test



1. Firstly, the payload, container body and mechanism will be built and initially tested. Independently, the electronics circuits will be soldered, programmed and debugged.
2. When all of the components are working properly, they will be put together and tested as a whole device to ensure that all mechanism and electronics work together properly.
3. Once the integration test are complete, we will use simulation mode to ensure that all algorithms are implemented properly.
4. Finally, the environmental tests will be conducted to make sure that our CanSat can work within specified environment.

Subsystem Level Testing Plan (1/4)

Subsystem	Test case	Acceptance criteria	Test result
FSW	Microcontroller is correctly initialized	Debugger is connecting	Tested successfully
	Debug logs can be sent via serial interface	Serial interface transmits data	Tested successfully
	After initialization FSM jumps to state selector	Debug log indicates FSM entered state selector	TBT
	FSW can progress through mission states	Each state sends signed debug log	TBT
	FSW Can survive power cuts	FSM starts with last state, sensor calibration data is remembered, date and time is correct	Tested successfully
Mechanical	Payload must maintain its structural integrity after experiencing shock and vibrations	No cracks or loose bolts must be detected after a shock test, vibration test and simulated landing	Partially tested
	The heatshield opens immediately after the payload is released	The heatshield spokes move freely	Partially tested
	The mechanism detaches the heatshield assembly	The heatshield assembly detaches easily after the mechanism is activated	TBT
	Stable descent on heatshield is maintained	The main camera faces in one direction for the whole descent	TBT
	The pitot tube is not destroyed upon landing	The pitot tube must be operational after a simulated landing	TBT
	The Egg Compartment can protect the hen's egg from impact	The egg isn't cracked after a simulated landing	TBT
	Payload parachute opens after heatshield assembly detachment	Heatshield is detached, parachute cords are not tangled	TBT
	Main electronics board is isolated from vibrations	Electronics does not reset, all communications interfaces are working	TBT

Subsystem Level Testing Plan (2/4)

Subsystem	Test case	Acceptance criteria	Test result
Sensors	Sensors and devices are initialized, communication is possible	Correct sensor/device data can be read via debug logs	Tested successfully
	Altitude can be properly calculated	Altitude calculated from test values corresponds to expected altitude	TBT
	Temperature measuring	Temperature is correctly measured	TBT
	Pressure measuring	Pressure is correctly measured	TBT
	Voltage measuring	Voltage is correctly measured	Tested successfully
	GPS data is available	GPS data can be read by FSW	TBT
	GPS NMEA frame is correctly decoded	GPS time, latitude, longitude, altitude and sats are available	TBT
	Sensors can be calibrated and return zero values when stationary on ground	Values read from sensors are zeroed	TBT
CDH	File System is properly initialized on SD Card	File system initialization success, SD card can be read by a windows laptop	Tested successfully
	SD Card is ready for writing data	File system returns success when saving files	Tested successfully
	SD Card is accessible, data can be read	File system allows fore reading files, data in files are correct	Tested successfully
	Data saved on SD Card survives resets and power downs	Files are consistently available	Tested successfully
	Real-Time Clock measures time	Times of CanSat and GS are in-sync during the tests	Tested successfully

Subsystem	Test case	Acceptance criteria	Test result
EPS	Pressure sensor is powered	Proper voltage on necessary pins is measured	Tested successfully
	Temperature sensor is powered	Proper voltage on necessary pins is measured	Tested successfully
	GPS sensor is powered	Proper voltage on necessary pins is measured	Tested successfully
	Voltage sensor is powered	Proper voltage on necessary pins is measured	Tested successfully
	Camera is powered	Proper voltage on necessary pins is measured	Tested successfully
	Processor is powered	Proper voltage on necessary pins is measured	Tested successfully
	microSD card is powered	Proper voltage on necessary pins is measured	Tested successfully
	PCB doesn't have shortcuts	PCB has no shorts to ground or power lines	Tested successfully
	PCB is properly soldered	Joints are properly soldered, there are no cold joints	Tested successfully
	Battery withstands high current	Battery works correctly under high current	Tested successfully
	CanSat can operate for two hours	CanSat does not turn off when left in IDLE state for two hours	TBT
	Voltage regulators and DC converters work properly	Proper voltage on all power lines under load	TBT
	IMU is powered	Proper readings from the sensor given the positioning	Tested successfully
	XBee is powered	Proper voltage on necessary pins is measured	Tested successfully

Subsystem	Test case	Acceptance criteria	Test result
Descent control	FSW can control deployment mechanisms	FSW can control parachute and heatshield	TBT
	State machine detects change in flight phases	State is changed when appropriate conditions are met	TBT
	Aerobraking surface maintain correct descent velocity	CanSat descends at rate between 10 to 30 m/s	TBT
	Parachute flight maintain correct descent velocity	CanSat descends at less than 5 m/s	TBT
Radio communication	XBee Radio can receive data	Data sent to XBee can be read via debug logs	TBT
	XBee Radio can transmit data	Data sent from FSW can be read on PC with XBee	TBT
	XBee radios NETID/PANID is set to their team number	NETID/PANID is set to 2044	TBT
	Open air radio range testing	Radio range is sufficient for double of the planned flight altitude	TBT
	Ground level radio range testing	Radio range is sufficient for the flight altitude	TBT
	GCS can connect with the CanSat	The connection is reached and stable	TBT
	GCS can transmit and receive data	Data is successfully transmitted and received	Tested successfully
	GCS plotting data	Data is plotted in real time	TBT

System	Test overview
Descent testing	The CanSat will be lifted using an airplane and then released. The stability of flight will be checked. Both cameras stability will be verified. Descent rate of the probe with heatshield open shall be between 10 to 30 m/s. The descent rate after releasing the parachute at 100m shall be less than 5 m/s.
Communications	We will conduct field tests to check the range of radio communication and bandwidth. The range will be checked on a football field by increasing distance between probe and ground station.
Mechanisms	Tests of Mechanical and Descent Control Subsystem, such as release mechanism of heat shield and parachute, and drop test will be performed with the CanSat at Integrated Level.
Deployment	To make sure that each deployment subsystem works according to our assumptions, the Payload drop test will be performed. Additionally, stationary tests of parachutes opening and heatshield release will be done.
Simulation	After sending two commands SIMULATION ENABLE and SIMULATION ACTIVE pressure data will be sent to the container. Recorded data frames will be checked and compared with our prepared mission state diagram

Test	Test overview
Drop test	The CanSat parachute mounting will be mounted to a non-stretching cord and then dropped, letting the probe free fall from altitude of 61 cm to simulate 30 Gs shock. After the test, the mechanical and electronic systems will be checked. During the test the communications will be maintained. All of them shall survive and work properly, data packets should not be lost.
Thermal test	As a thermal chamber, an oven will be used. It will be heated to 60°C. Once hot, the CanSat will be placed inside for 2 hours. During the whole test telemetry shall be received constantly to verify if all systems work in required temperature range.
Vibration test	The CanSat will be attached to an orbit sander. Vibrations (up to 233 Hz) must not affect the operation of all systems and all mechanical components must avoid damage. The accelerometer data shall be collected throughout the test.
Fit Check	A shoulder of 136 mm diameter x 50 mm length will be built. The test will verify, if the CanSat fits inside and can be easily deployed during the flight (it must not get stuck).
Vacuum test	The CanSat will be placed in a hermetic chamber connected to a vacuum cleaner. The pressure value will decrease and the altitude value will increase. The test will show how the altitude value change when the pressure value changes.

FSW:

Test No.	Test Description	RN	Pass Criteria
1	Microcontroller is correctly initialized	-	Debugger is connecting
2	Debug logs can be sent via serial interface	-	Serial interface transmits data
3	After initialization FSM jumps to state selector	-	Debug log indicates FSM entered state selector
4	FSW can progress through mission states	-	Each state sends signed debug log
5	FSW Can survive power cuts	F2	FSM starts with last state, sensor calibration data is remembered, date and time is correct

Mechanics:

Test No.	Test Description	RN	Pass Criteria
6	Payload must maintain its structural integrity after experiencing shock and vibrations	M3	No cracks or loose bolts must be detected after a shock test, vibration test and simulated landing
7	The heatshield opens immediately after the payload is released	C3	The heatshield spokes move freely
8	The mechanism detaches the heatshield assembly	C5	The heatshield assembly detaches easily after the mechanism is activated
9	Stable descent on heatshield is maintained	SN8	The main camera faces in one direction for the whole descent
10	The pitot tube is not destroyed upon landing	-	The pitot tube must be operational after a simulated landing
11	The Egg Compartment can protect the hen's egg from impact	M8	The egg isn't cracked after a simulated landing
12	Payload parachute opens after heatshield assembly detachment	C5	Heatshield is detached, parachute cords are not tangled
13	Main electronics board is isolated from vibrations	-	Electronics does not reset, all communications interfaces are working

SSD:

Test No.	Test Description	RN	Pass Criteria
14	Sensors are initialized, communication is possible	-	Correct sensor/device data can be read via debug logs
15	Altitude can be properly calculated	-	Altitude calculated from test values corresponds to expected altitude
16	Temperature measuring	SN3	Temperature is correctly measured
17	Pressure measuring	SN2	Pressure is correctly measured
18	Voltage measuring	SN6	Voltage is correctly measured
19	GPS data is available	-	GPS data can be read by FSW
20	GPS NMEA frame is correctly decoded	X5	GPS time, latitude, longitude, altitude and sats are available
21	Sensors can be calibrated and return zero values when stationary on ground	G1	Values read from sensors are zeroed

CDH:

Test No.	Test Description	RN	Pass Criteria
22	File System is properly initialized on SD Card	-	File system initialization success, SD card can be read by a laptop with Windows
23	SD Card is ready for writing data	-	File system returns success when saving files
24	SD Card is accessible, data can be read	-	File system allows for reading files, data in files are correct
25	Data saved in backup registers survives resets and power downs	F2	Variables are consistently available.
26	Real-Time Clock measures time	-	Time of CanSat and GS is in-sync during the tests

EPS:

Test No.	Test Description	RN	Pass Criteria
27	Pressure sensor is powered	SN2	Proper voltage on necessary pins is measured
28	Temperature sensor is powered	SN3	Proper voltage on necessary pins is measured
29	GPS sensor is powered	X5	Proper voltage on necessary pins is measured
30	Voltage sensor is powered	SN6	Proper voltage on necessary pins is measured
31	Camera is powered	SN7	Proper voltage on necessary pins is measured
32	Processor is powered	-	Proper voltage on necessary pins is measured
33	MicroSD card is powered	-	Proper voltage on necessary pins is measured
34	PCB doesn't have shortcuts	-	PCB has no shorts to ground or power lines
35	PCB is properly soldered	-	Joints are properly soldered, there are no cold joints
36	Battery withstands high current	-	Battery works correctly under high current
37	CanSat can operate for two hours	E5	CanSat does not turn off, when left in IDLE state, for two hours
38	Voltage regulators and DC converters work properly	-	Proper voltage on all power lines under load
39	IMU is powered	SN4, SN5	Proper readings from the sensor given the positioning
40	Xbee is powered	X1	Proper voltage on necessary pins is measured

Descent control:

Test No.	Test Description	RN	Pass Criteria
41	FSW can control deployment mechanisms	C5	FSW can control parachute and heatshield release
42	State machine detects change in flight phases	C2, C5, C6, C7	State is changed when appropriate conditions are met
43	Aerobraking surface maintains correct descent velocity	C10	CanSat descends at rate between 10 to 30 m/s
44	Parachute flight maintains correct descent velocity	C11	CanSat descends at less than 5 m/s

Radio communication:

Test No.	Test Description	RN	Pass Criteria
45	XBee Radio can receive data	X4	Data sent to XBee can be read via debug logs
46	XBee Radio can transmit data	G1	Data sent from FSW can be read on PC with Xbee
47	XBee radios NETID/PANID is set to their team number	X2	Payload NETID/PANID is read as 2044
48	Open air radio range testing	-	Radio range is sufficient for double of the flight altitude
49	Ground level radio range testing	-	Radio range is sufficient for the flight altitude
50	GCS can connect with the CanSat	G2, G15	The connection is reached and stable
51	GCS can transmit and receive data	G1, G3	Data is successfully transmitted and received
52	GCS plotting data	G6	Data is plotted in real time, no freezing occurs

Tests at integrated level:

Test No.	Test Description	RN	Pass Criteria
53	Descent testing: CanSat dropped from a plane will descent with velocity required by mission guide	SN1, M6, M7	Descent velocities read from sensors classified against an FSW state will be in the required range
54	Communications: Maximum radio range will be found. Test data will be send to GS	-	Minimum radio range must exceed 1.5 km. Format of data frame will be compared to mission guide
55	Mechanism 1: electromagnets can release the heatshield and open the parachute.	C5, M7	The heatshield is released immediately upon release mechanism activation. The parachute opens freely.
56	Mechanism 2: Heatshield opens after release from the rocket, mylar streamer develops	C3, C4, M5	The heatshield opens immediately, mylar streamer opens immidieatly
57	Mechanism 3: Egg box, that shields the egg from cracking	C5, M8	The egg is intact after a simulated landing test
58	Mechanism 4: The camera is pointed in one direction	SN7, SN8, SN9	The camera records the horizon in one direction for the whole duration of the flight.
59	Deployment: all stages of deployment will be tested during trial mission	M3	The trial mission goes along the mission guide. All requirements are meet
60	Simulation: obtained data will be compared to mission state diagram	G3, G12, X4, F4	Obtain results must be received in correct format. States should change in accordance of mission state diagram. Frequency of send data will be checked.
61	Electrical 1: Audio beacon test	C7	The speaker produces 92dB of sound pressure when integrated into the body of CanSat. Measured from 0.5m

Environmental tests:

Test No.	Test Description	RN	Pass Criteria
62	Drop test: CanSat will be connected to the cord and dropped from 61 cm	S7, S11, M3, M4	Probe must be fully intact and must still transmit data after drop test
63	Thermal test: CanSat will be placed in oven for two hours in 60 Celsius degree	M3, SN3	Probe must be fully intact and transmit data without loses during test
64	Vibration test: CanSat will be held on orbital sander	S6, M3, M4	Probe must be fully intact and transmit data without loses during test
65	Fit Check: CanSat will be placed in a tube with diameter of 136mm, total length except nosecone will not exceed 350mm, the nosecone has dimensions specified by mission guide.	S2, S3, S4, S5	Probe must fit the tube without any problems, the length must not exceed 350mm
66	Vacuum test: CanSat will be placed in vacuum chamber	M3, SN2	Probe must be fully intact and transmit data without loses during test

Remaining requirements:

RN	Description	Evidence of compliance [Page]
C1	The Cansat shall function as a nose cone during the rocket ascent portion of the flight.	10, 12, 45
C8	The Cansat shall carry a provided large hens egg with a mass range of 51 to 65 grams	38, 50
C9	0 altitude reference shall be at the launch pad.	83, 96
C12	Cost of the Cansat shall be under \$1000. Ground support and analysis tools are not included in the cost of the Cansat. Equipment from previous years shall be included in this cost, based on current market value.	144
S1	The Cansat mass shall be 900 grams +/- 10 grams without the egg being installed.	52-57, 58
S5	Nose cone shoulder length shall be a minimum of 50 mm	12, 15, 41
S8	The Cansat shall perform the function of the nose cone during rocket ascent.	10, 45
S9	The rocket airframe can be used to restrain any deployable parts of the Cansat but shall allow the Cansat to slide out of the payload section freely.	10, 15, 45
S10	The rocket airframe can be used as part of the Cansat operations.	10, 45
M1	No pyrotechnical or chemical actuators are allowed.	38
M2	Mechanisms that use heat (e.g., nichrome wire) shall not be exposed to the outside environment to reduce potential risk of setting the vegetation on fire.	40
M9	If the nose cone is to be considered as part of the heat shield, the documentation shall identify the configuration.	40

Remaining requirements:

RN	Description	Evidence of compliance [Page]
M10	After the Cansat has separated from the rocket and if the nose cone portion of the Cansat is to be separated from the rest of the Cansat, the nose cone portion shall descend at less than 10 meters/second using any type of descent control device.	48
E1	Lithium polymer batteries are not allowed.	75
E2	Battery source may be alkaline, Ni-Cad, Ni-MH or Lithium. Lithium polymer batteries are not allowed. Lithium cells must be manufactured with a metal package similar to 18650 cells. Coin cells are allowed.	75
E3	Easily accessible power switch is required	13, 42, 72
E4	Power indicator is required	13, 42, 72
X3	XBEE radios shall not use broadcast mode.	65
SN5	Cansat shall measure its rotation rate during descent.	25
G4	Configuration states such as zero altitude calibration shall be maintained in the event of a processor reset during launch and mission.	84
G5	Each team shall develop their own ground station.	89
G7	All telemetry shall be displayed in engineering units (meters, meters/sec, Celsius, etc.) and the units shall be indicated on the displays.	96, 97
G8	Teams shall plot each telemetry data field in real time during flight.	96, 97
G9	The ground station shall include one laptop computer with a minimum of two hours of battery operation, XBEE radio and an antenna.	89, 91

Remaining requirements:

RN	Description	Evidence of compliance [Page]
G10	The ground station must be portable so the team can be positioned at the ground station operation site along the flight line. AC power will not be available at the ground station operation site.	91
G11	The ground station software shall be able to command the payload to operate in simulation mode by sending two commands, SIMULATION ENABLE and SIMULATION ACTIVATE.	96
G13	The ground station shall use a table top or handheld antenna.	94
G14	Because the ground station must be viewed in bright sunlight, the displays shall be designed with that in mind, including using larger fonts (14 point minimum), bold plot traces and axes, and a dark text on light background theme.	91
F1	The flight software shall maintain a count of packets transmitted which shall increment with each packet transmission throughout the mission. The value shall be maintained through processor resets.	66, 68, 84
F3	The Cansat shall have its time set to within one second UTC time prior to launch.	66, 83
F5	In simulation mode, the flight software shall use the radio uplink pressure values in place of the pressure sensor for determining the payload altitude.	85
F6	The payload flight software shall only enter simulation mode after it receives the SIMULATION ENABLE and SIMULATION ACTIVATE commands.	85

Simulation mode is used mostly for testing the Flight Software. It helps testing general logic and corner cases.

In this mode, the probe swaps actual readings of **pressure sensor** for those received from ground station.

It doesn't affect other sensors readings - the values, other than the pressure and altitude will be actual sensor readings.

Simulation sequence:

- Ground Station sends a sequence of two commands (SIM ENABLE and SIM ACTIVATE)
- FSW enters simulation mode
- GS sends air pressure values at a one second interval as barometric pressure sensor commands, using a provided CSV file
- FSW uses received values instead of real data from pressure sensor
- To disable simulation mode, GS sends SIM DISABLE
- FSW goes back to normal mode

Mission Operations & Analysis

Jakub Kaszowski

Overview of Mission Sequence of Events (1/3)

Our team acquired experience thanks to participation in the previous competition editions. For this reason, we expanded our team with new members to share our knowledge and make the design better.

Role	Person
Mission Control Officer (MCO)	Jakub Kaszowski
Ground Station Crew (G)	Julia Szymańska, Tymoteusz Puślecki, Filip Ziolo
Recovery Crew (R)	Maciej Sikorski, Szymon Jutrzenka
CanSat Crew (C)	Paweł Rak, Maksymilian Tara, Hubert Kulik

Each role's responsibilities will be attached in the remaining two slides of this section.

Antenna construction and ground system setup

We want to make our design as simple as possible, to mitigate the risk of running out of time to prepare the setup during launch day. The laptop will be properly connected to XBee Radio via dedicated USB adapter and connected to the dipole antenna. CanSat will communicate with ground station via its onboard antenna, connected before probe check-in.

CanSat assembly and test:

MicroSD card will be inserted, electronics will be mounted, parachute will be attached, and heatshield stowed. The final communication tests will be performed right before check-in.

Overview of Mission Sequence of Events (2/3)

Arrival

- Arriving at launch site (whole team)
- Check for any damages that could appear during the travel (C)



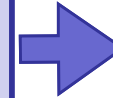
Setting up Ground Station

- Ground control station assembly (G)
- Antenna assembly (G)



Preparations

- Battery charge control (C)
- CanSat assembly and test (C)
- CanSat communication and sensors test (C/G)
- Mounting heatshield (C)
- Weight and size final check (MCO/C)
- Turning in the CanSat for inspection (MCO/C)

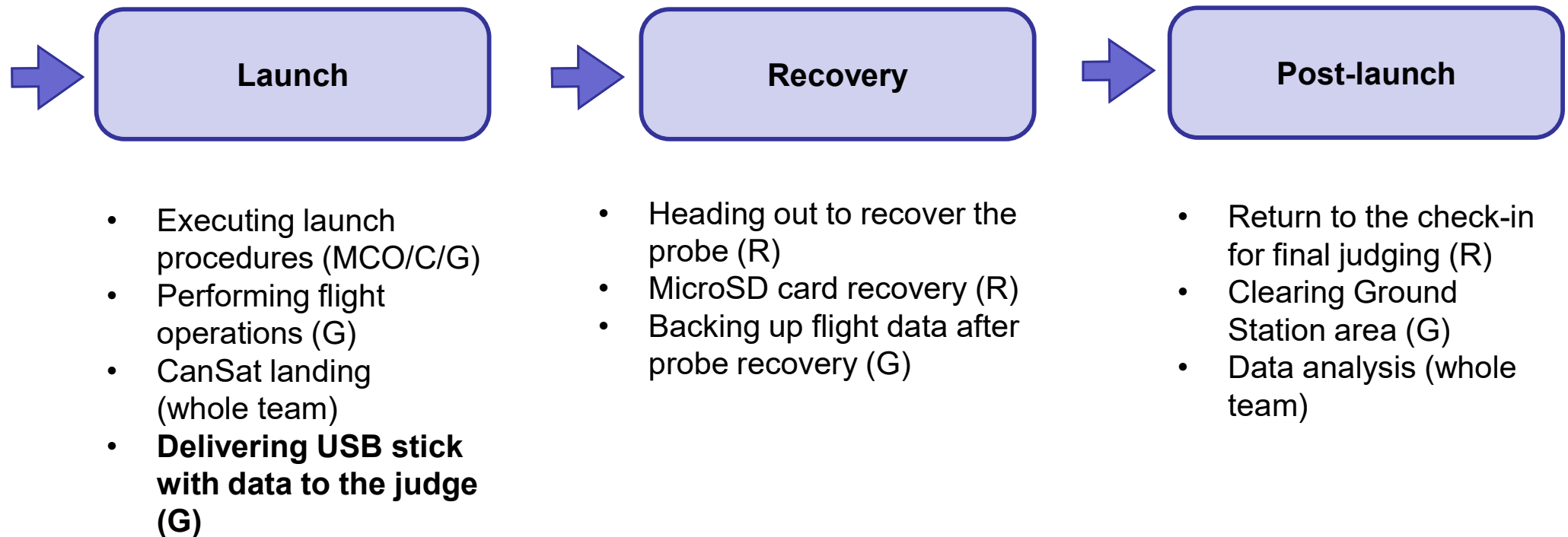


Pre-launch

- Moving GCS to dedicated position (G)
- Mounting CanSat in the rocket (MCO/G)
- Communication check (G)
- Sensors calibration (G)
- Safety check (whole team)



Overview of Mission Sequence of Events (3/3)



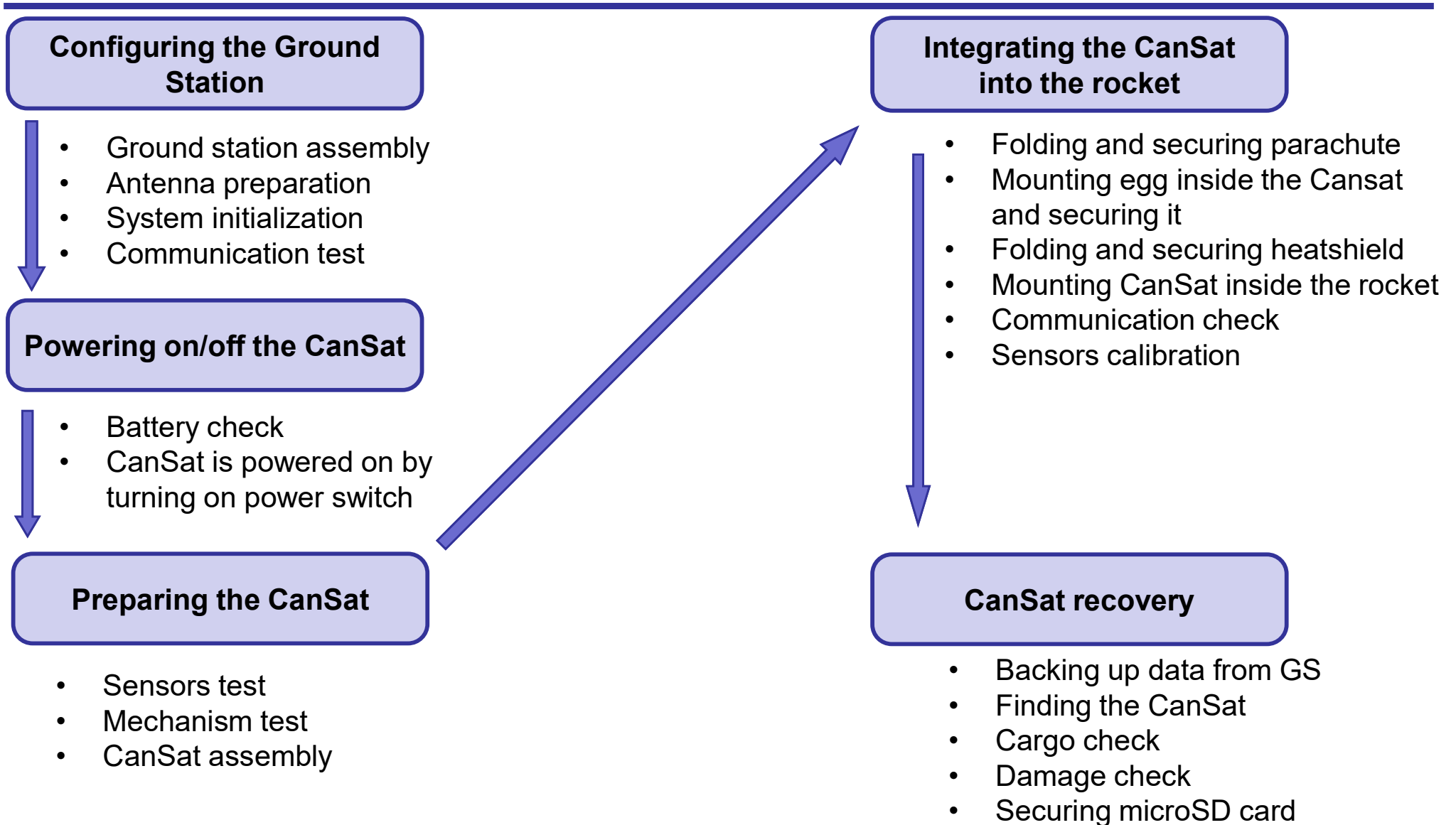
Development	Content
- The Mission Operations Manual will be developed using our experience from previous years and based on tests that are yet to be carried. - The final version of the operations manual will be ready after completion of Critical Design Review.	- Knowing the overview of the Mission Sequence of Events, some of the crucial points of the Mission Operation Manual can be indicated. - The diagram depicting them is presented on the Mission Rehearsal Activities slide.
Execution	
- Final version of the Mission Operation Manual will include not only Mission Sequence, but also team members' roles, Cansat Competition Guide and safety instructions. - Mission Operation Manual will be assembled into a three-ring binder. - The launch preparation, launch and removal procedures will be provided by the organizers.	

Probe Recovery

- Probe's color will be fluorescent orange so that it can be easily detected.
- An attached parachute will be in purple in color and a streamer will be attached to the probe.
- Landing zone will be determined by observing the descent and GPS location data.
- Speaker will be activated after touchdown.

Labeling

Payload will be labeled with the team email address, contact informations and phone number.



Requirements Compliance

Jakub Kaszowski

Current state of design meets all requirements although some changes may appear before construction in order to improve already working systems.

Tests are planned out and scheduled however the CanSat parts are still being produced. Only electronics elements were tested.

Further test are prepared in order to check if there is anything to improve before final design, even though constructing the probe as it is, should suffice to all the tasks mentioned in Mission Guide.

Following slides provide information about design compliance to requirements based on Mission Guide.

Operational Requirements (1/2)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
C1	The Cansat shall function as a nose cone during the rocket ascent portion of the flight.	Comply	10, 12, 45	-
C2	The Cansat shall be deployed from the rocket when the rocket motor ejection charge fires.	Comply	10, 46	-
C3	After deployment from the rocket, the Cansat shall deploy its heat shield/aerobraking mechanism.	Comply	10, 46	-
C4	A silver or gold mylar streamer of 50 mm width and 1.5 meters length shall be connected to the Cansat and released at deployment. This will be used to locate and identify the Cansat.	Comply	10, 38	-
C5	At 100 meters, the Cansat shall deploy a parachute and release the heat shield.	Comply	10, 48, 49	-
C6	Upon landing, the Cansat shall stop transmitting data.	Comply	81, 83	-
C7	Upon landing, the Cansat shall activate an audio beacon.	Comply	81, 83	-
C8	The Cansat shall carry a provided large hen's egg with a mass range of 51 to 65 grams	Comply	38, 50	-

Operational Requirements (2/2)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
C9	0 altitude reference shall be at the launch pad.	Comply	83, 96	-
C10	During descent with the heat shield deployed, the descent rate shall be between 10 to 30 m/s.	Comply	10, 35, 36, 46	-
C11	At 100 meters, the Cansat shall have a descent rate of less than 5 m/s.	Comply	10, 35, 36, 49	-
C12	Cost of the Cansat shall be under \$1000. Ground support and analysis tools are not included in the cost of the Cansat. Equipment from previous years shall be included in this cost, based on current market value.	Comply	144	-

Structural Requirements (1/2)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
S1	The CanSat mass shall be 900 grams +/- 10 grams without the egg being installed.	Comply	52 – 57, 58	-
S2	Nose cone shall be symmetrical along the thrust axis.	Comply	12, 41	-
S3	Nose cone radius shall be exactly 71 mm	Comply	12, 41	-
S4	Nose cone shoulder radius shall be exactly 68 mm	Comply	12, 15, 41	-
S5	Nose cone shoulder length shall be a minimum of 50 mm	Comply	12, 15, 41	-
S6	Cansat structure must survive 15 Gs vibration	Comply	51	-

Structural Requirements (2/2)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
S7	Cansat shall survive 30 G shock.	Comply	51	-
S8	The Cansat shall perform the function of the nose cone during rocket ascent	Comply	10, 45	-
S9	The rocket airframe can be used to restrain any deployable parts of the Cansat but shall allow the Cansat to slide out of the payload section freely.	Comply	10, 15, 45	-
S10	The rocket airframe can be used as part of the Cansat operations.	Comply	10, 45	-
S11	Nose cone shoulder length shall be a minimum of 50 mm	Comply	38, 42, 43, 51	-
S12	All electronics and mechanical components shall be hard mounted using proper mounts such as standoffs, screws, or high performance adhesives.	Comply	51	-

Mechanical Requirements (1/2)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
M1	No pyrotechnical or chemical actuators are allowed.	Comply	38	-
M2	Mechanisms that use heat (e.g., nichrome wire) shall not be exposed to the outside environment to reduce potential risk of setting the vegetation on fire	Comply	40	-
M3	All mechanisms shall be capable of maintaining their configuration or states under all forces	Comply	51	-
M4	Spring contacts shall not be used for making electrical connections to batteries. Shock forces can cause momentary disconnects.	Comply	51	-
M5	The Cansat shall deploy a heat shield after deploying from the rocket.	Comply	10, 48, 49	-
M6	The heat shield shall be used as an aerobrake and limit the descent rate to 10 to 30 m/s.	Comply	36, 35, 46	-

Mechanical Requirements (2/2)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
M7	At 100 meters, the Cansat shall release a parachute to reduce the descent rate to less than 5 m/s.	Comply	36, 35, 48, 49	-
M8	The Cansat shall protect a hens egg from damage during all portions of the flight.	Comply	50	-
M9	If the nose cone is to be considered as part of the heat shield, the documentation shall identify the configuration.	Not applicable	40	The nosecone is NOT a part of the heatshield
M10	After the Cansat has separated from the rocket and if the nose cone portion of the Cansat is to be separated from the rest of the Cansat, the nose cone portion shall descend at less than 10 meters/second using any type of descent control device.	Not applicable	48	The nosecone DOES NOT separate from the CanSat

Electrical Requirements

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
E1	Lithium polymer batteries are not allowed.	Comply	75	-
E2	Battery source may be alkaline, Ni-Cad, Ni-MH or Lithium. Lithium polymer batteries are not allowed. Lithium cells must be manufactured with a metal package similar to 18650 cells. Coin cells are allowed.	Comply	75	-
E3	Easily accessible power switch is required	Comply	13, 42, 72	-
E4	Power indicator is required	Comply	13, 42, 72	-
E5	The Cansat shall operate for a minimum of two hours when integrated into the rocket.	Comply	77	-

Communications Requirements

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
X1	XBee radios shall be used for telemetry. 2.4 GHz Series radios are allowed. 900 MHz XBee radios are also allowed.	Comply	60, 65	-
X2	XBee radios shall have their NETID/PANID set to their team number	Comply	65	-
X3	XBee radios shall not use broadcast mode.	Comply	65	-
X4	The Cansat shall transmit telemetry once per second.	Comply	65	-
X5	The Cansat telemetry shall include altitude, air pressure, temperature, battery voltage, Cansat tilt angles, air speed, command echo, and GPS coordinates that include latitude, longitude, altitude and number of satellites tracked.	Comply	66, 67, 68	-

Sensor Requirements

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
SN1	Cansat shall measure its speed with a pitot tube during ascent and descent.	Comply	17, 23	-
SN2	Cansat shall measure its altitude using air pressure.	Comply	17, 19	-
SN3	Cansat shall measure its internal temperature.	Comply	20	-
SN4	Cansat shall measure its angle stability with the aerobraking mechanism deployed.	Comply	24	-
SN5	Cansat shall measure its rotation rate during descent.	Comply	25	-
SN6	Cansat shall measure its battery voltage	Comply	22	-
SN7	The Cansat shall include a video camera pointing horizontally.	Comply	26	-
SN8	The video camera shall record the flight of the Cansat from launch to landing.	Comply	26	-
SN9	The video camera shall record video in color and with a minimum resolution of 640x480.	Comply	26	-

Ground Station Requirements (1/3)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
G1	The ground station shall command the Cansat to calibrate the altitude to zero when the Cansat is on the launch pad prior to launch.	Comply	96	-
G2	The ground station shall generate csv files of all sensor data as specified in the Telemetry Requirements section.	Comply	96	-
G3	Telemetry shall include mission time with 1 second or better resolution.	Comply	96	-
G4	Configuration states such as zero altitude calibration shall be maintained in the event of a processor reset during launch and mission.	Comply	84	-
G5	Each team shall develop their own ground station.	Comply	89	-
G6	All telemetry shall be displayed in real time during descent on the ground station.	Comply	96,97	-
G7	All telemetry shall be displayed in engineering units (meters, meters/sec, Celsius, etc.) and the units shall be indicated on the displays.	Comply	96,97	-
G8	Teams shall plot each telemetry data field in real time during flight.	Comply	96,97	-

Ground Station Requirements (2/3)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
G9	The ground station shall include one laptop computer with a minimum of two hours of battery operation, XBEE radio and an antenna.	Comply	89, 91	-
G10	The ground station must be portable so the team can be positioned at the ground station operation site along the flight line. AC power will not be available at the ground station operation site.	Comply	91	-
G11	The ground station software shall be able to command the payload to operate in simulation mode by sending two commands, SIMULATION ENABLE and SIMULATION ACTIVATE.	Comply	96	-
G12	When in simulation mode, the ground station shall transmit pressure data from a csv file provided by the competition at a 1 Hz interval to the CanSat.	Comply	96	-

Ground Station Requirements (3/3)

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
G13	The ground station shall use a tabletop or handheld antenna.	Comply	94	Handheld antenna is used
G14	Because the ground station must be viewed in bright sunlight, the displays shall be designed with that in mind, including using larger fonts (14 point minimum), bold plot traces and axes, and a dark text on light background theme.	Comply	91	-
G15	The ground system shall count the number of received packets. Note that this number is not equivalent to the transmitted packet counter, but it is the count of packets successfully received at the ground station for the duration of the flight.	Comply	96	-

Flight Software Requirements

Rqmt Num	Requirement	Comply / No Comply / Partial	Slide	Notes
F1	The flight software shall maintain a count of packets transmitted which shall increment with each packet transmission throughout the mission. The value shall be maintained through processor resets.	Comply	66, 68, 84	-
F2	The Cansat shall maintain mission time throughout the whole mission even with processor resets or momentary power loss.	Comply	63, 72, 83	-
F3	The Cansat shall have its time set to within one second UTC time prior to launch.	Comply	66, 83	-
F4	The flight software shall support simulated flight mode where the ground station sends air pressure values at a one second interval using a provided flight profile csv file.	Comply	85	-
F5	In simulation mode, the flight software shall use the radio uplink pressure values in place of the pressure sensor for determining the payload altitude.	Comply	85	-
F6	The payload flight software shall only enter simulation mode after it receives the SIMULATION ENABLE and SIMULATION ACTIVATE commands.	Comply	85	-

Management

Jakub Kaszowski

We don't expect **any** problem with missing components

Mechanics:

All mechanical elements are obtained. The CanSat prototype exist.
3D printer material is prepared for eventual changes

Electronics:

All sensors, actuators, electronic components, PCBs are already procured, shipped and assembled

Ground Station:

All Ground Station components are prepared for the tests.

Software:

No procurements are required

Construction – components and electronics					
Component	Model/Material	Quantity	Price per unit [USD]	Total [USD]	Status
Structural Ring	ASA	1	1.84	1.84	Actual
Aerodynamic Fairings	PET	1	5	5	Estimated
Shoulder	ASA	1	1.68	1.68	Actual
Nose cone Fairing	Nylon PA 6	1	4.95	4.95	Actual
Pitot tube	PET-G	1	0.04	0.04	Estimated
Pitot mounting struts	steel	2	0.1	0.2	Estimated
Pitot dampening springs	steel	2	0.5	1	Actual
Detachable Structural Ring	ASA	1	2	2	Actual
Camera Glass	Glass	1	0.5	0.5	Estimated
Detach mechanism structure	ASA	1	1.2	1.2	Actual
Parachute compartment structure	ASA	1	2.64	2.64	Actual
Auxiliary Camera Mount	PET-G	1	0.08	0.08	Actual
Long Brackets	Steel	2	0.2	0.4	Estimated
Short Brackets	Steel	4	0.2	0.8	Estimated
Parachute mounting U-Bolt	Steel	1	1	1	Estimated

Construction – components and electronics					
Component	Model/Material	Quantity	Price per unit [USD]	Total [USD]	Status
LED holders	Aluminium	6	0.05	0.3	Estimated
Main Struts	Carbon fiber	4	10	40	Actual
Spring Distancer	PET	4	0.05	0.2	Actual
Dampening spring	Steel	8	0.5	4	Actual
Egg Box	ASA	1	3	3	Actual
Foam insert	Foam	1	1	1	Estimated
Egg Box dooor	ASA	1	0.4	0.4	Actual
Pitot air pipe	Plastic	1	0.5	0.5	Actual
50mm x 2m mylar streamer	PP	2	1	2	Actual
Heatshield material	Ripstop nylon	1	6	6	Actual
Parachute material	Ripstop nylon	1	10	10	Actual
Nylon thread	Nylon	4m	0.3	1.2	Actual
M3 x 5.5 mm Brass inserts	Brass	16	0.1	1.6	Actual
M3 x 8 mm screw	Steel	16	0.1	1.6	Actual

Construction – components and electronics					
Component	Model/Material	Quantity	Price per unit [USD]	Total [USD]	Status
M3 5.5 2.4 ISO Nuts	Steel	6	0.05	0,3	Actual
Board	-	1	2	2	Actual
Intergrated circuits	-	1	24	24	Actual
Passive components	-	-	4	4	Actual
Connectors	-	4	6	24	Actual
XBee 900MHz	XBP9B-XCST-001	1	54	54	Actual
BLDC	PM3505	1	30	30	Actual
Camera module	ESP32-CAM	1	10	10	Actual
Main Camera PCB	PCB	1	3	3	Actual
Battery	Li-ion 14500	1	7.5	7.5	Actual
Secondary Camera	Adafruit 3202	1	16	16	Actual
GPS	MTK3339	1	38	38	Actual
Battery	Li-ion 18650	1	7.5	7.5	Actual
Pitot's tube sensor	ABPDJTT001PDSA3	1	12	12	Actual

Construction – components and electronics					
Component	Model/Material	Quantity	Price per unit [USD]	Total [USD]	Status
Speaker	57CS50G-75ND	1	3	3	Actual
Electromagnets	JF-0530B	2	5	10	Actual
LED Stripe	-	0.4	1.5	0.6	Estimated
Battery holder	-	1	1	1	Actual

Total cost [USD]
342.03

The maximum cost of a CanSat is specified as \$1000. The cost of our CanSat is well within this limit with plenty of room to spare.

We reuse a BLDC PM3505 motor as it is the best choice for our application and it is hard to get. Our motor, although used, is in an excellent condition The price given in the budget is a price we found for this motor on e-commerce sites.

Prices were converted from Polish złoty to US dollars using current exchange rate, which was USD/PLN = 4.00 at the time.

Ground Control Station					
Component	Model/Material	Quantity	Price [PLN/USD]	Total [PLN/USD]	Status
XBee USB Adapter	XBee Adapter USB v2.2 - DFRobot DFR0174	2	105.00 / 26.25	210.00 / 52.50	Actual
XBee Radio	XBP9B-XCST-001	2	140.93 / 35.23	281.86 / 70.46	Estimated
RPSMA XBee	DELTACO SMA-FM500	1	100.00 / 25.00	100.00 / 25.00	Actual
Antenna	Interline HORIZON 900 8V	1	400.00 / 100.00	400.00 / 100.00	Actual
RPSMA Antenna	DELTACO SMA-FM500	1	100.00 / 25.00	100.00 / 25.00	Actual
Total [PLN/USD]				1 091.86 / 272.96	

Computer for Ground Station is not included, as we are going to use one of our own.

Prototyping	
Carbon rods	200.00/50.00
Heatshield and Parachute material	100.00/25.00
3D printer filament	500.00/125.00
Electronics	600.00 / 150.00
Total [PLN/USD]	1400.00/ 350.00

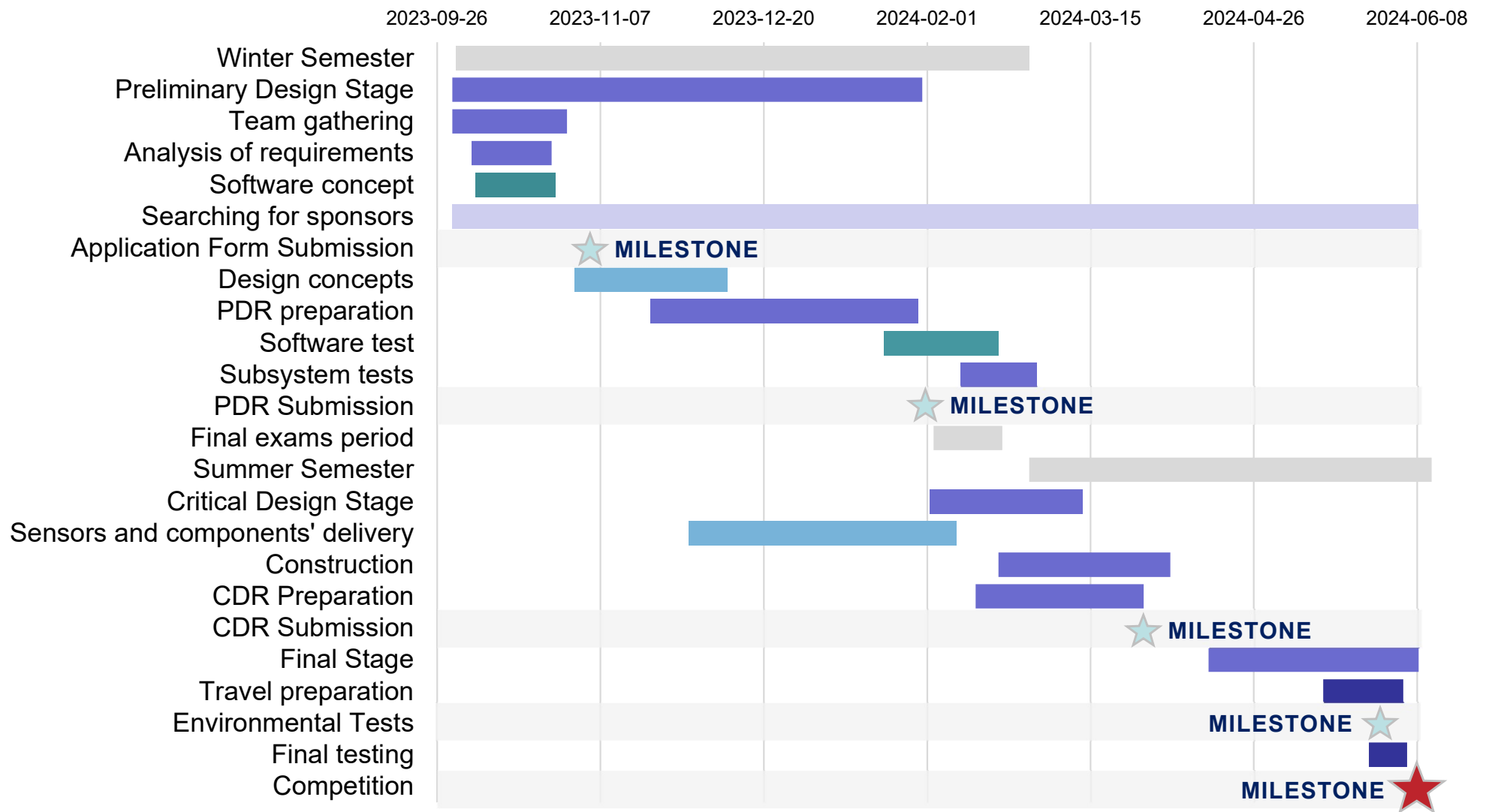
We lowered prototyping costs thanks to a barter deal with sponsor.

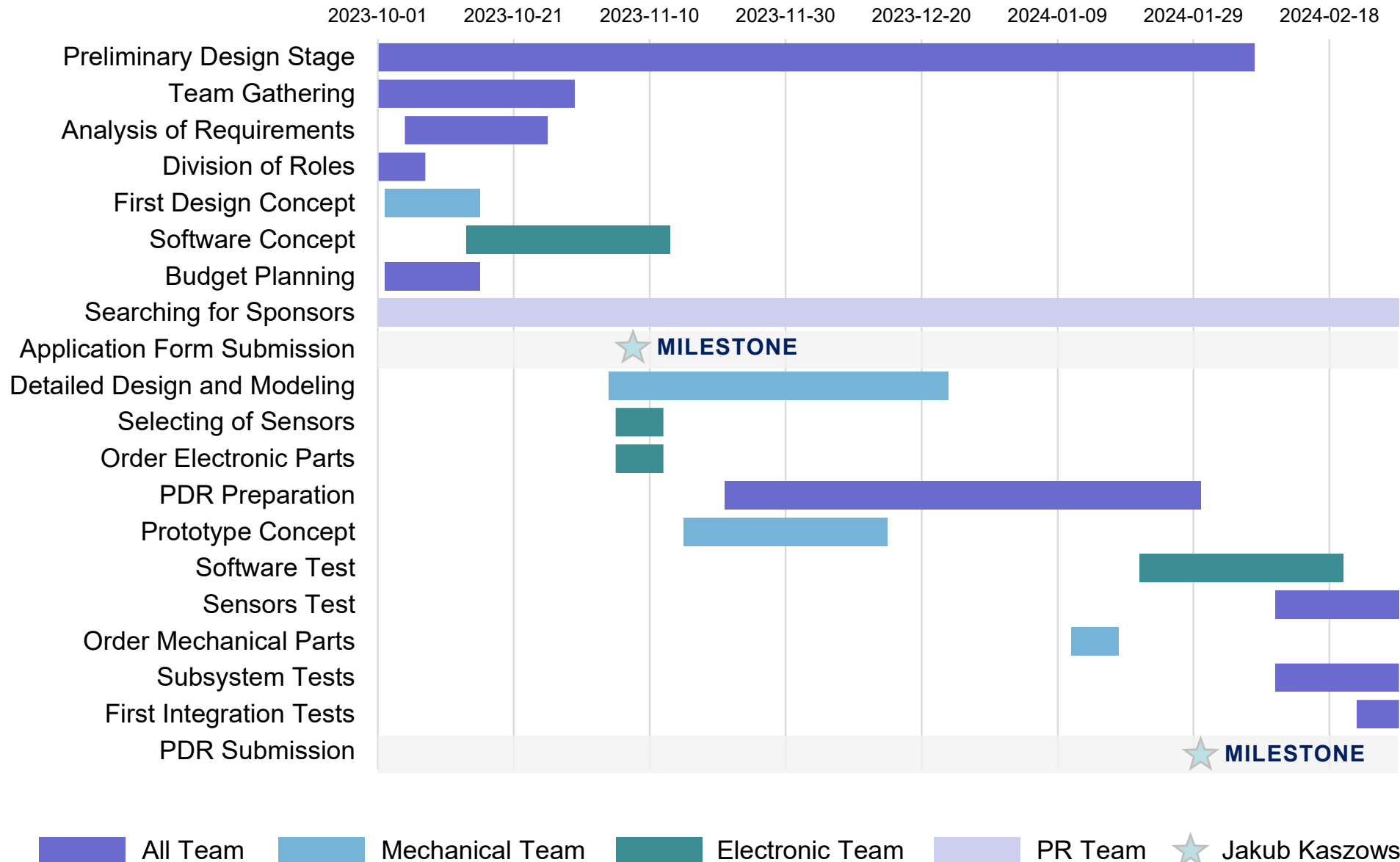
Attending cost	Quantity	Price [PLN/USD]	Total [PLN/USD]
Travel	10	4 900.00/1 225.00	49 000.00/ 12 250.00
Accommodation and food	10	5 000.00/ 1 250.00	50 000.00/12 500.00
T-shirts	10	59.00/14.75	590.00/147.50
Registration	1	794.00/198.50	794.00/198.50
Car rent + fuel	2	6 850.00/1 713.00	13 700.00/3 425.00
Total cost			114 084.00/28 521.00

Overall mission cost [PLN/USD]	117943.96/29 485.99
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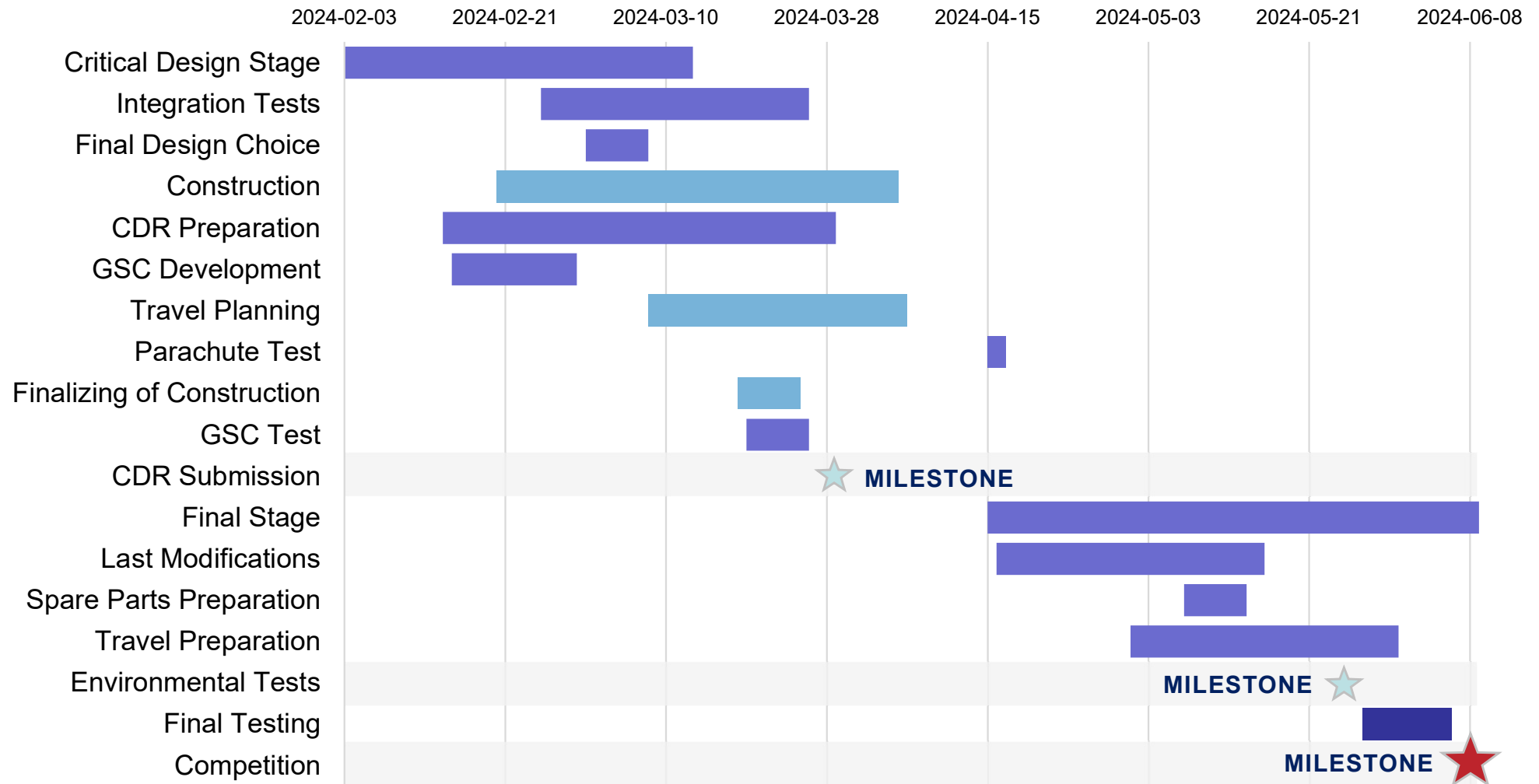
Sources of income [PLN/USD]	
University	49 000.00/ 12 250.00
Sponsors	10 000.00/ 2 500.00

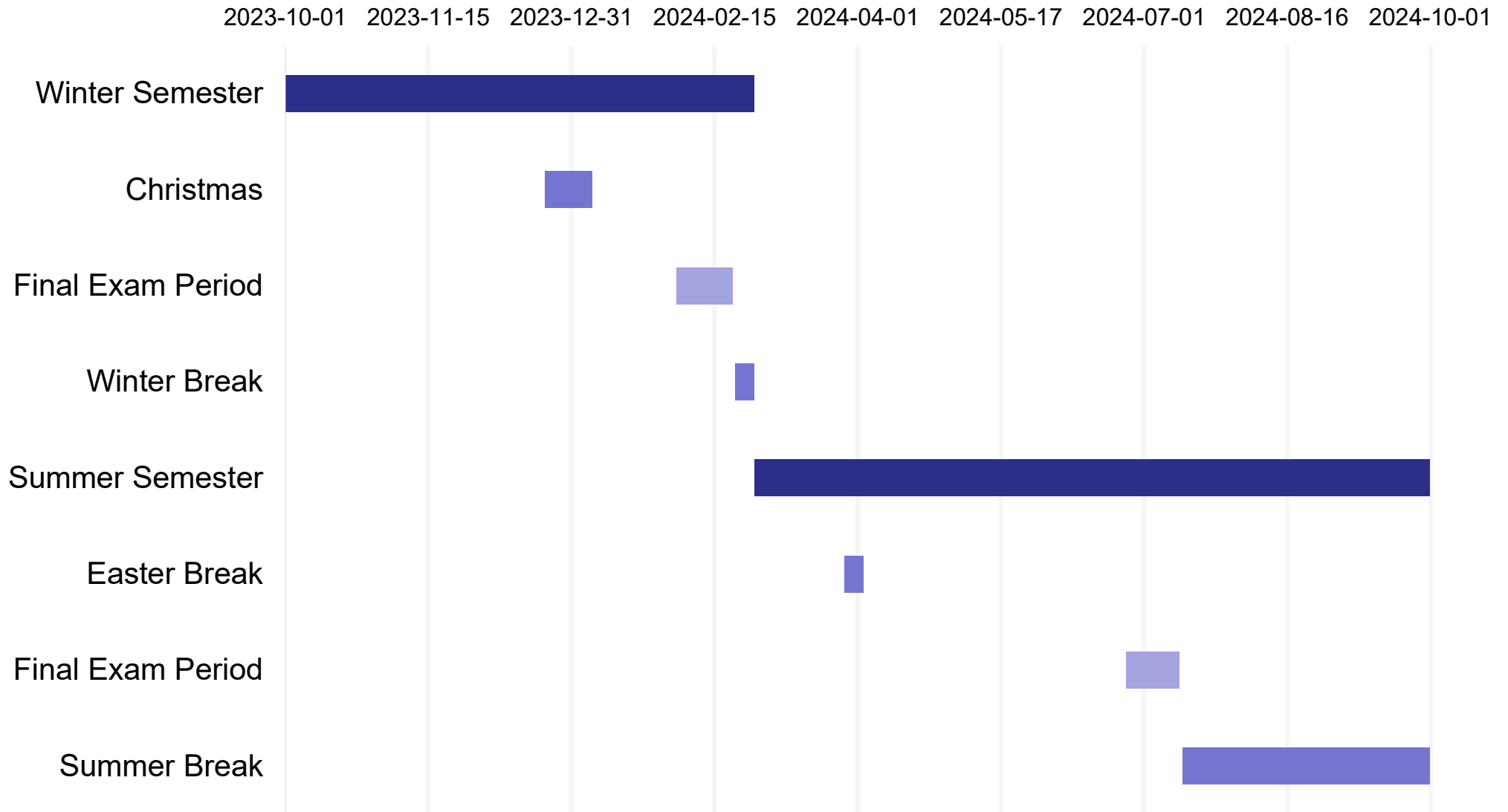
Some prices are estimates and are subject to change. We are still talking to sponsors.
The final mission cost **includes** the cost of the probe.





Detailed Program Schedule (2/4)





In order to estimate our total progress we split the whole mission into 5 parts:
Electronics, Mechanics, Software & Ground Station, Integration & Test, Management
The progress in a part was estimated by members who were working on the given part

Part	Estimated progress
Electronics	65%
Mechanics	90%
Software & Ground Station	90% , 90%
Integration & Test	50%
Management	60%

The overall progress is estimated as the average

Total progress: 71%

- We are planning to take CanSat hardware as parts of our hand luggage in order to make sure nothing gets lost on the way. The same holds for any necessary equipment.
- Non-essential subsystems and tools will be shipped in checked baggage, as they could be easily replaced in case the luggage went missing. We don't take any restricted parts to USA.
- CanSat will be transported to the launch site by car together with the team.

Major accomplishments
- Teamwork has been organized and conducted according to principles of Scrum methodologies including weekly meetings in our laboratory. - Most of the tasks are finished on schedule. - There is a lot of initiative on the part of team members.
Major unfinished work
- A fully functioning prototype is yet to be finished. - Pitot tube with its sensor need calibration.
Testing to complete
- Environmental tests need to be done. - Simulated landing with egg inside is yet to be completed.
Flight software status
- FSW system will be tested soon. - The state machine concept has already been created. It will be tested in two weeks. - Camera, memory card logging and heatshield release mechanism support are implemented.